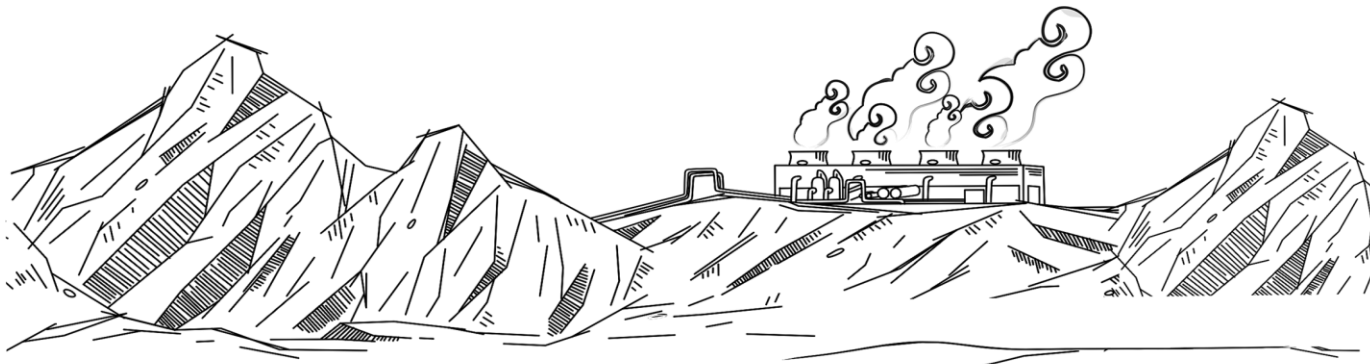


GEOFYSICAL EXPLORATION METHODS

TRAINING SISTEM,
TEKNOLOGI DAN REVIEW
DOKUMEN FS
GEOTHERMAL

Angga Bakti Pratama S.Si, M.T.
KK Geothermal – FTTM ITB



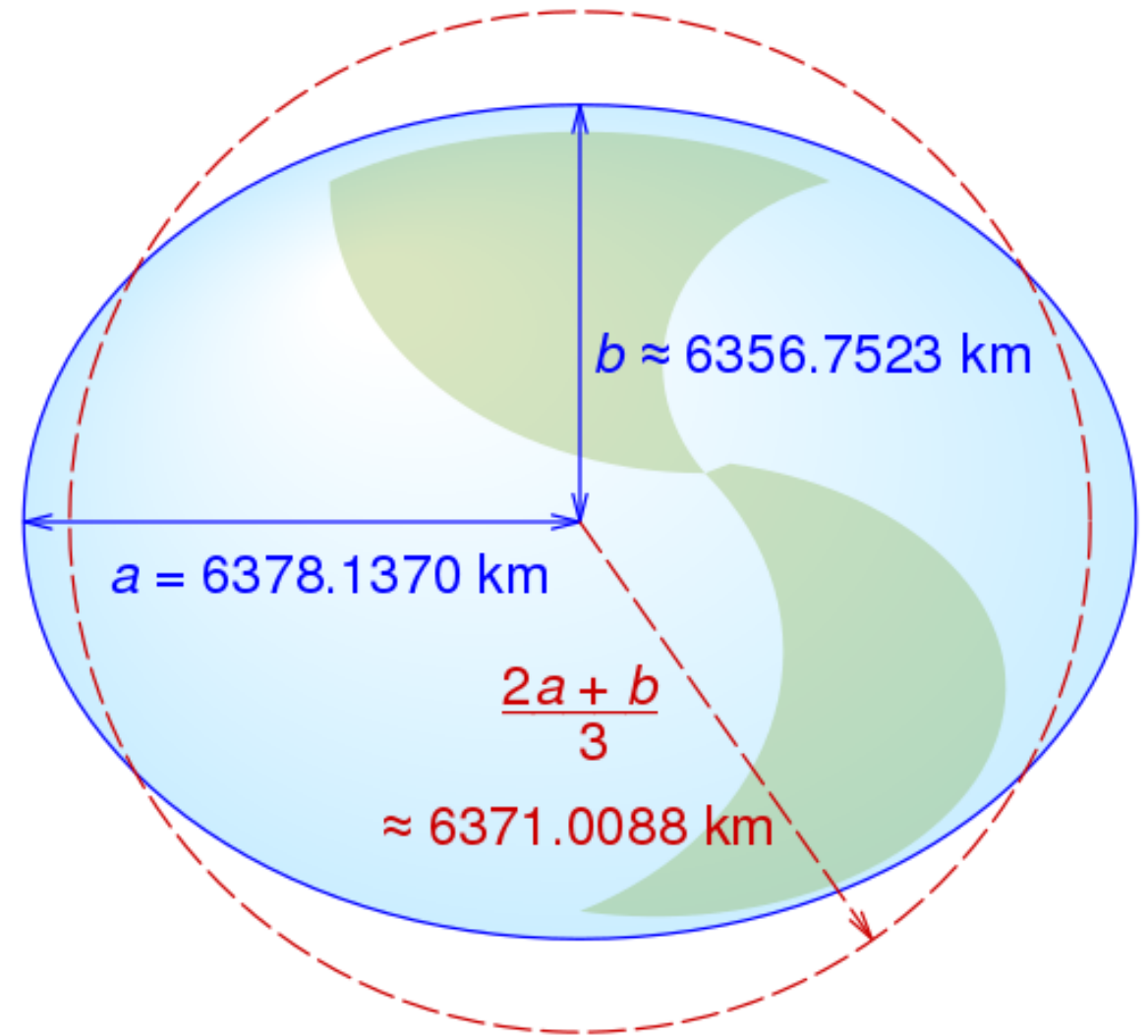
Outline

- Intro
- How geophysics works
- Role geophysics in geothermal exploration
- Methods in geothermal exploration



INTRODUCTION

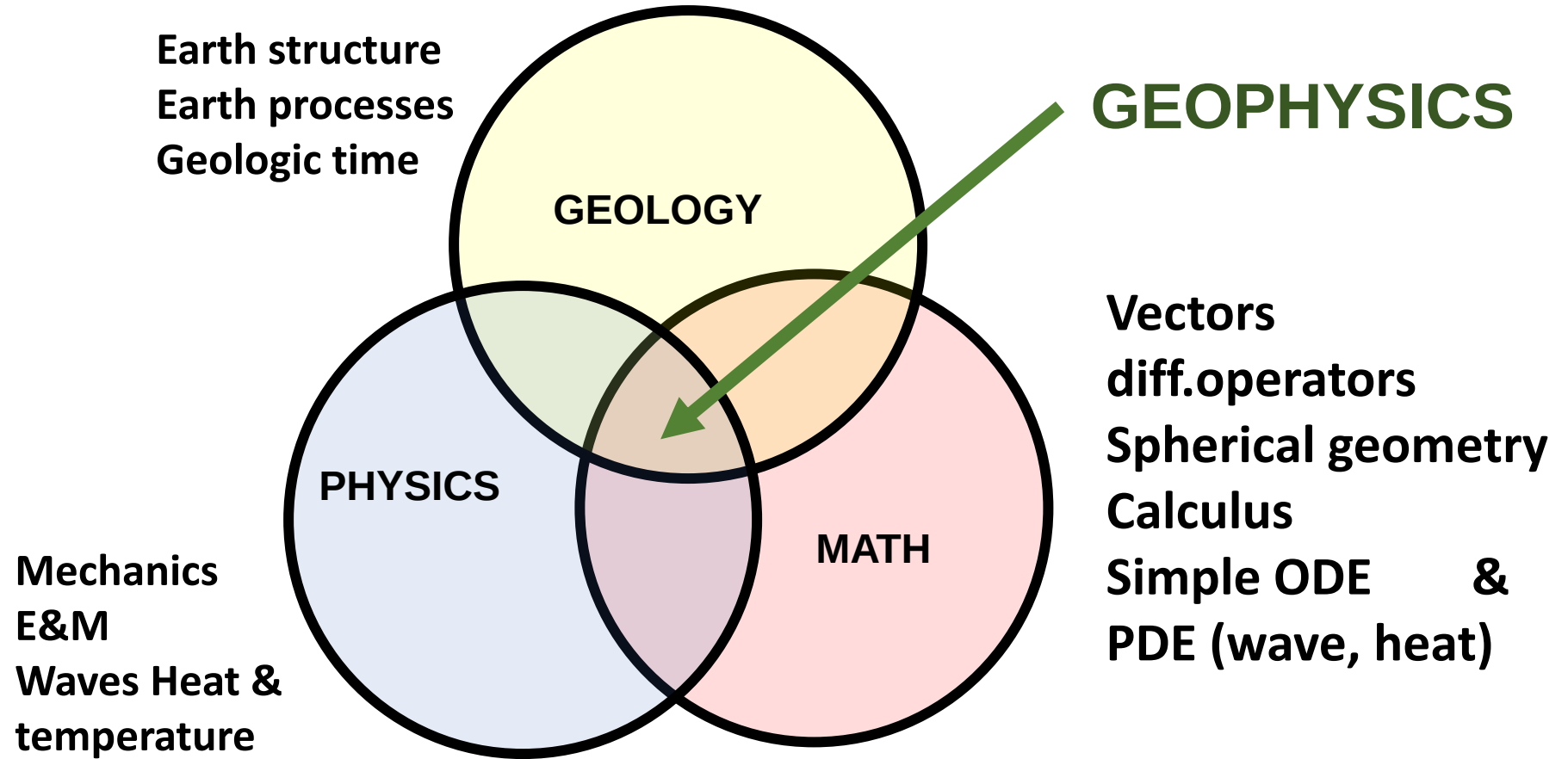
- How can we obtain information on the Earth's Interior?
 - Radius of earth: ~6400 km
 - Deepest well only reach 12 km



INTRODUCTION

- Deepest well in the world “Kola Superdeep Borehole” with 12226 m.

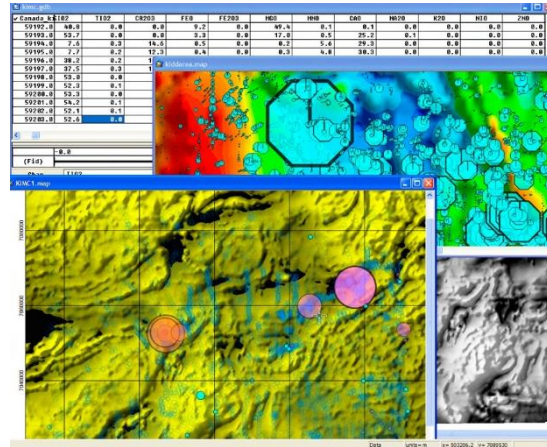




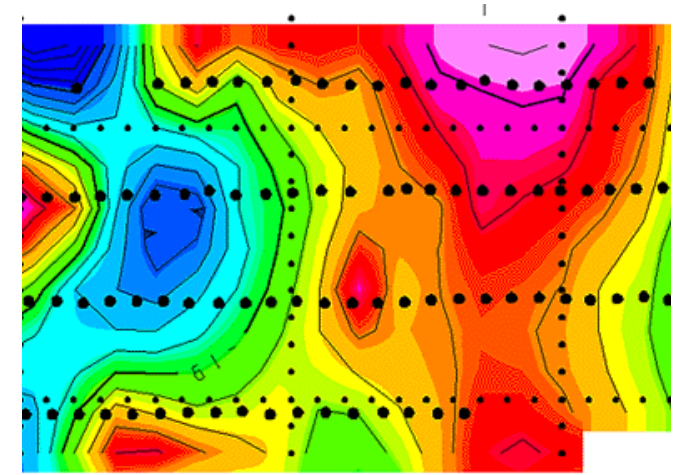
HOW'S GEOPHYSICS WORKS



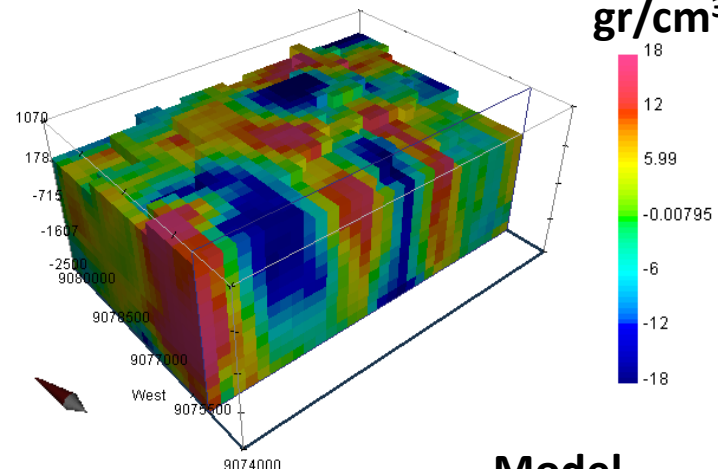
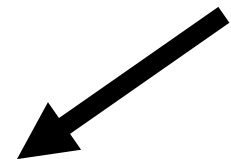
Field Survey



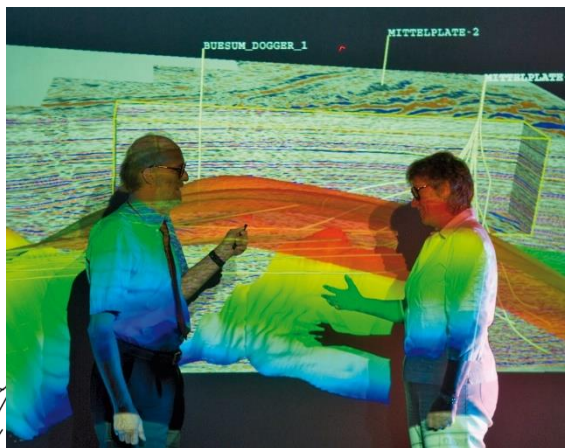
Data Processing



Data



Model

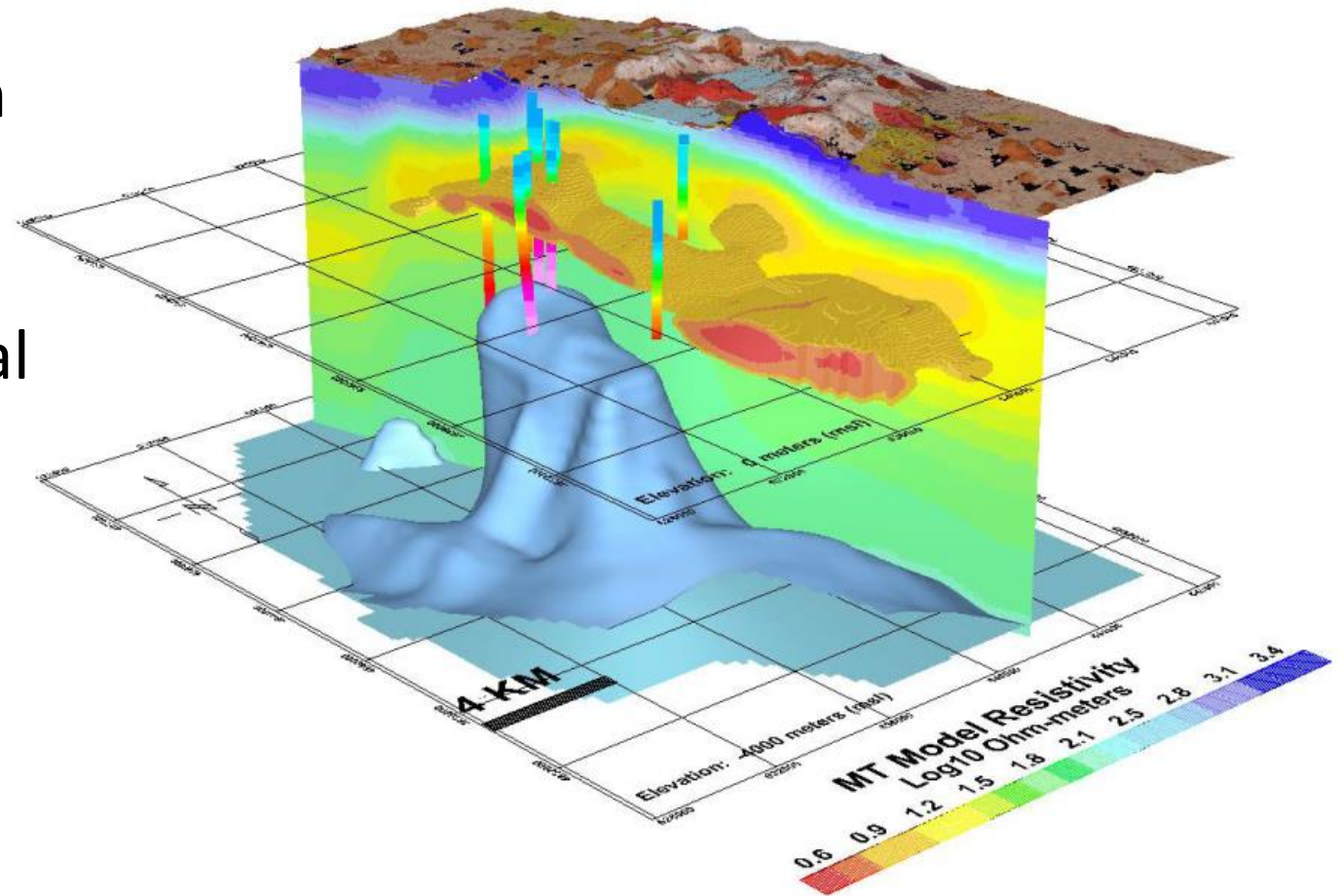


Interpretation



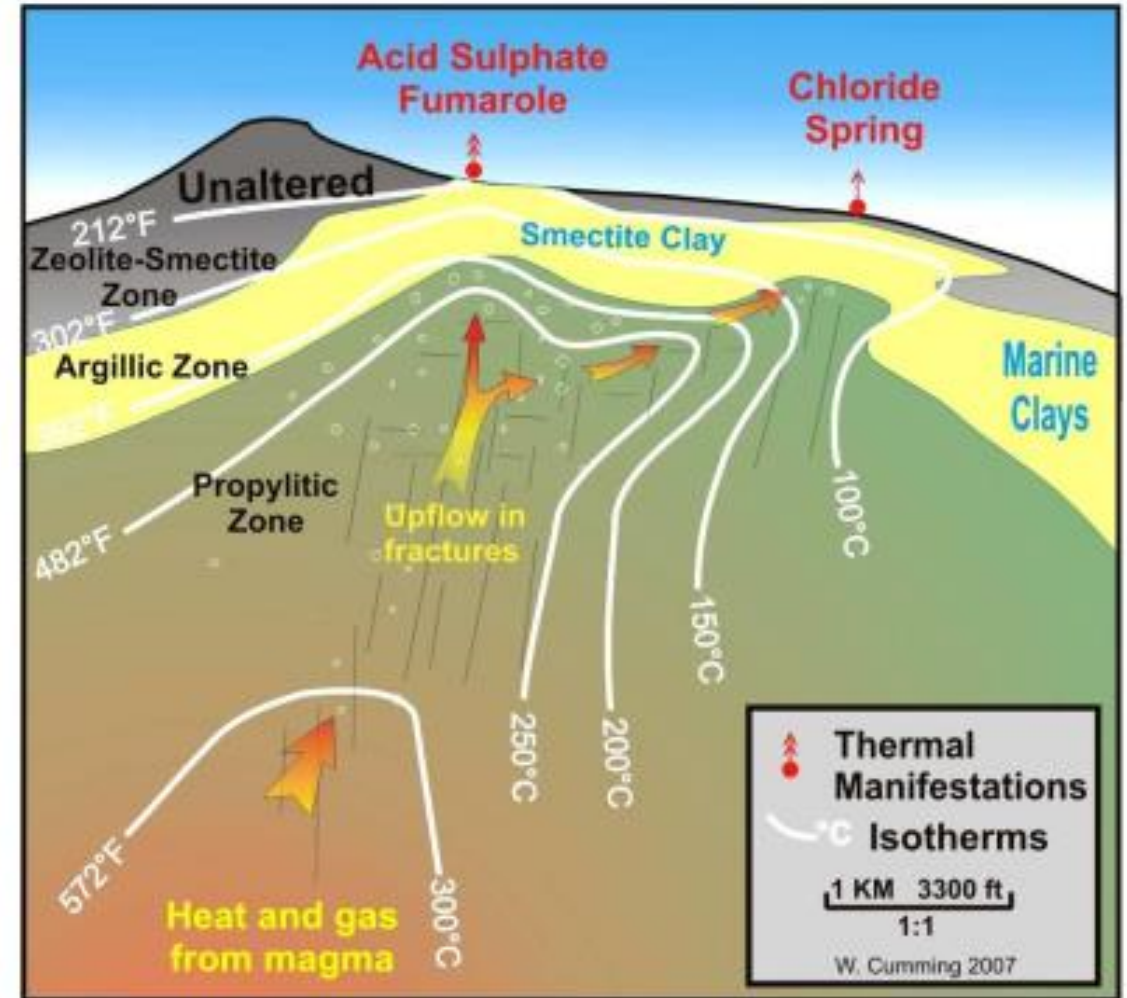
GEOPHYSICIST ROLE

- Using concept in physics with understanding of earth to image subsurface
- Define subsurface geothermal features boundaries of reservoir
- Identify drilling targets



GEOTHERMAL SYSTEM

- Elements of a geothermal system
 - Heat Source
 - Reservoir
 - Caprock
 - Recharge system
- Role of Geophysics
 - Geometry
 - Lithology
 - Conceptual Model
 - Resource Estimation



Parameter Fisis

- Suseptibilitas Magnetik
- Densitas
- Konduktivitas Listrik
- Permittivitas Dielektrik
- Temperatur
- Kecepatan Gelombang

Metode Geofisika

- Magnetik
- Gaya Berat (gravity)
- Geolistrik
- Magnetotelurik
- TDEM
- Self Potensial
- Seismik



GEOFYSICAL EXPLORATION IN GEOTHERMAL

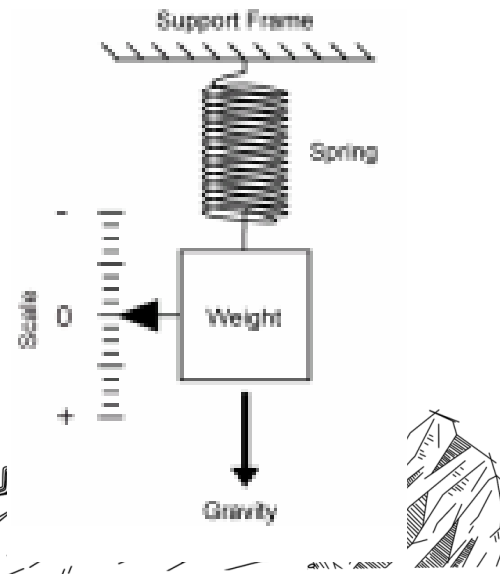
- Preliminary
 - Gravity
 - Magnetic
 - Geo-electrical Resistivity (sounding and mapping)
- Advanced
 - Magnetotellurics (MT)
 - TDEM
 - Micro-Earthquake (MEQ)



GRAVITY

Mengukur **variasi medan gaya berat** di permukaan bumi pada setiap titik amat, digunakan untuk mendeteksi kontras rapat massa sumber anomali terhadap lingkungannya (**kontras densitas**).

Batuan sumber panas dan daerah akumulasinya di bawah permukaan bumi → perbedaan densitas antara massa batuan di sekitarnya.

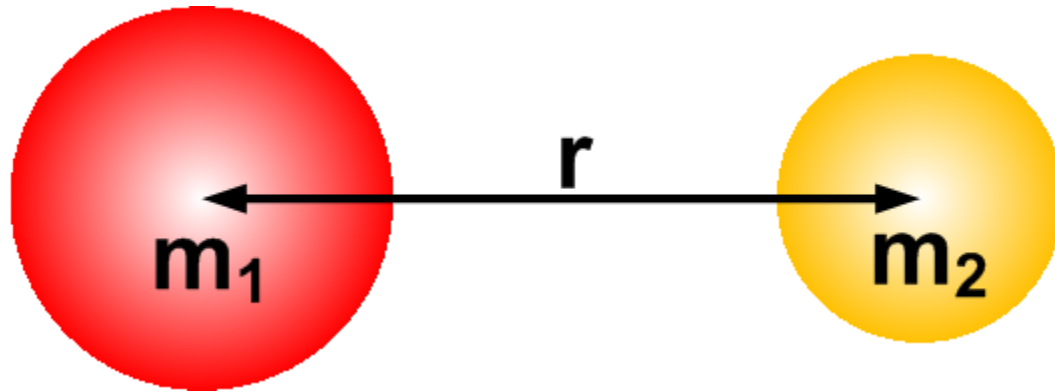


Analisa hasil survey gaya berat:

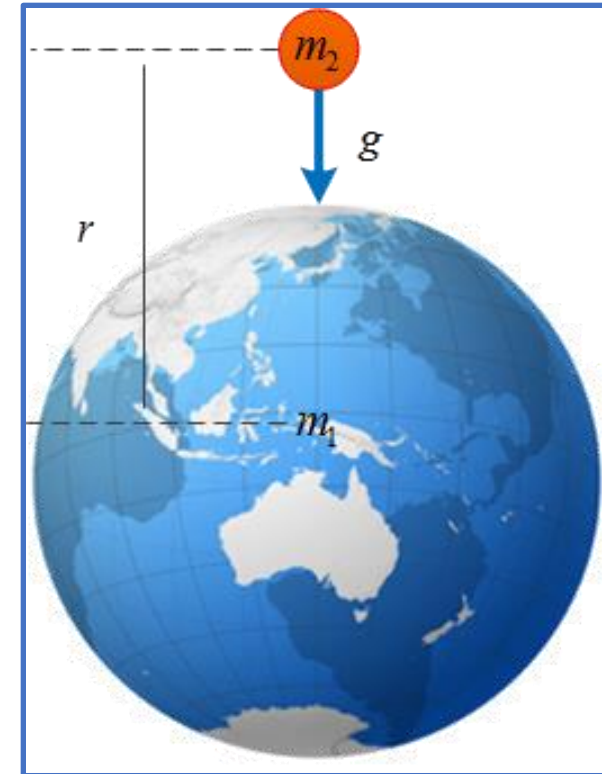
- Struktur batuan dasar
- Keberadaan sesar sebagai pengontrol sistem geothermal
- Batuan intrusi sebagai sumber panas sistem geothermal



METODE GAYA BERAT (GRAVITY)

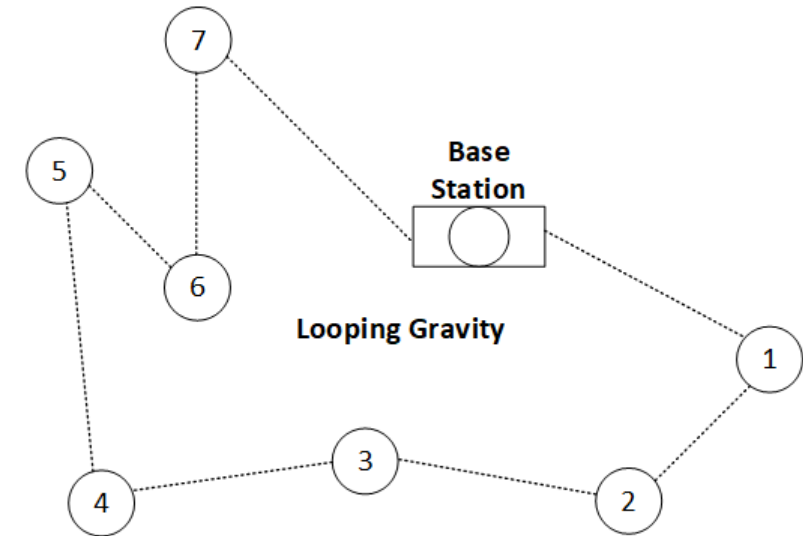


$$F = \frac{Gm_1m_2}{r^2} \longrightarrow \cancel{mg} = \frac{G\cancel{m_1}m_2}{r^2}$$

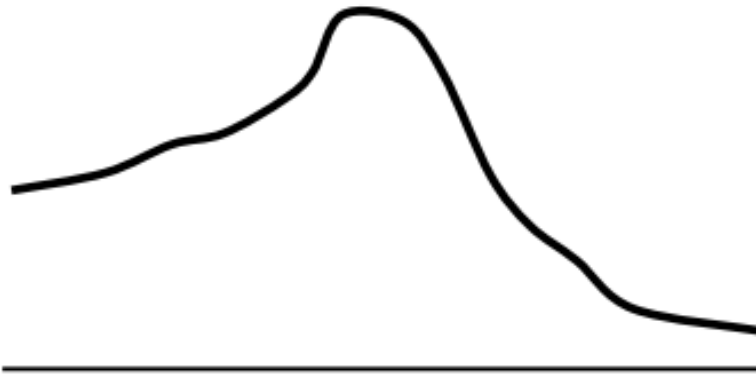


MEASUREMENT PROCEDURE

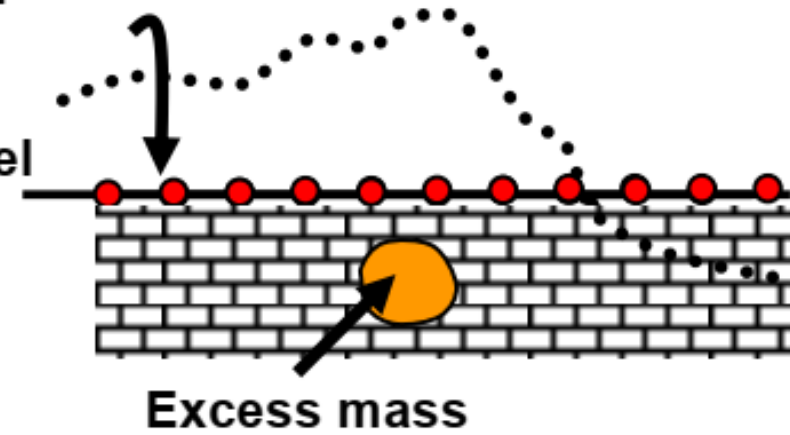
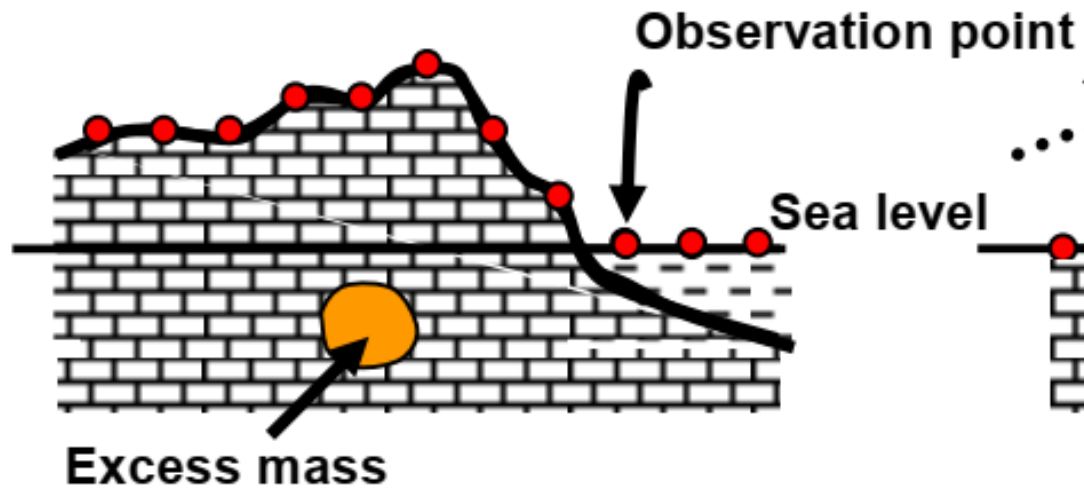
- Start gravity measurement at base station
- Gravity Measurement at stations distributed over the survey area
- End gravity measurement at base station



Observed gravity anomaly



**Bouguer gravity anomaly
due to the excess mass**



ANOMALI GRAVITASI

- **Gravity anomaly** = difference between observed gravity dan theoretical gravity
- **Observed gravity**= measured gravity corrected from factors affecting measurement (tide and drift)
- **Theoritical gravity** = normal gravity or corrected factors to meet measurement condition (topography)

$$g_{anomaly} = g_{observed} - g_{theory}$$



GRAVITY

Variasi Temporal :

- Instrument drift,
- Efek Tidal

Variasi Spasial :

- Variasi Lintang
- FAC
- Koreksi Bouguer
- Koreksi Medan (Terrain)

Bouguer Anomaly :

$$\Delta g = g_{\text{obs}} - g_n + \text{FAC} - \text{BC} + \text{TC}$$

g_n = normal gravity of the earth (Geoid)

$$g(\phi) = 978032.7 \left(1 + 0.0053024 \sin^2 \phi - 0.0000058 \sin^2 2\phi \right)$$

TAC = Free-air correction

BC = Bouguer correction

TC = Terrain correction

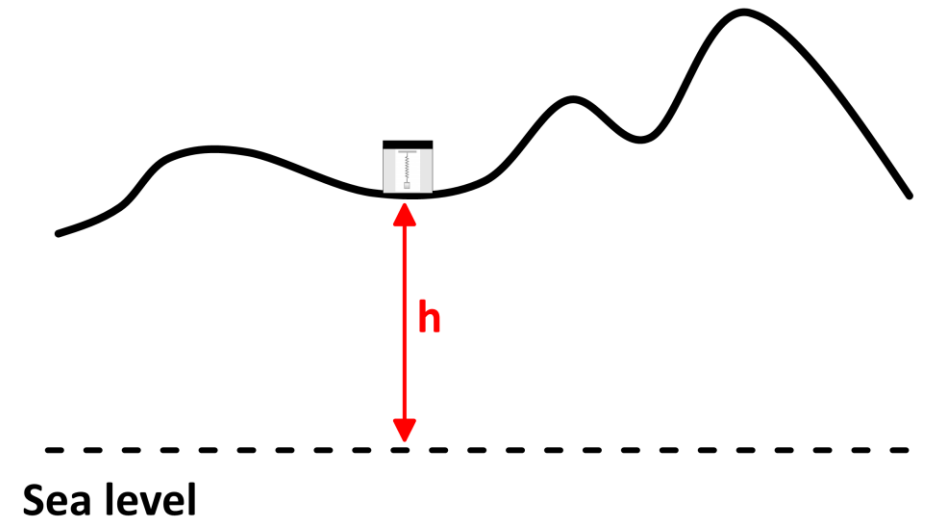


Koreksi Ketinggian !

Free Air Correction

- Correct the change in gravity due to change in the height of the observation point.

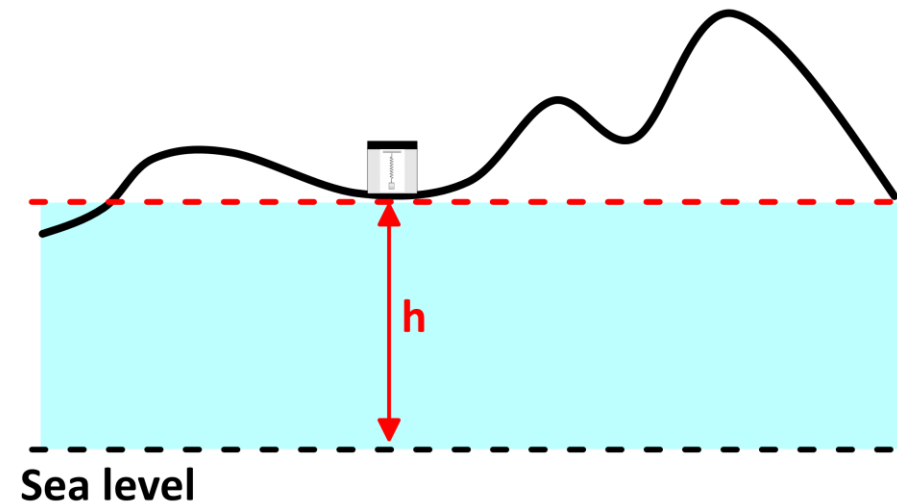
$$g_{fa} = -0.3086 \times h$$



Bouguer Correction

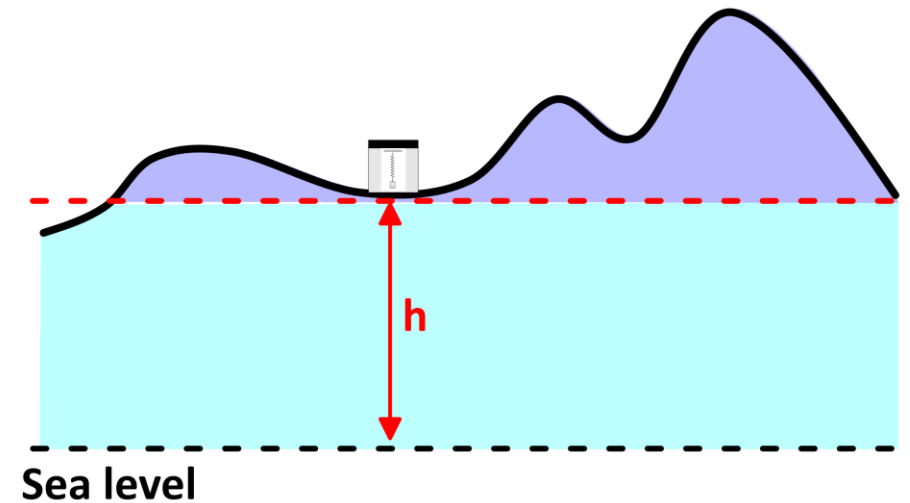
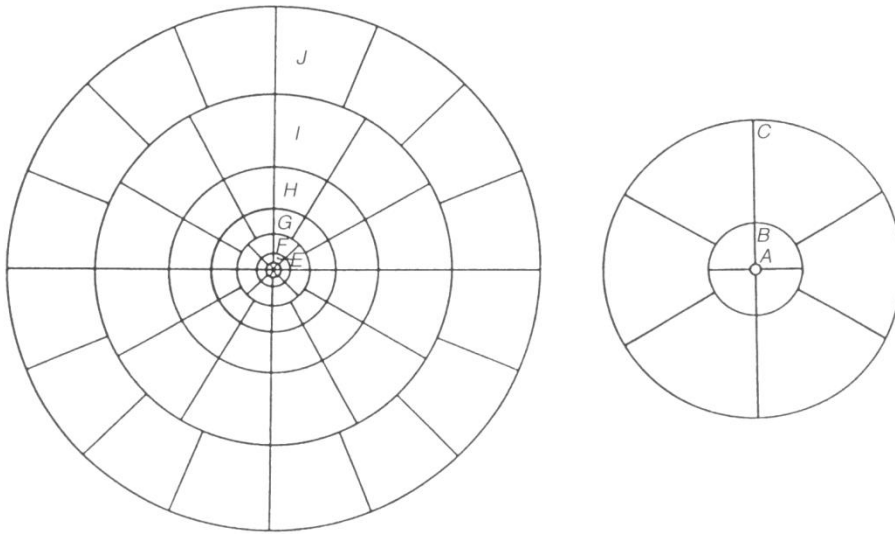
- Correct the gravity change contributed by the rock mass found between the sea level and the observation point.

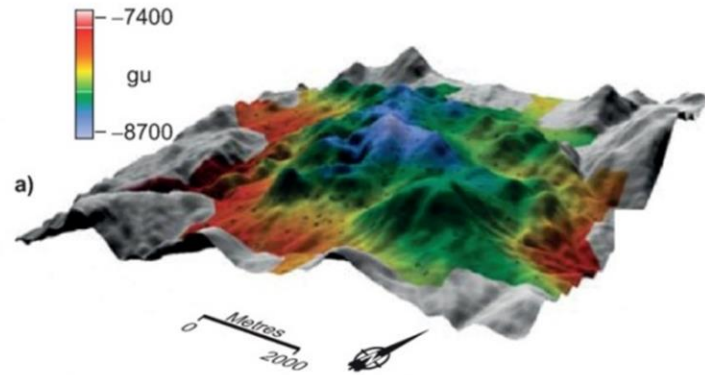
$$\begin{aligned} g_{BC} &= 2\pi G \rho h \\ &= 0.04193 \rho h \end{aligned}$$



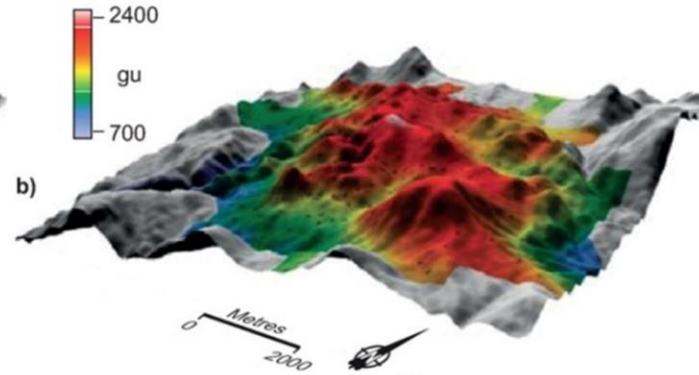
Terrain Correction

- Compensate the topographic irregularities existing around the observation point.

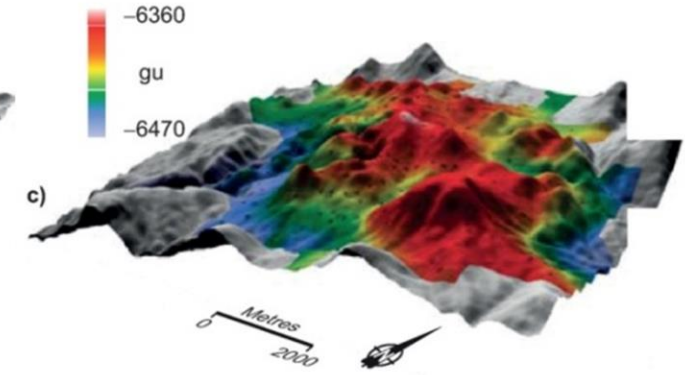




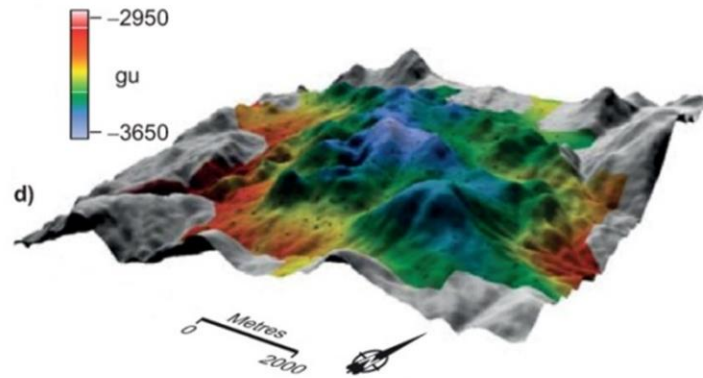
Data before height corrections



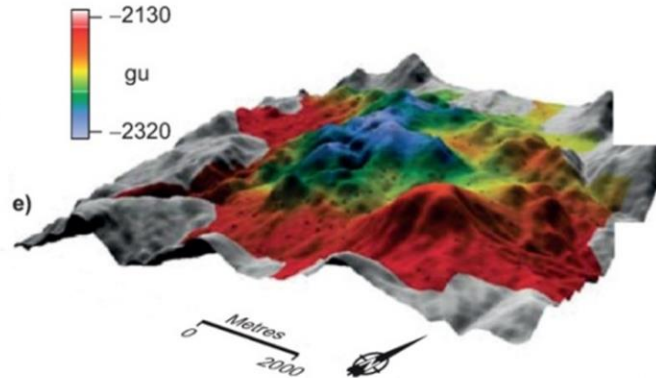
Free-air correction



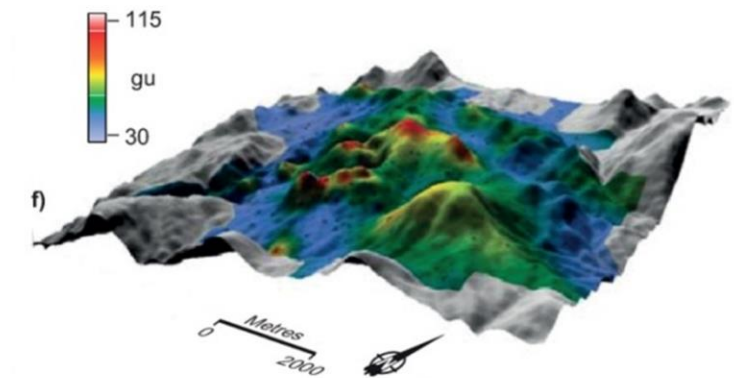
Free-air anomaly



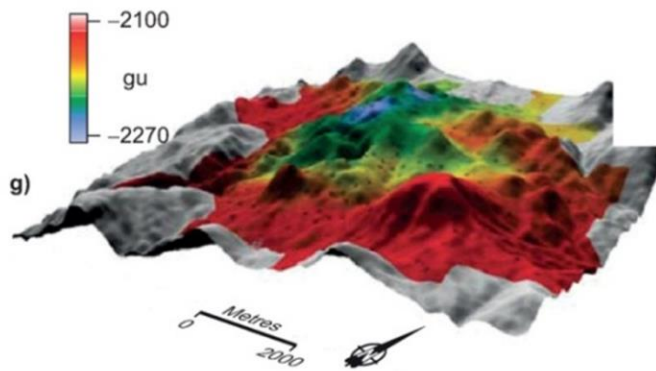
Bouguer correction



Partial Bouguer anomaly

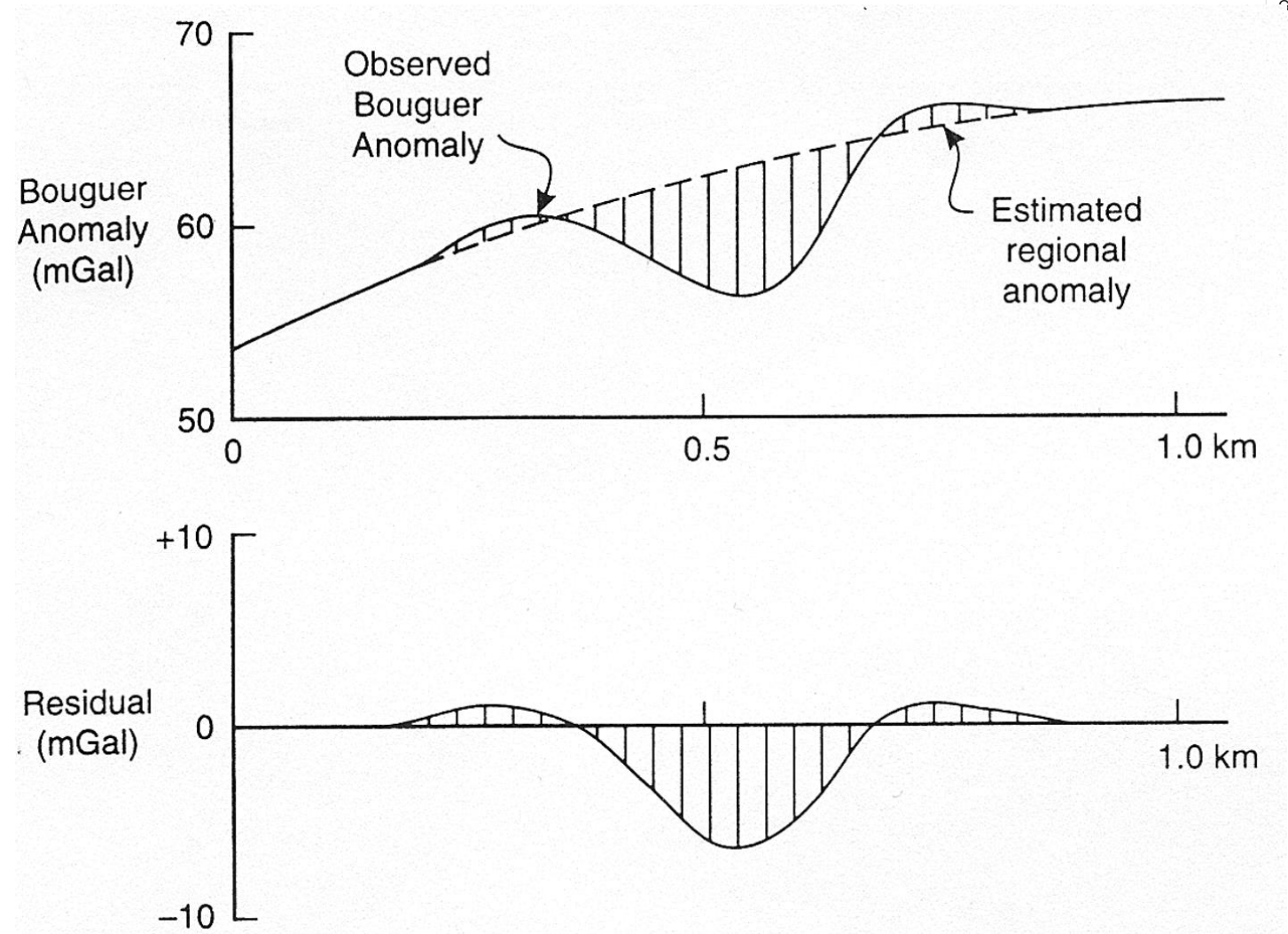


Terrain correction



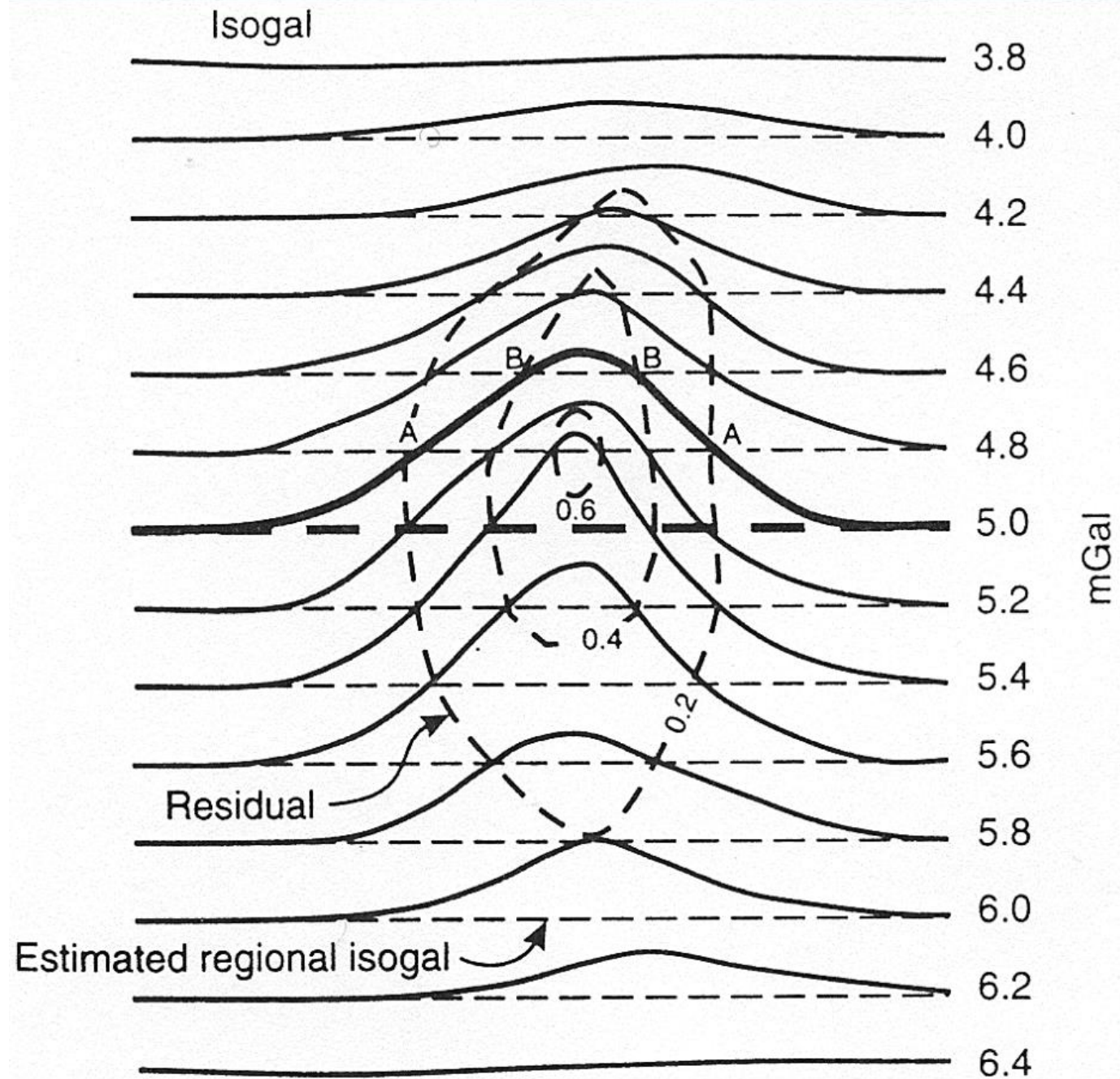
Complete Bouguer anomaly

Anomaly

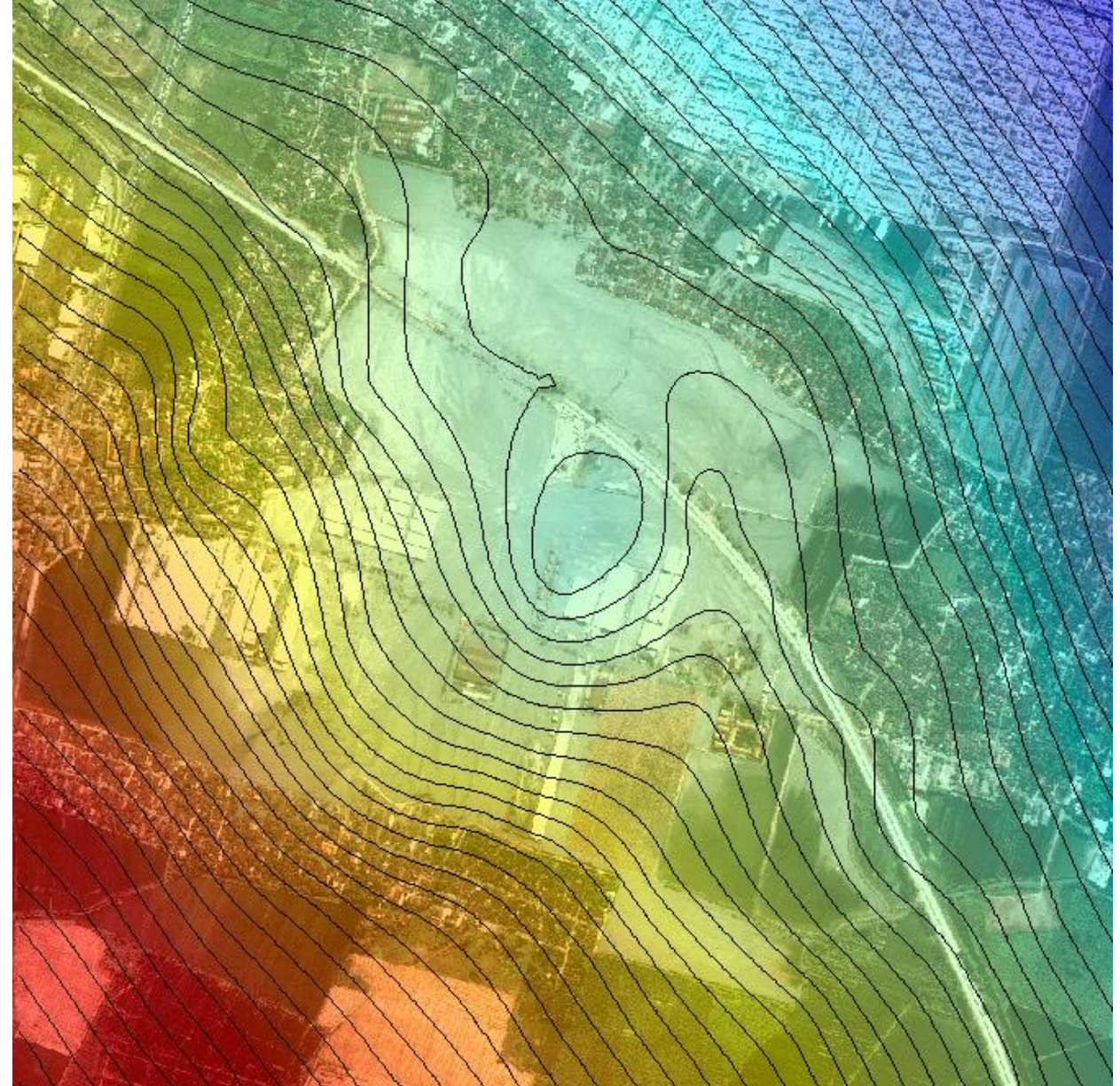




Anomaly



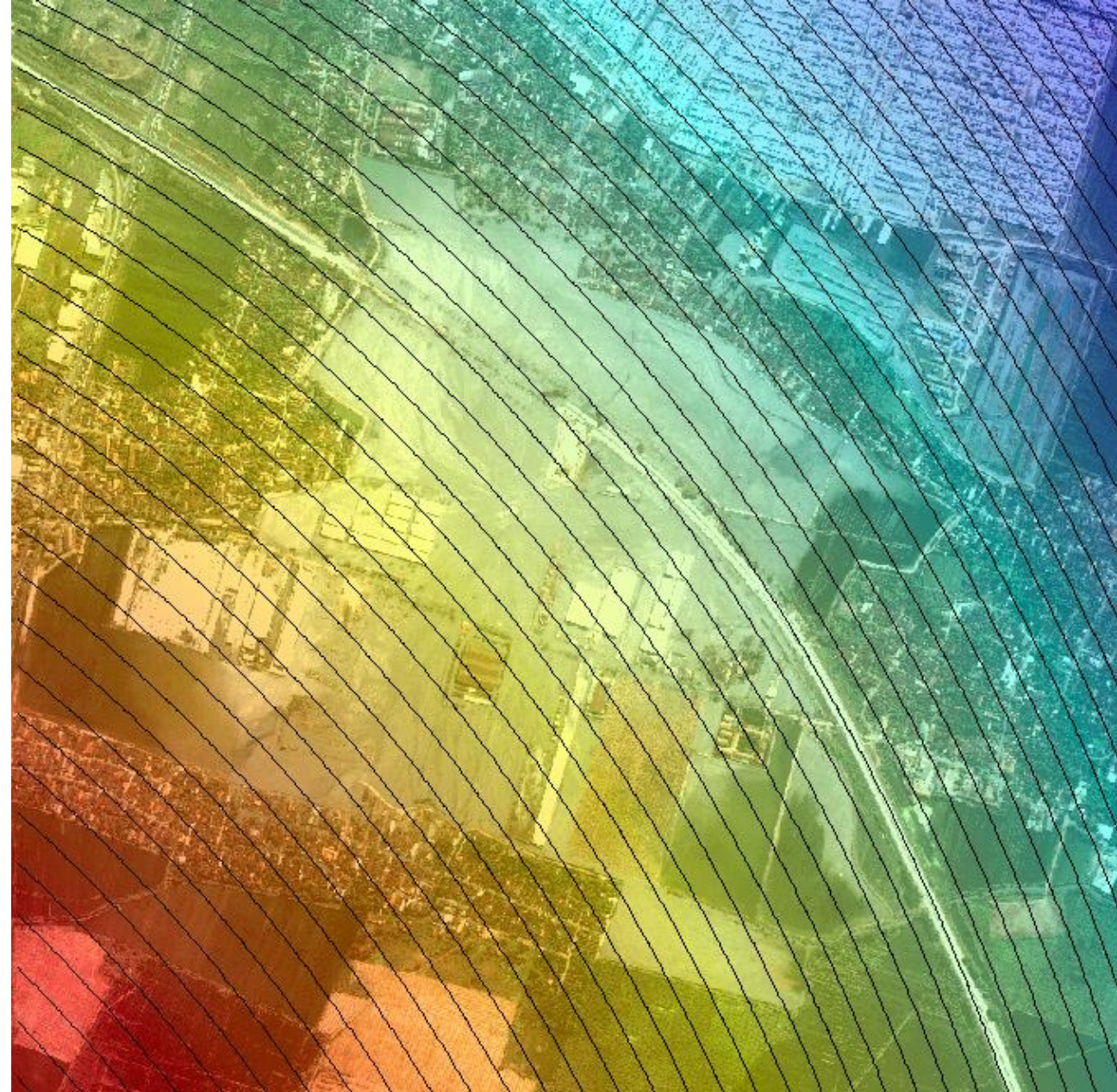
Bouguer Anomaly





Regional Anomaly

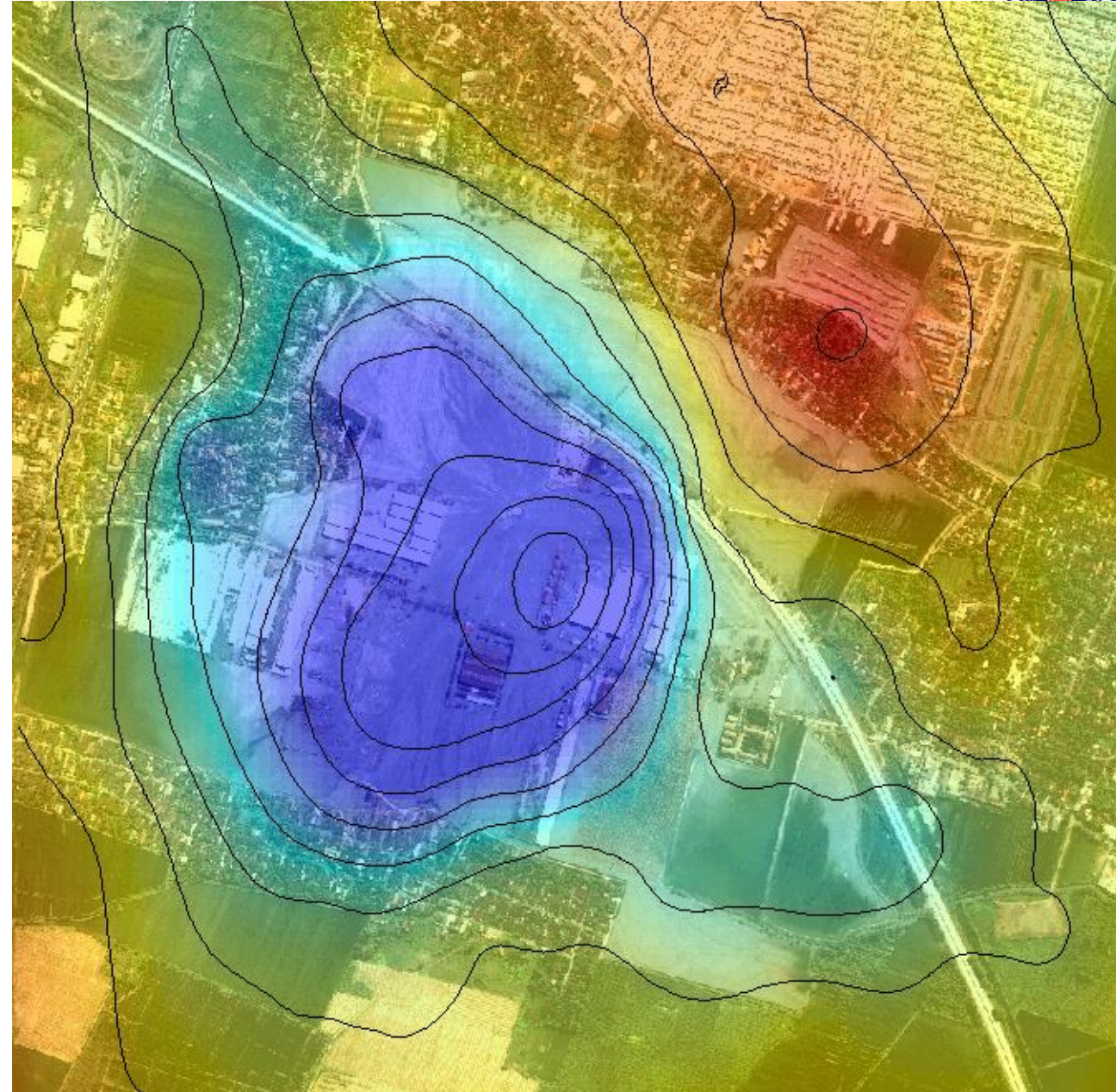
- Long wavelength
- Regional Trend
- Deep Sources





Residual Anomaly

- Short Wavelength
- Local Trend
- Shallow Sources



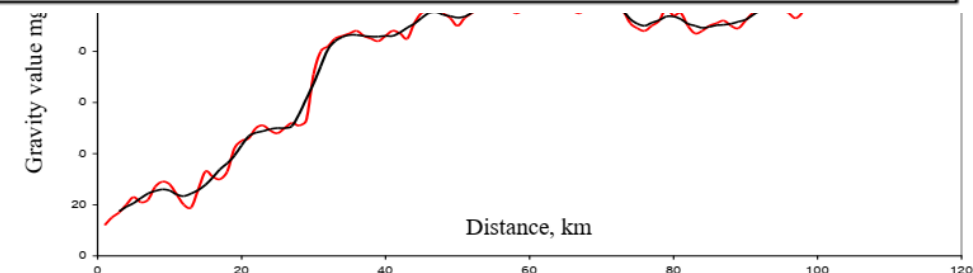
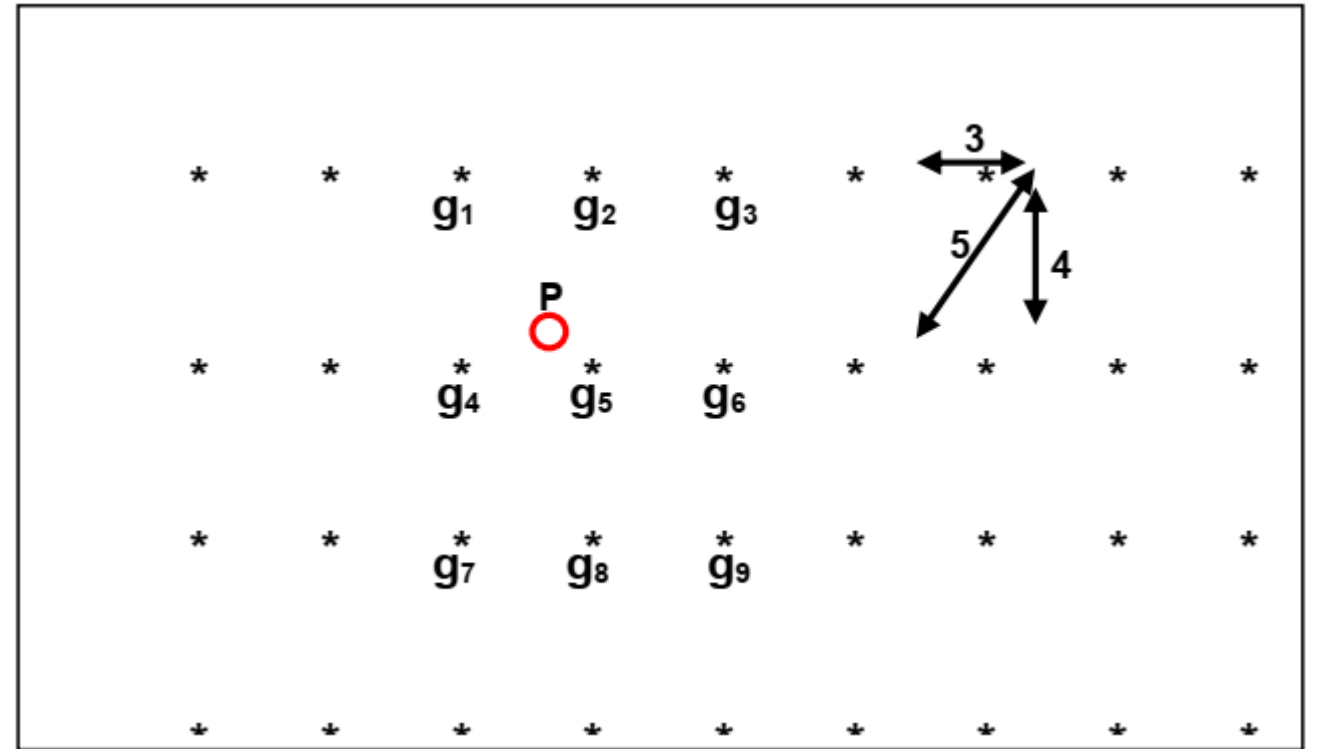
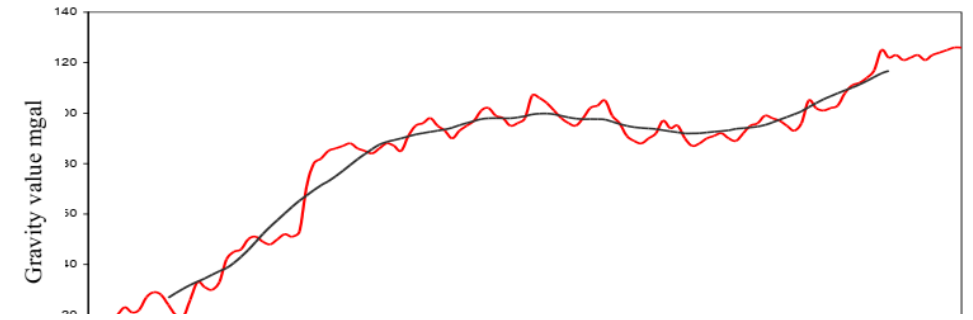
Anomaly Separation

- The Graphical Approach
- The Analytical Approach
 - Data Averaging
 - Polynomial Fitting
 - Upward Continuation
 - Second derivatives
 - Wavelength Filtering



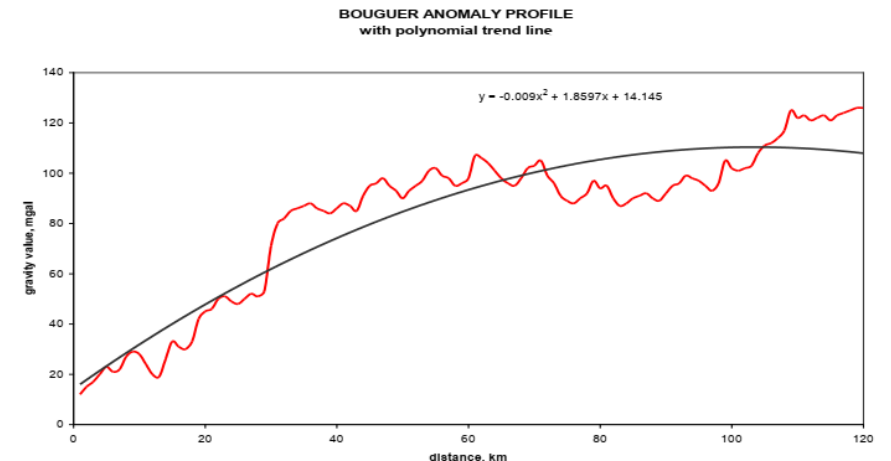
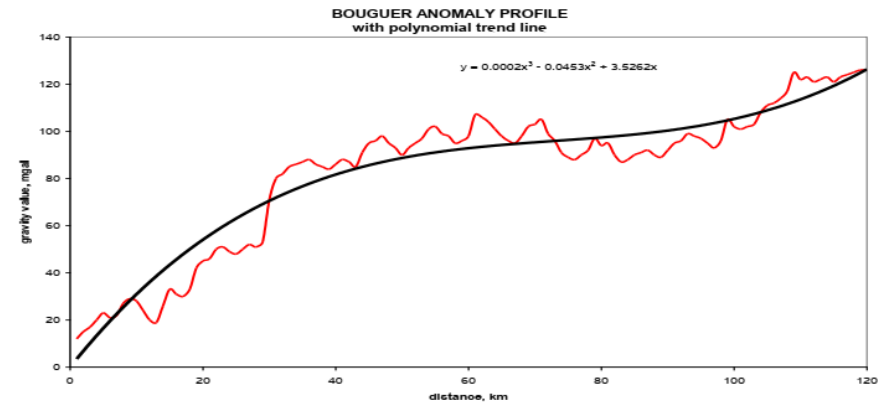
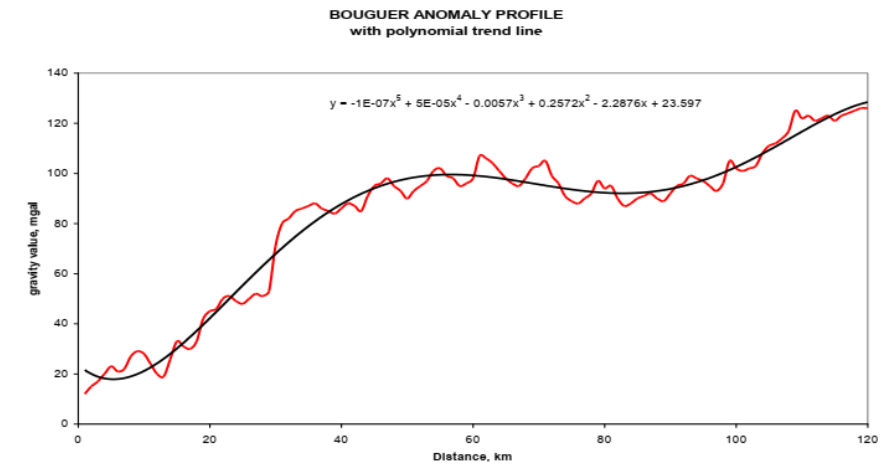
Data Averaging

- Griffin Method
- Moving-Average
- The Weighted Averages



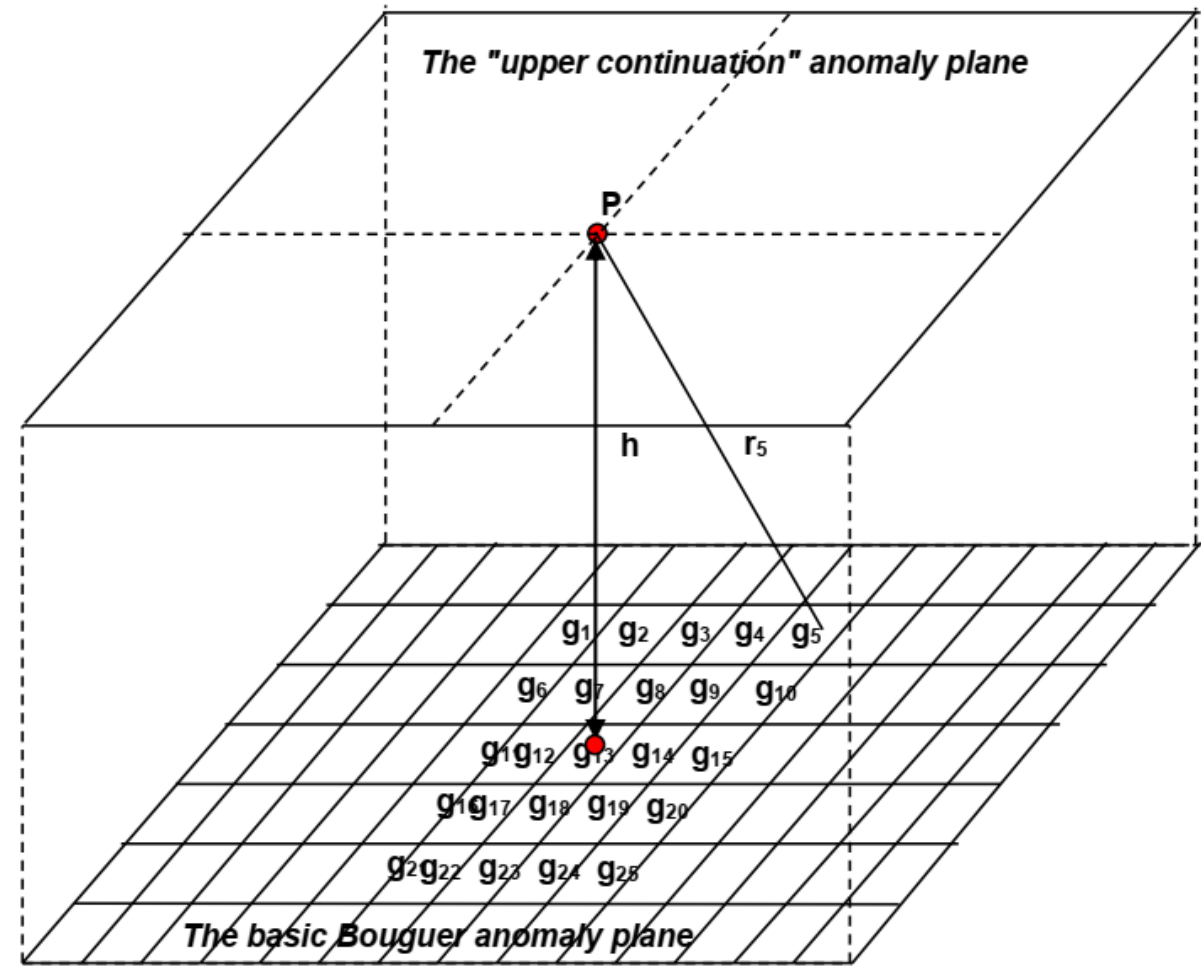
Polynomial Fitting

- Variation over different polynomial order
- Final choice depend on the degree of smoothing required



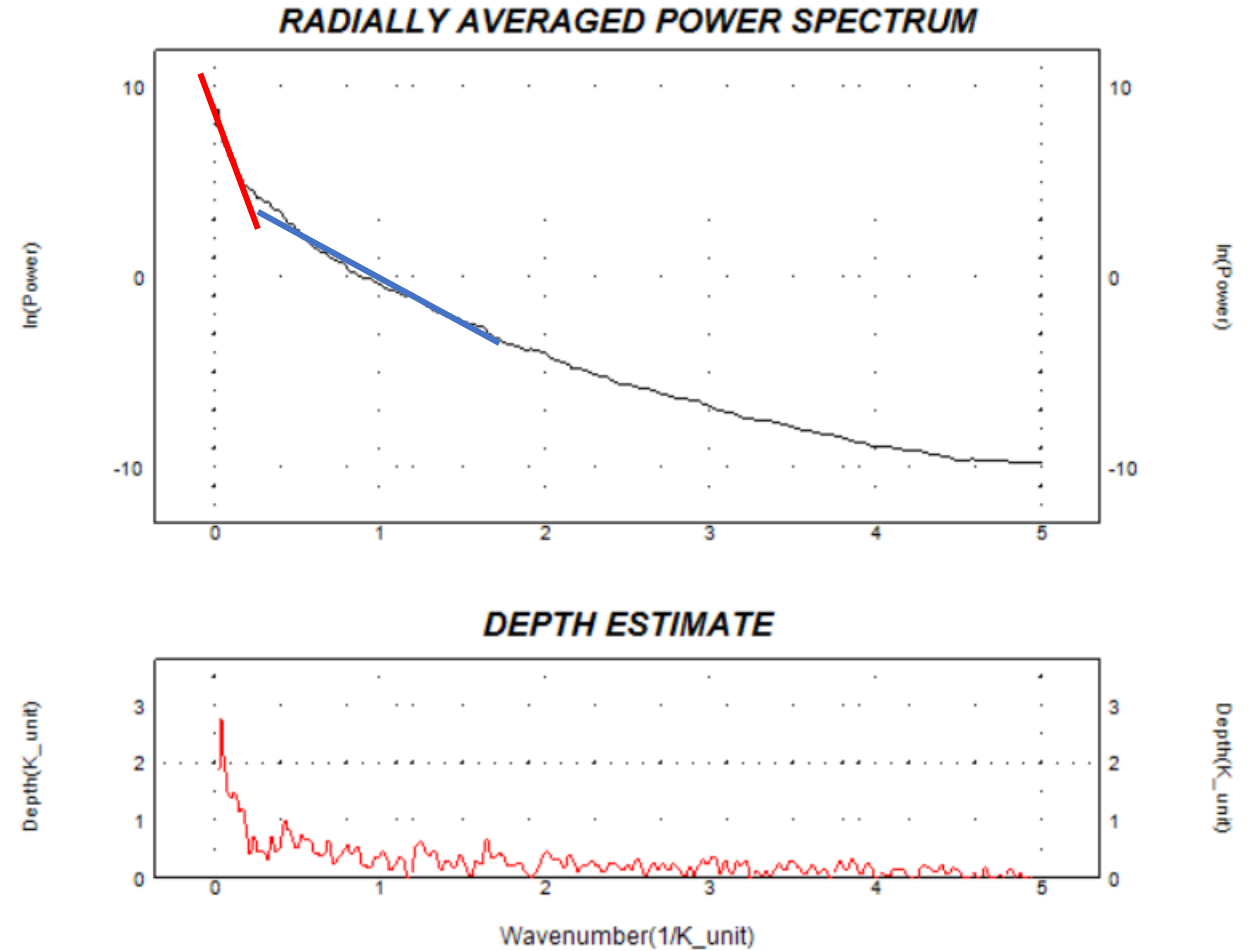
Upward Continuation

- Gravity anomaly can be projected

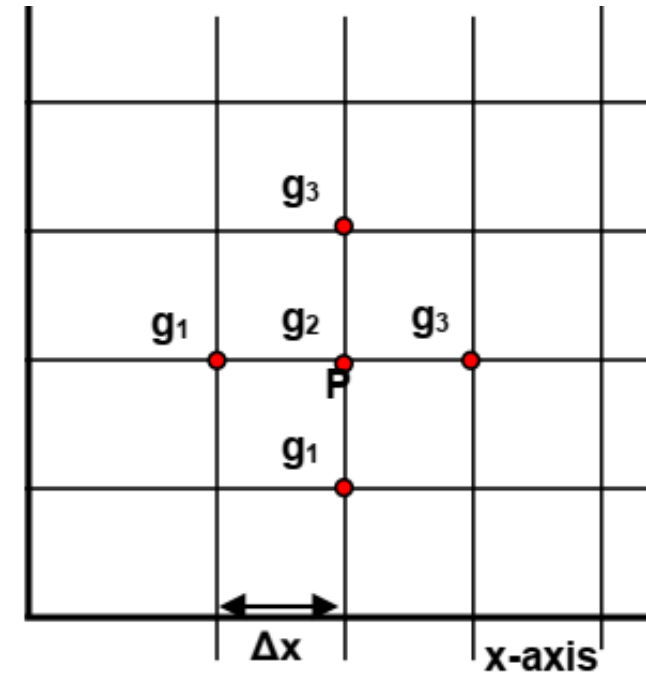
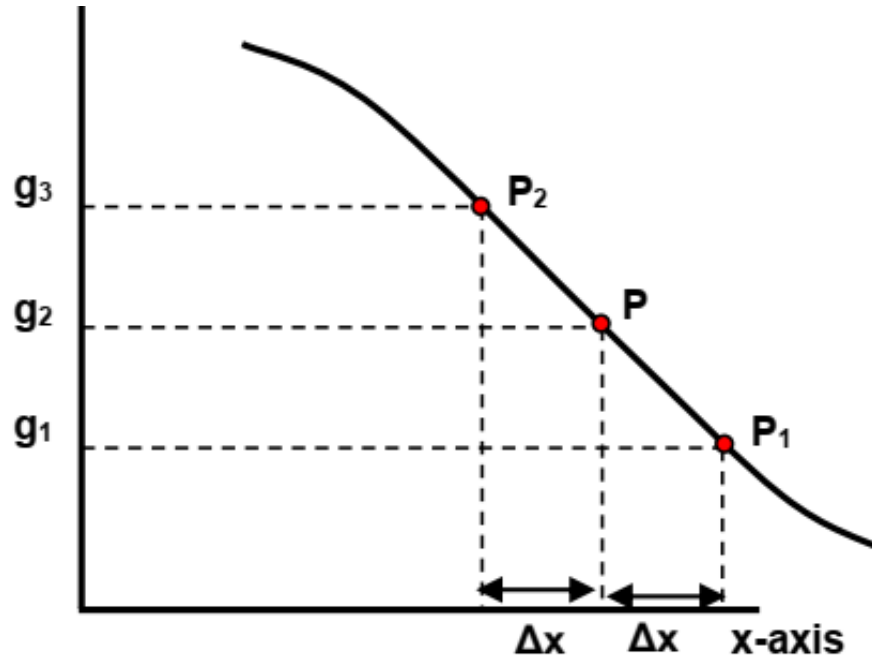


Wavelength Analysis

- Based on Fourier Spectral analysis theory



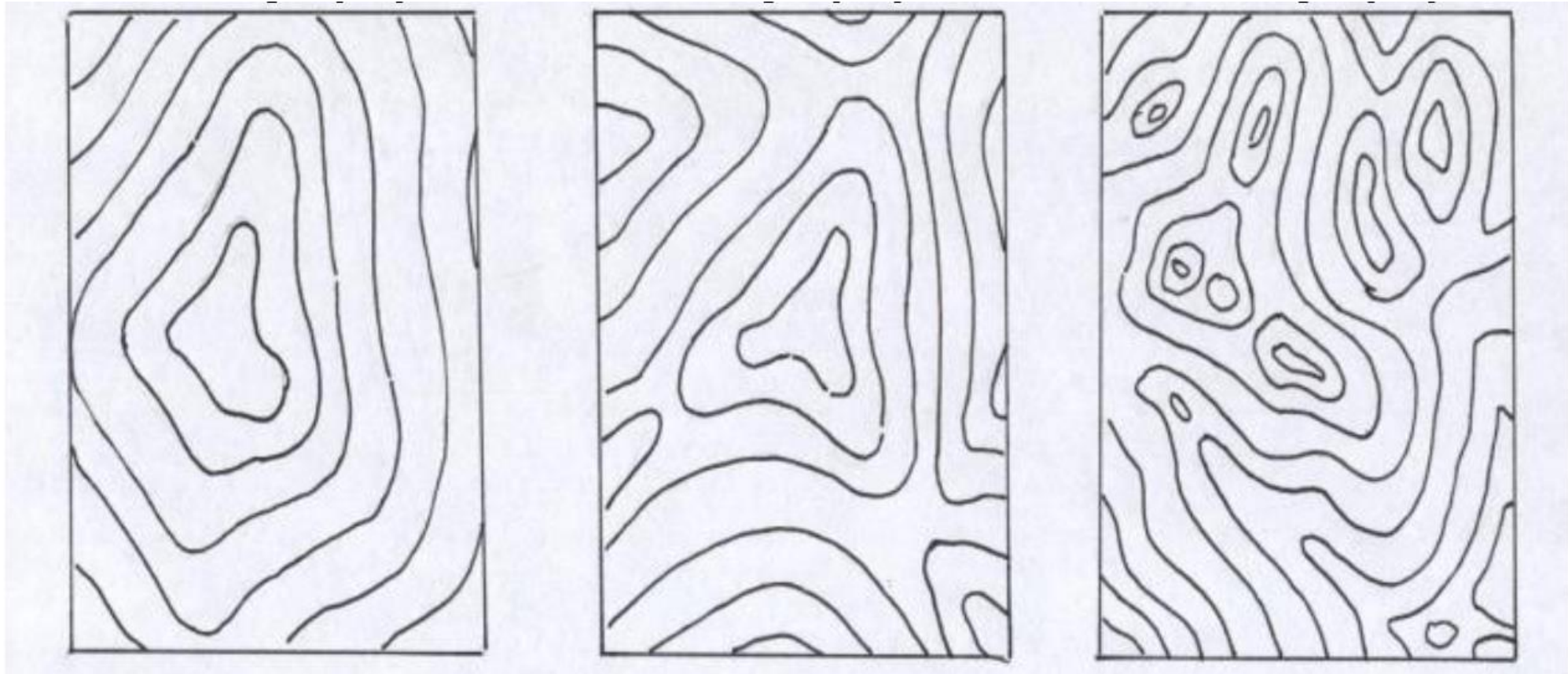
SECOND VERTICAL DERIVATIVE



$$(\partial^2 g / \partial x^2) = [2g_2 - g_1 - g_3]_x / (\Delta x)^2$$

$$(\partial^2 g / \partial y^2) = [2g_2 - g_1 - g_3]_y / (\Delta y)^2$$

$$(\partial^2 g / \partial z^2) = - [(\partial^2 g / \partial x^2) + (\partial^2 g / \partial y^2)] = \Sigma [g_i - 4g_2] / (\Delta s)^2$$



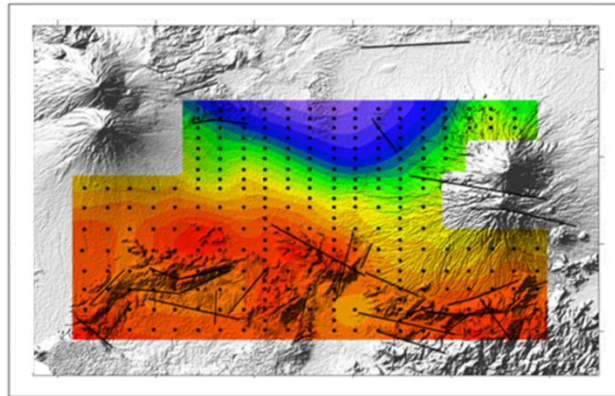
Bouguer Anomaly

Residual by Graphical Method

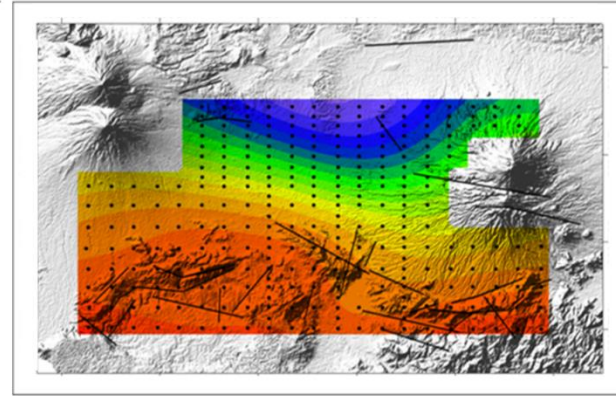
SVD



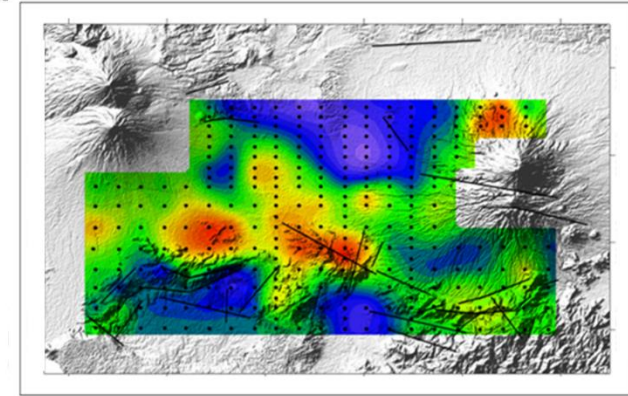
CASE STUDY GRAVITY (1)



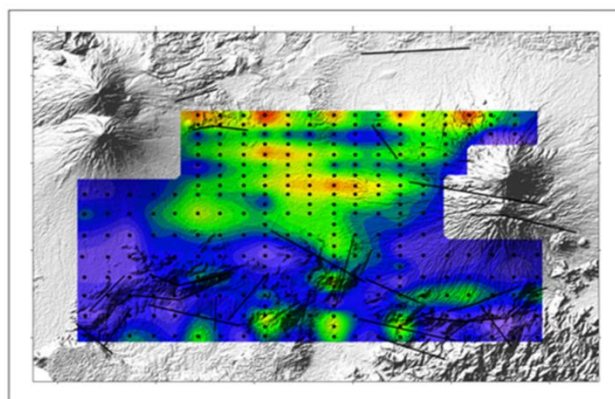
Peta Anomali Bouguer



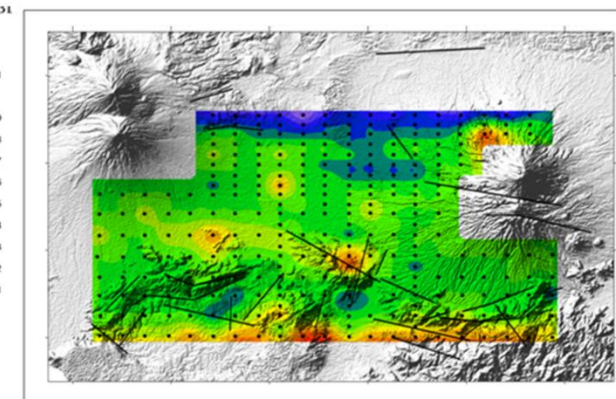
Peta Anomali Regional



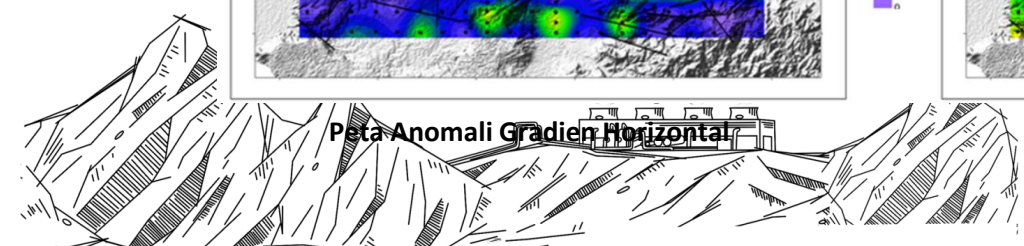
Peta Anomali Residual



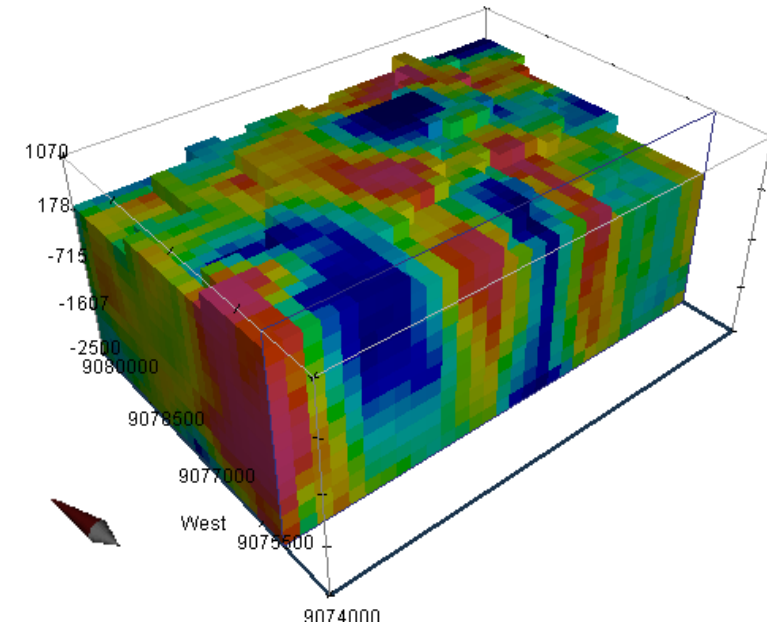
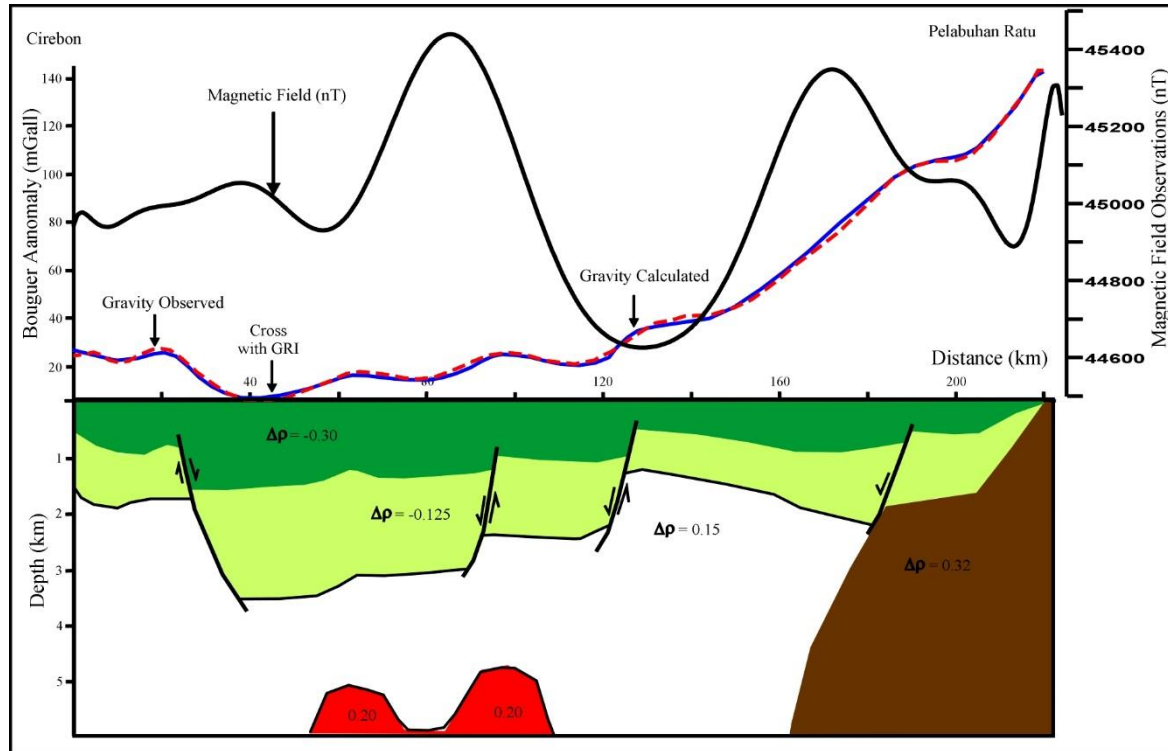
Peta Anomali Gradien Horizontal



Peta Anomali Gradien Vertikal



2D AND 3D MODEL



GEOMAGNETIC

Mengukur **variasi intensitas medan magnetik** di permukaan bumi yang disebabkan oleh adanya **variasi distribusi benda termagnetisasi** di bawah permukaan bumi.

$$\vec{M} = k\vec{H}$$

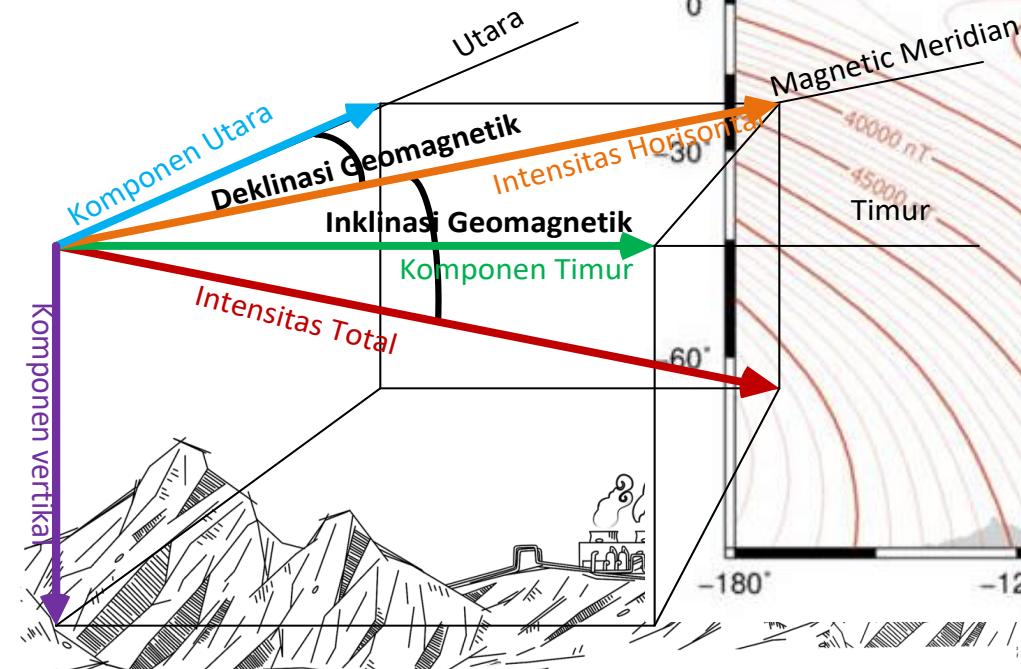
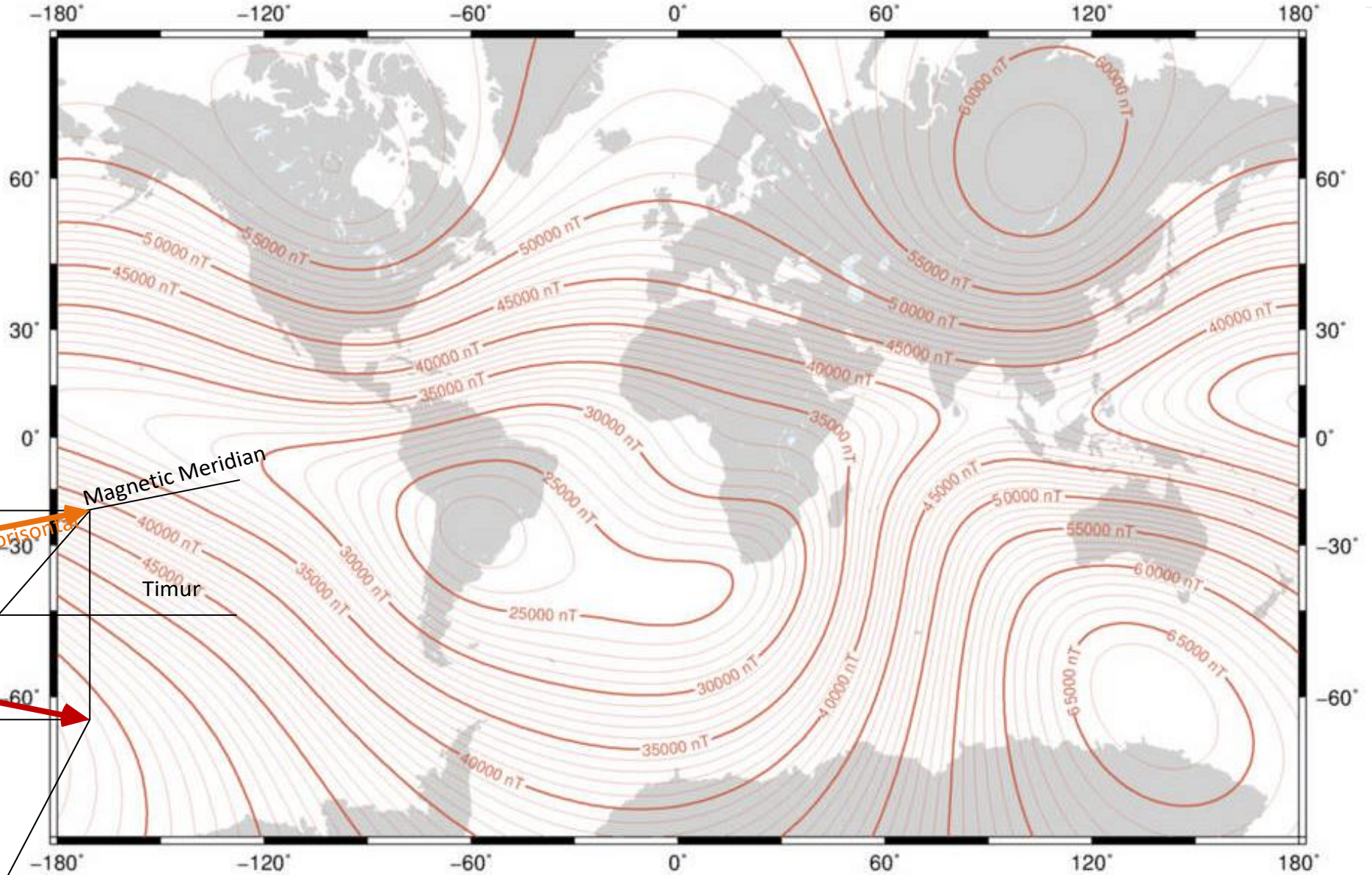
Variasi intensitas medan magnet $\rightarrow$ distribusi bahan magnetik di bawah permukaan $\rightarrow$ pendugaan geologi yang memungkinkan

Daerah geotermal $\rightarrow$ **alterasi hidrotermal** $\rightarrow$ mengubah sifat-sifat magnetik batuan baik melalui penghilangan kandungan besi secara menyeluruh maupun pengubahan komponen-komponen *magnetite* menjadi *hematite* $\rightarrow$ suseptibilitas magnetik (k) rendah $\rightarrow$ derajat magnetisasi negatif





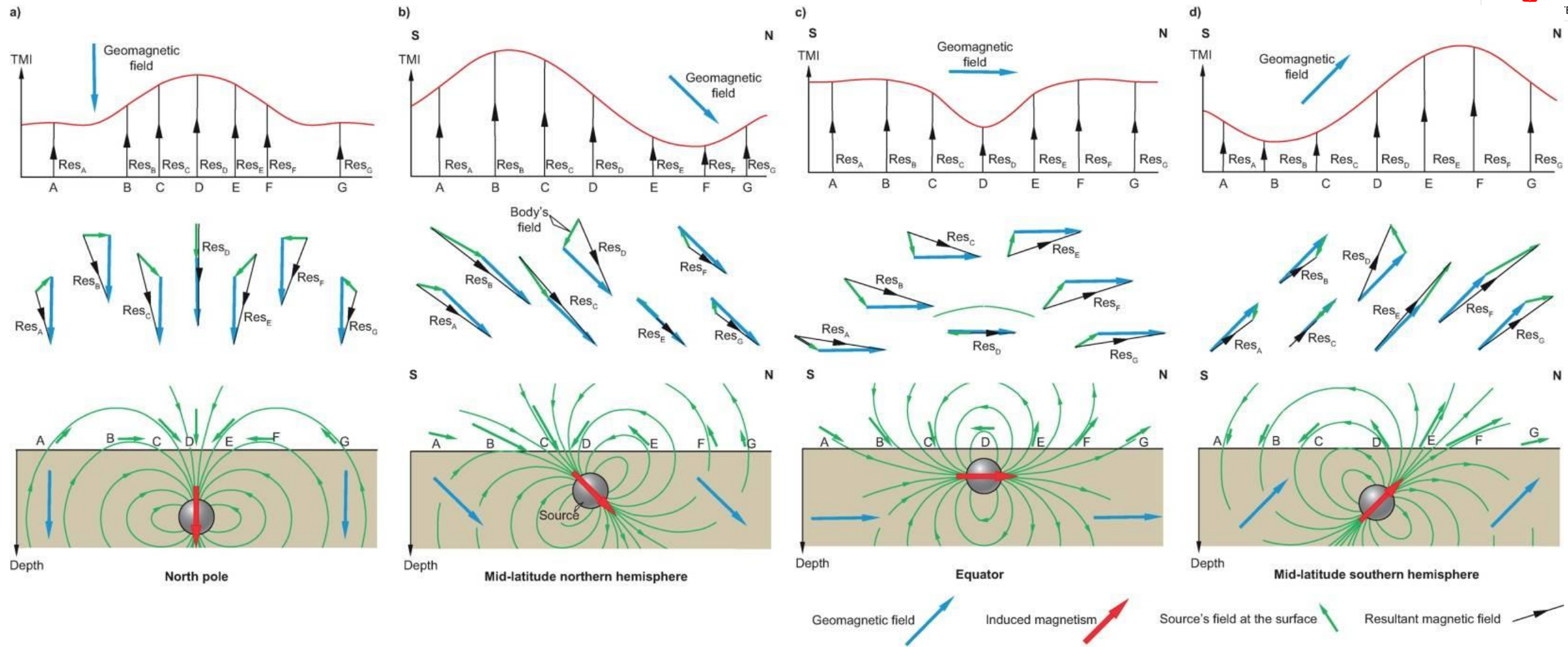
Total Force IGRF12



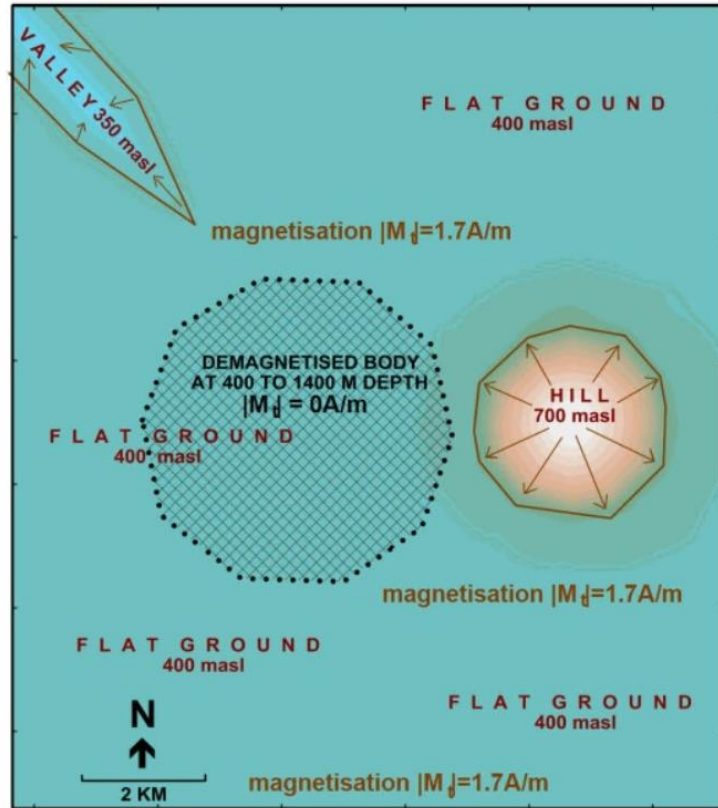
Data Reduction and Enhancement:

- Diurnal Correction
- Normal Field Correction
essentially a correction for variations in field with latitude and longitude – use IGRF to correct large scale surveys
- Elevation and Terrain Corrections
vertical gradient is a maximum at poles (0.03nT/m) minimum at equator (-0.015nT/m)
elevation correction not required for ground surveys terrain correction difficult since we need to know magnetic properties in surrounding terrain
- **Emphasizing shorter wavelength features.**
- **First vertical derivative emphasizes near surface features. It can be measured with gradiometer, or derived from corrected data**
- **Second vertical derivative emphasizes boundaries of target zones**

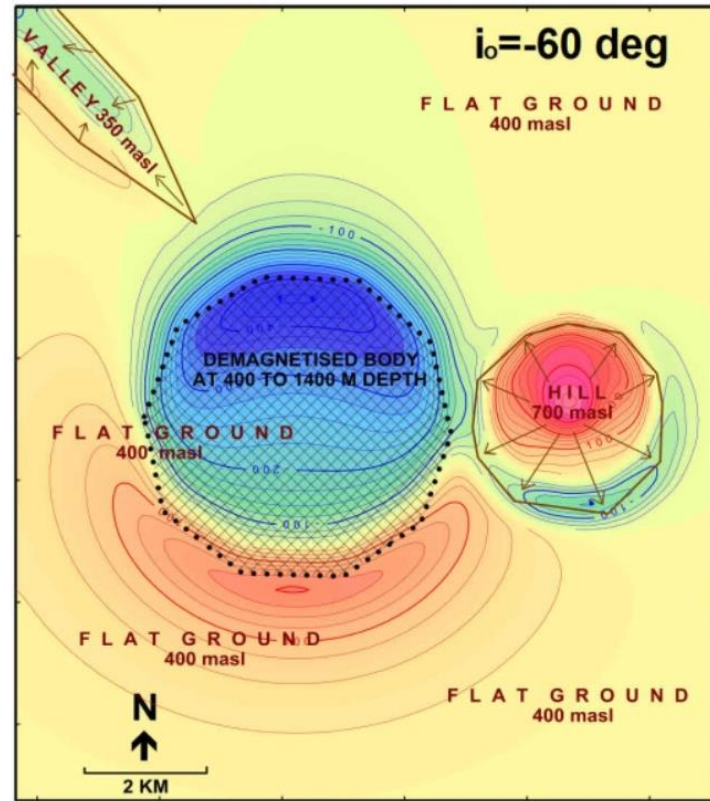




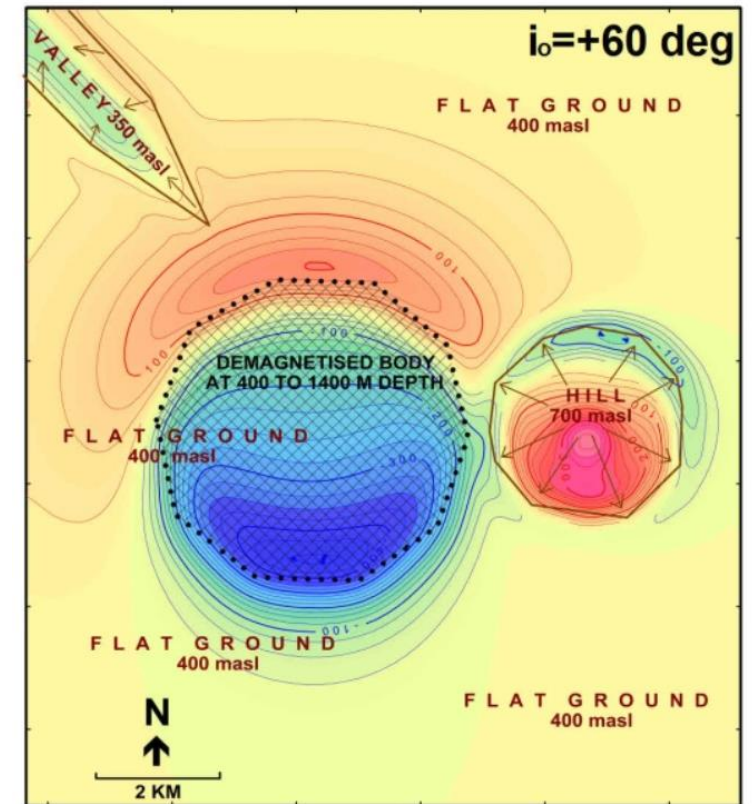
Dentith, 2014



(a) A simplified magnetic model of geothermal system

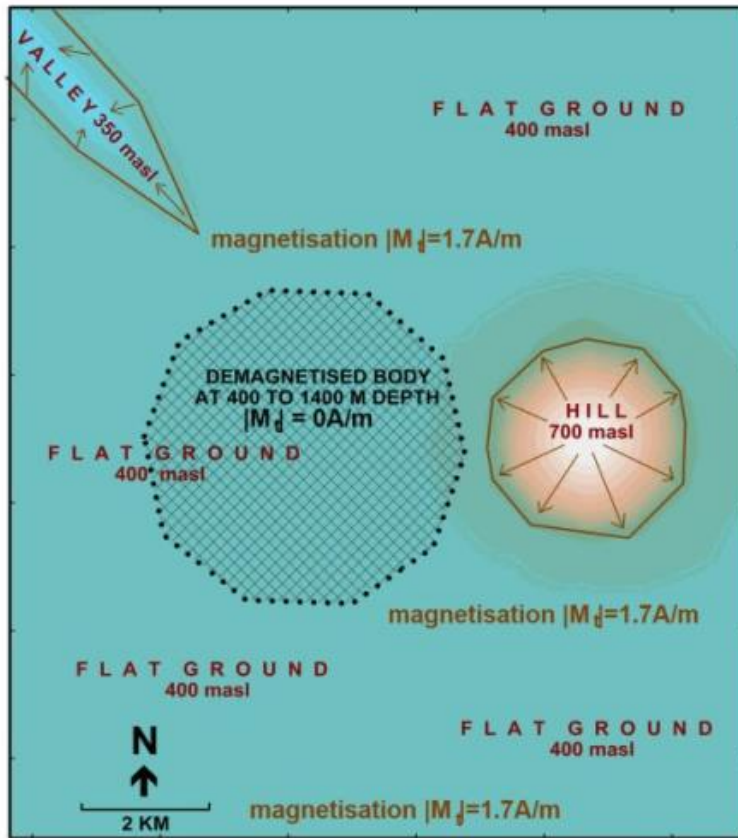


(b) Total force magnetic anomalies caused by the model in (a) at magnetic inclination -60 deg (southern hemisphere)

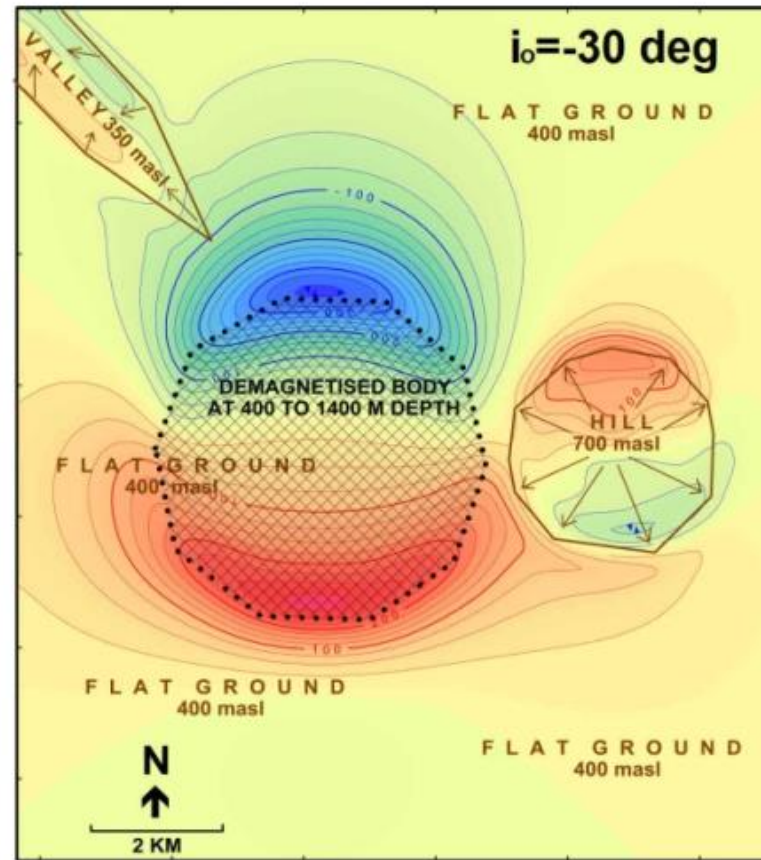


(c) Total force magnetic anomalies caused by the model in (a) at magnetic inclination +60 deg (northern hemisphere)

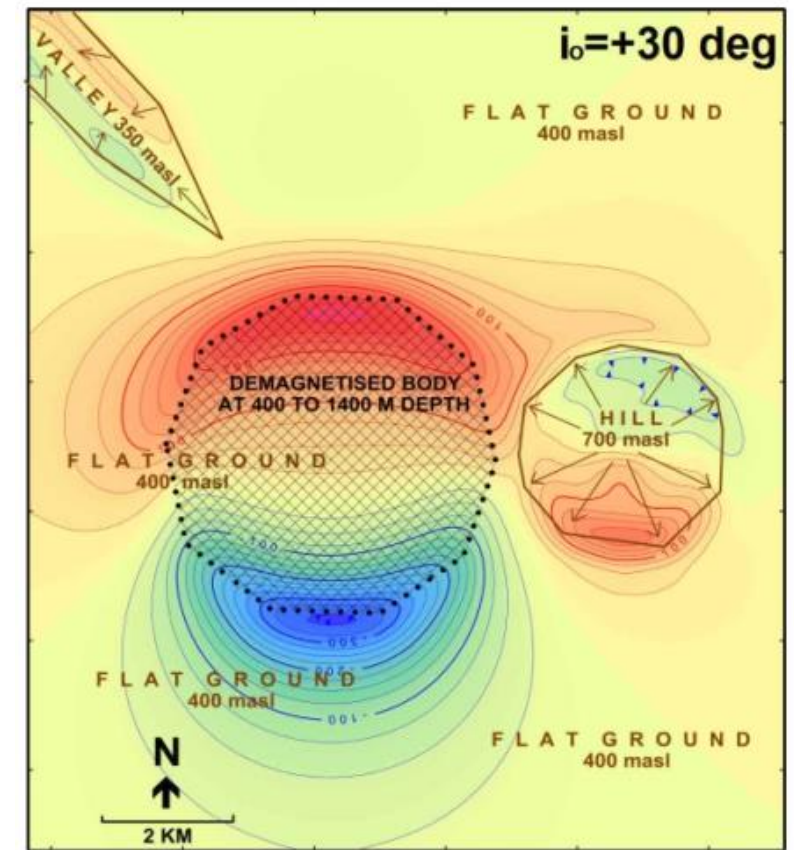
Soengkono, 2016



(a) A simplified magnetic model of geothermal system

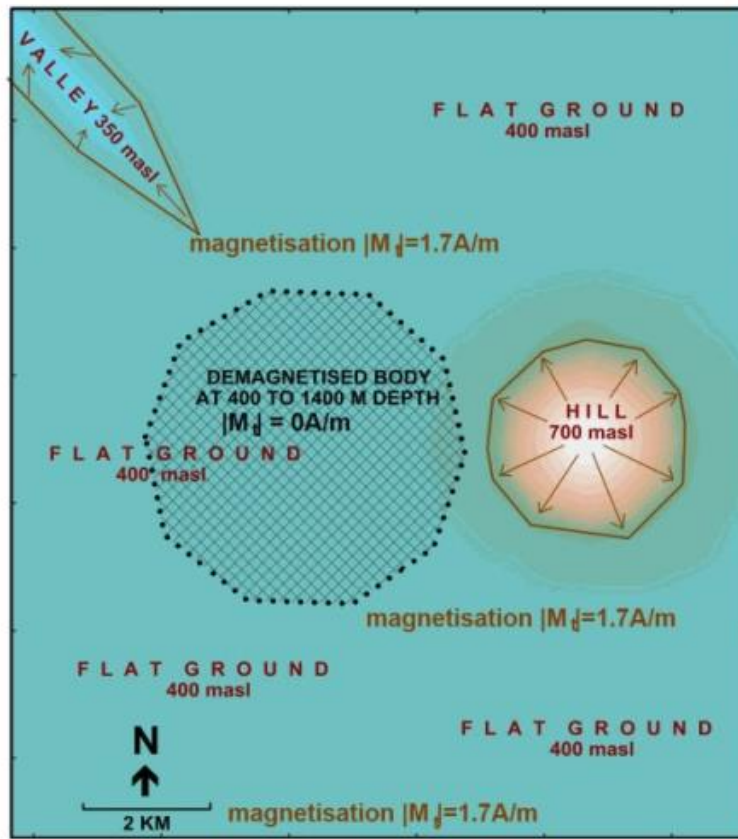


(b) Total force magnetic anomalies caused by the model in (a) at magnetic inclination -30 deg (southern hemisphere)

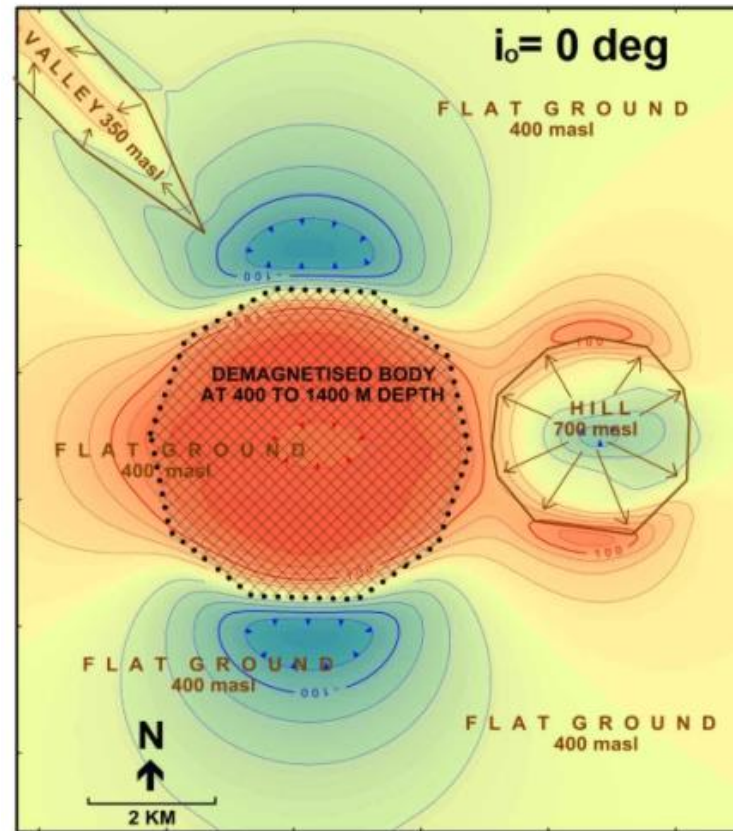


(c) Total force magnetic anomalies caused by the model in (a) at magnetic inclination +30 deg (northern hemisphere)

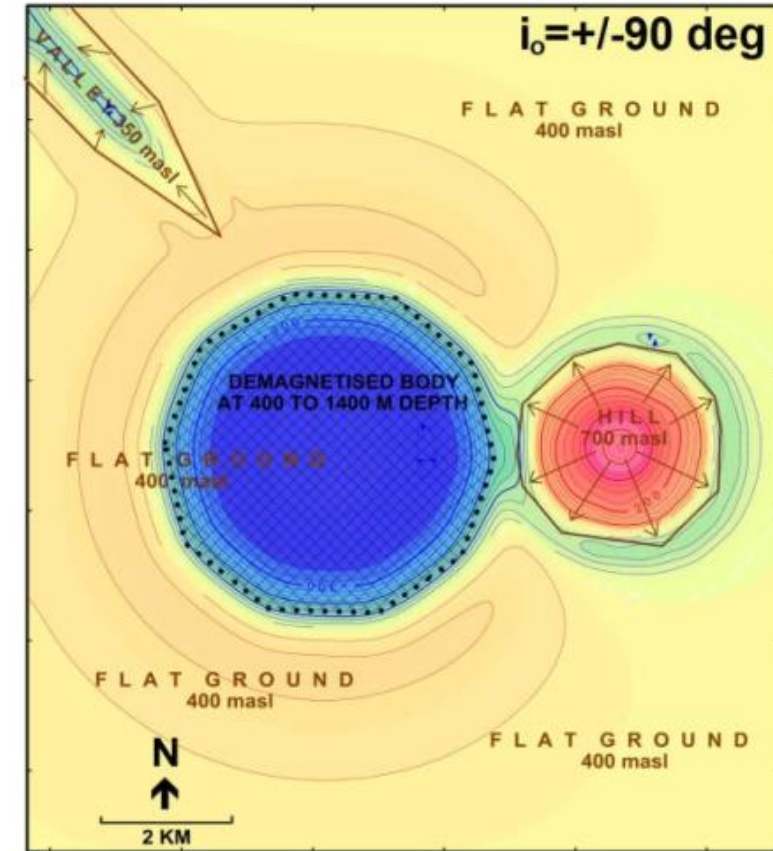
Soengkono, 2016



(a) A simplified magnetic model of geothermal system



(b) Total force magnetic anomalies caused by the model in (a) at magnetic inclination 0 deg (magnetic equator)



(c) Total force magnetic anomalies caused by the model in (a) at magnetic inclination +/- 90 deg (south/north magnetic pole)

Soengkono, 2016

Reduce to Pole

- Carried out in the Fourier Domain
- Simplify the magnetic maps
- Anomalies can be rotated by tectonism, reverse or normally polarized rocks
- Not always appropriate

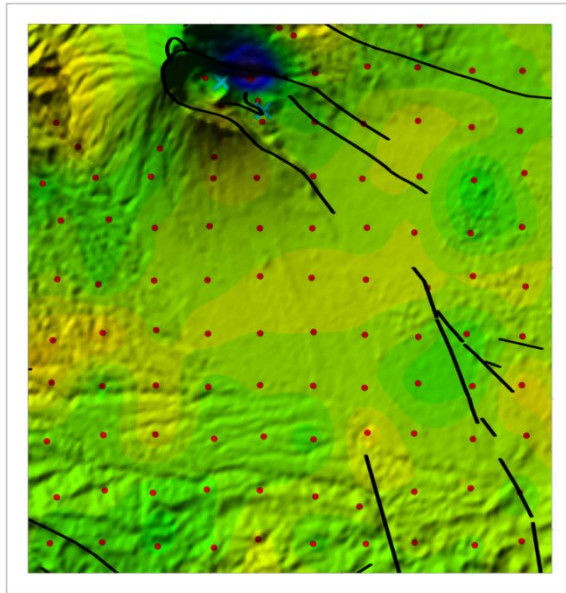
$$R(k_r) = |k_r|^2 \frac{F(k_r)k_r^2}{B^2}$$

$$k_r = \sqrt{k_x^2 + k_y^2}$$

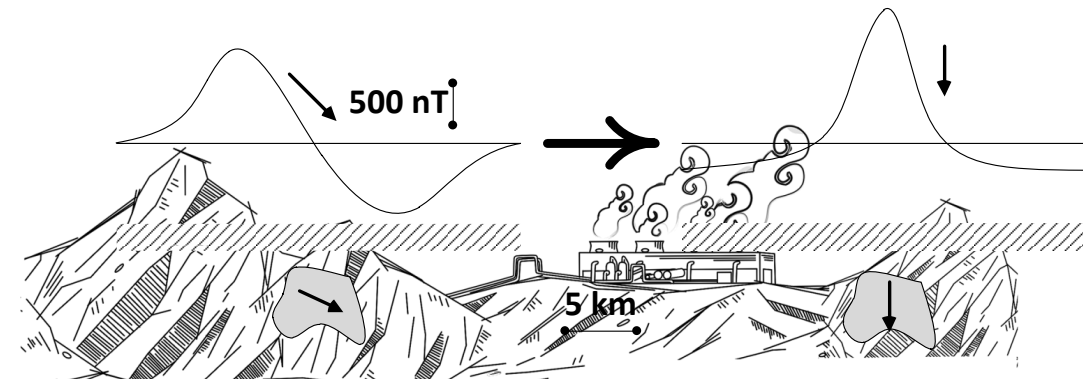
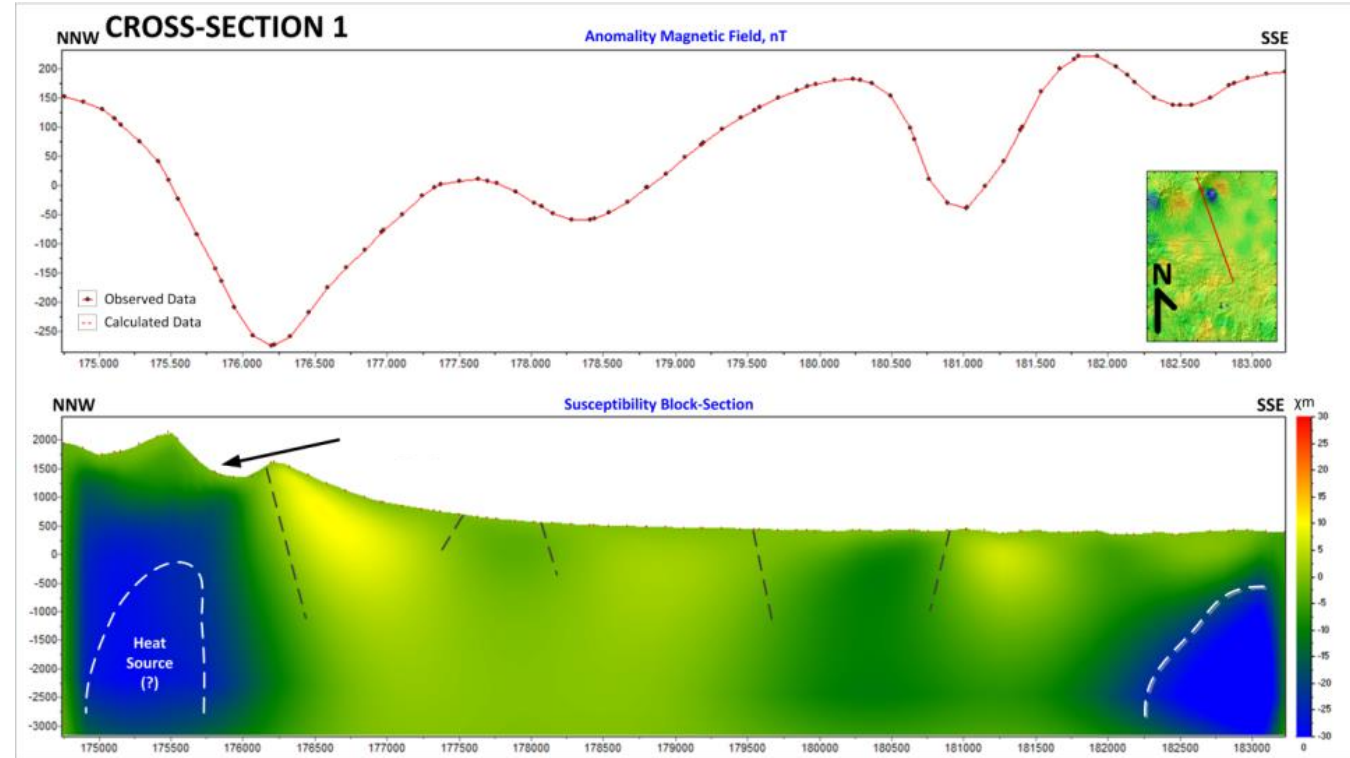
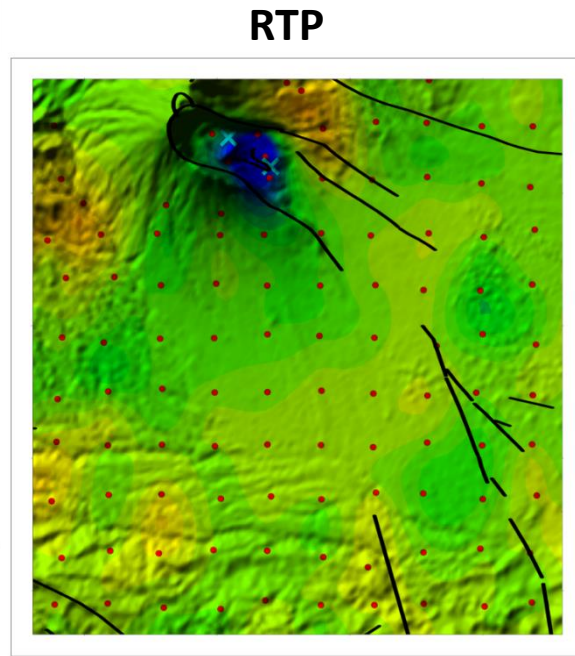
$$B = \frac{1}{|ik_x \cos I \cos D + ik_y \cos I \sin D + k_r \sin I|}$$



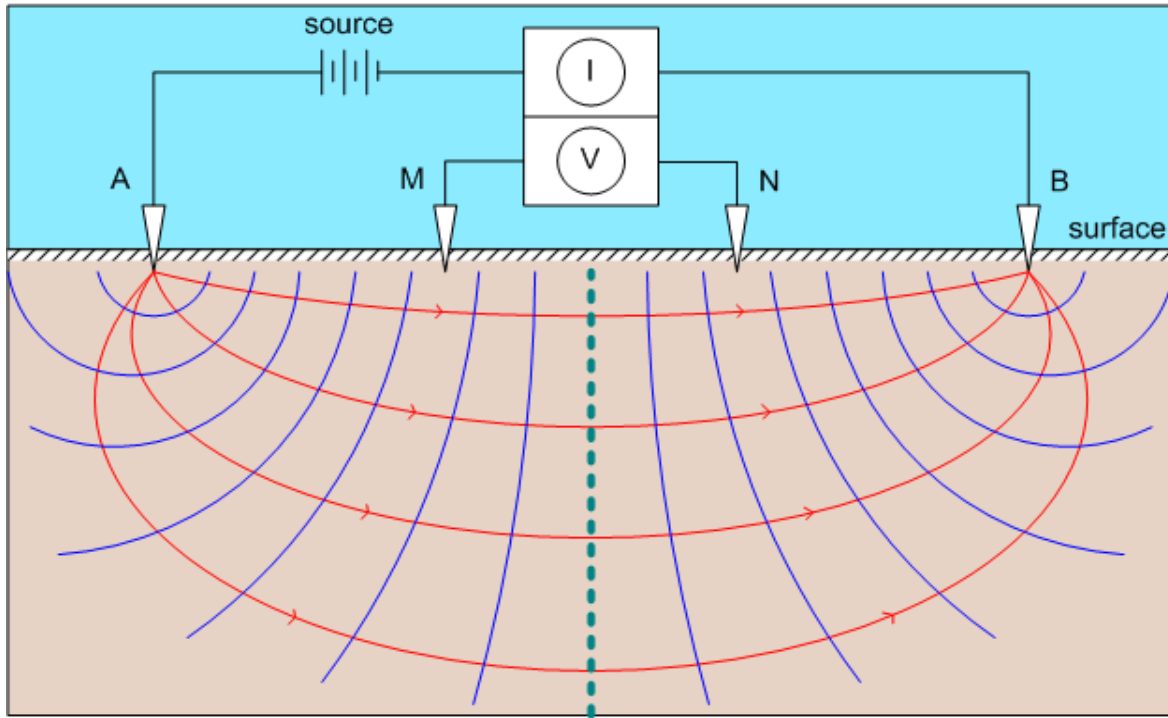
Case Study



Total Magnetic Anomaly



GEOLISTRIK/ DC RESISTIVITY



Metode geolistrik → mengukur sifat resistivitas batuan

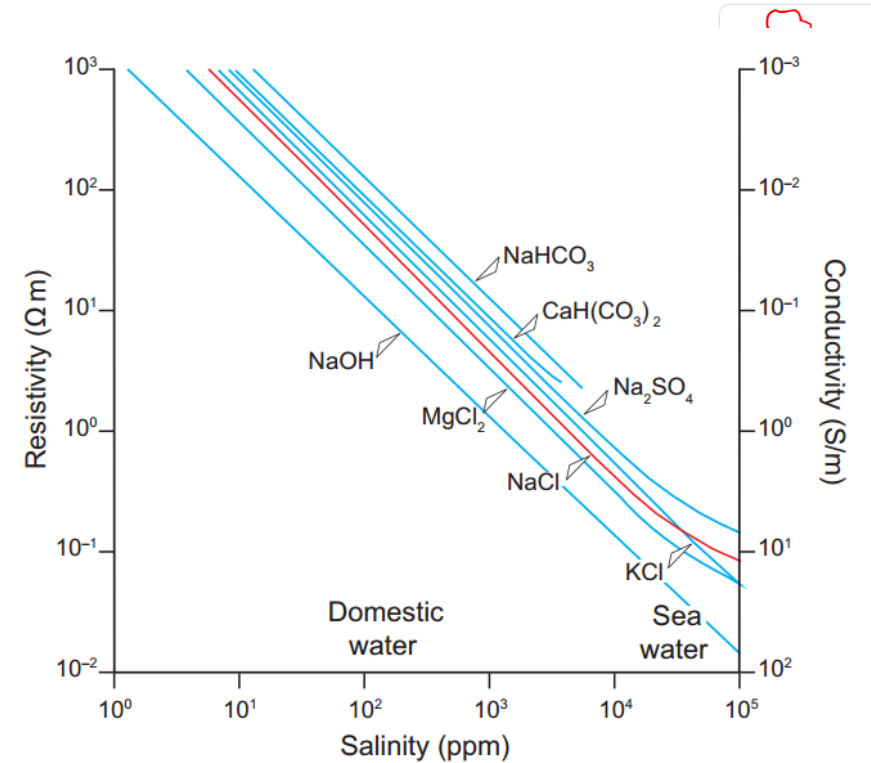
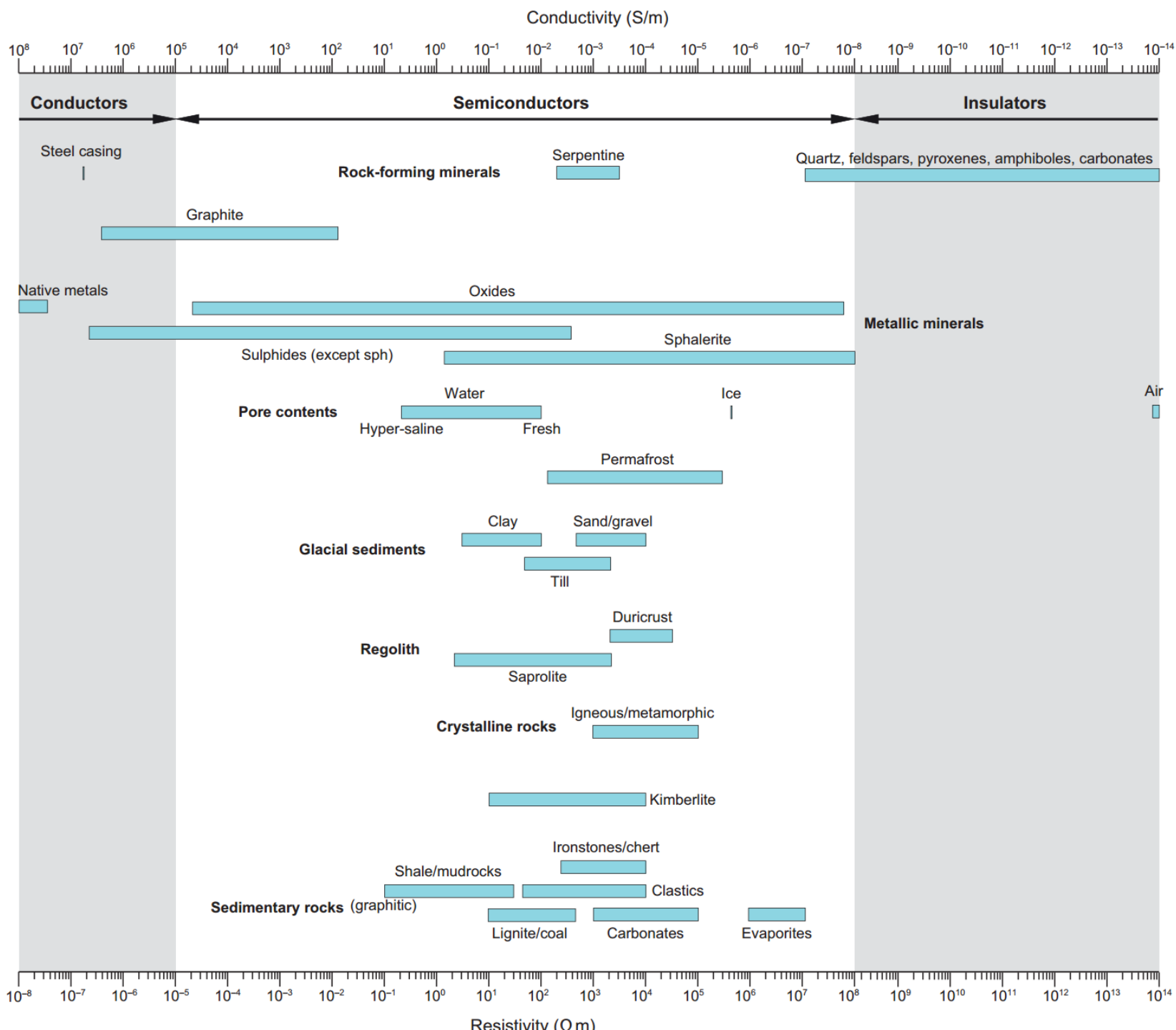


$$\rho_a = K \frac{\Delta V}{I}$$

Apparent Resistivity (Resistivitas Semu) mengukur nilai resistivitas rata-rata di bawah electrode.

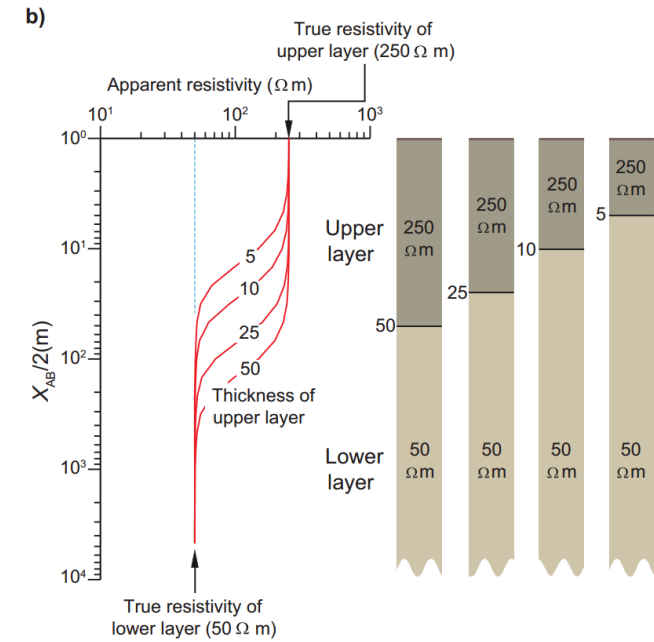
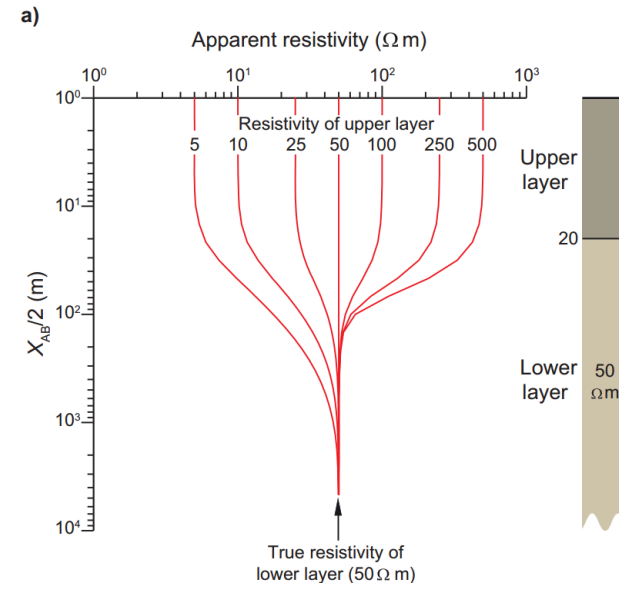
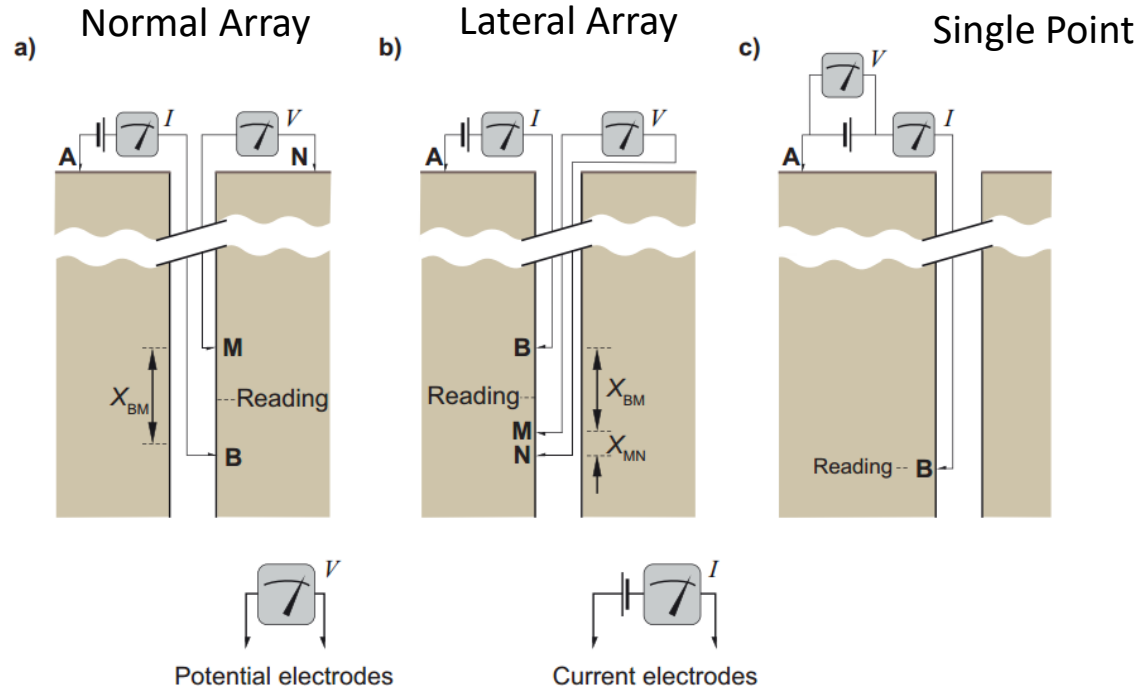
Jika kondisi di bawah permukaan homogen maka nilai tersebut menjadi “true” resistivity.

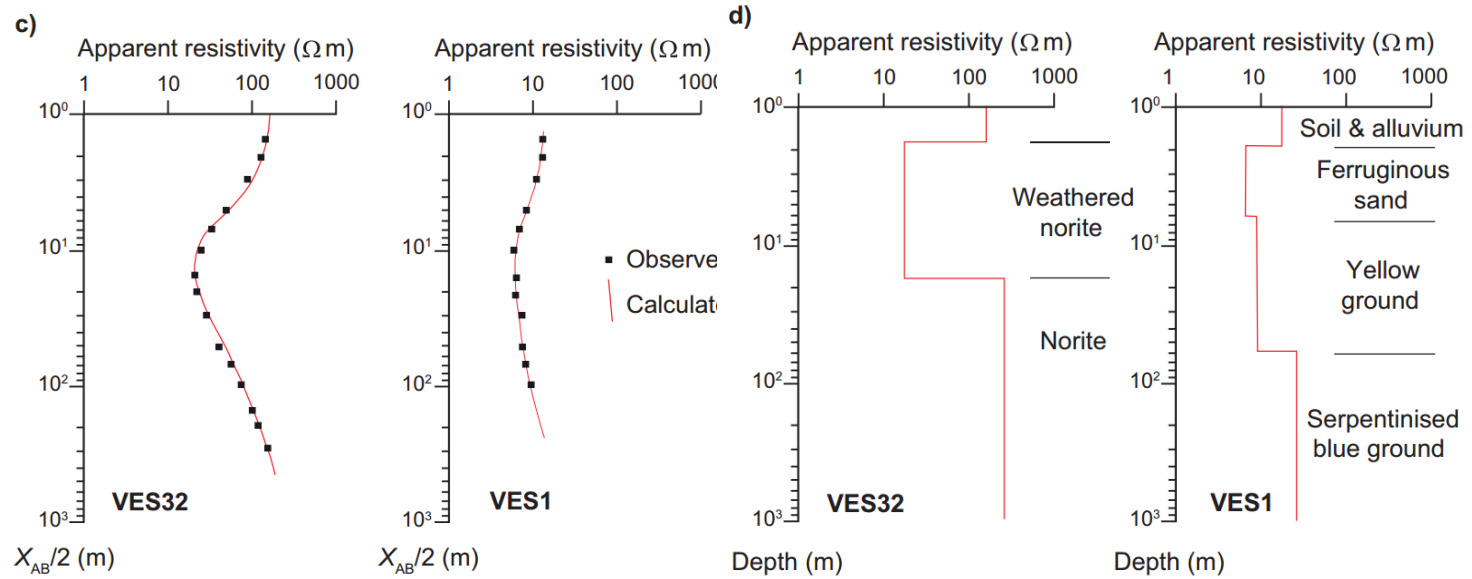
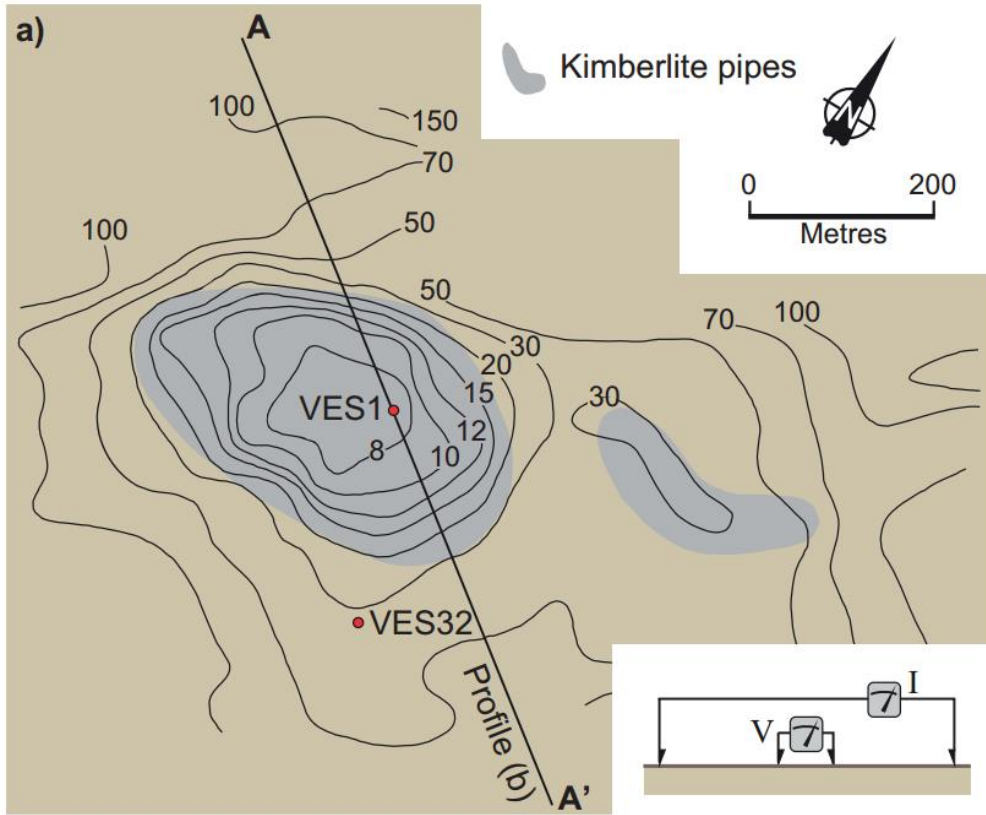


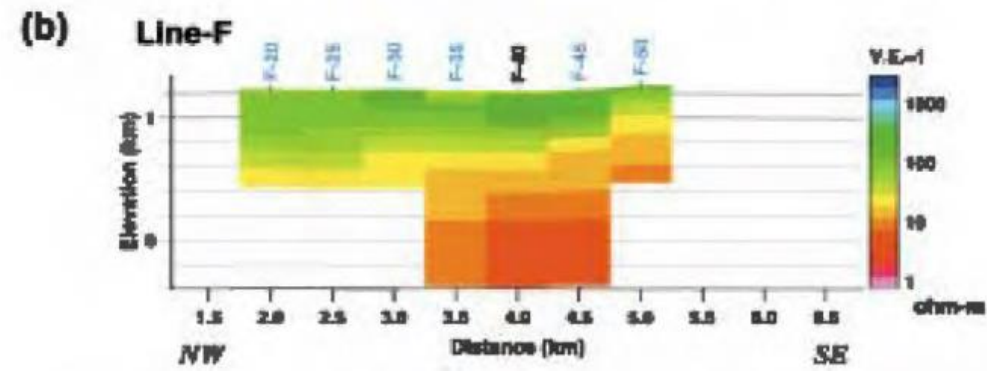
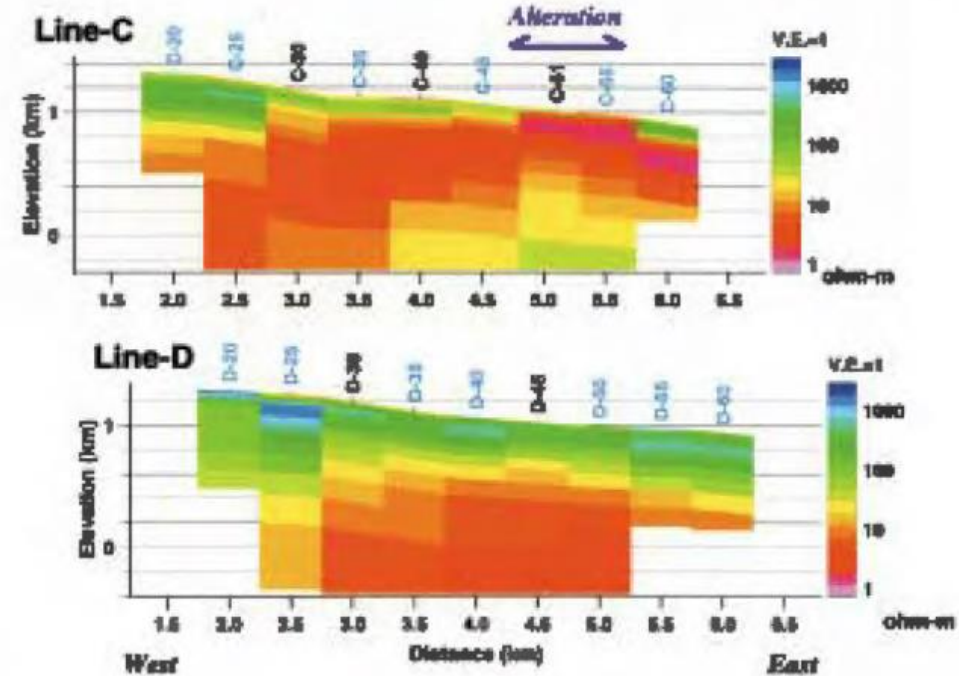
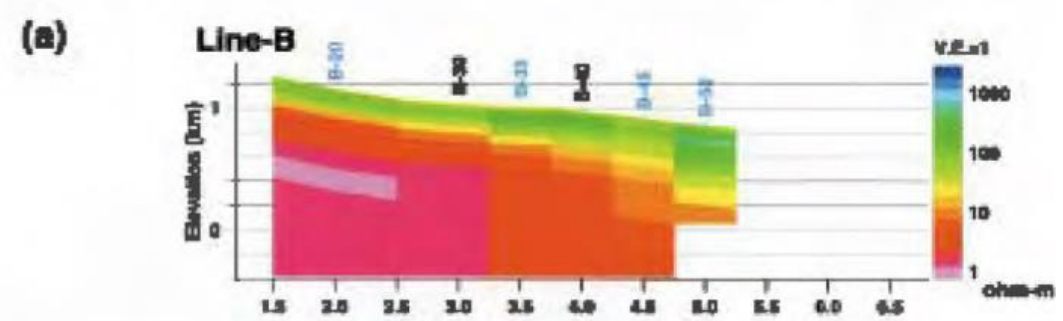
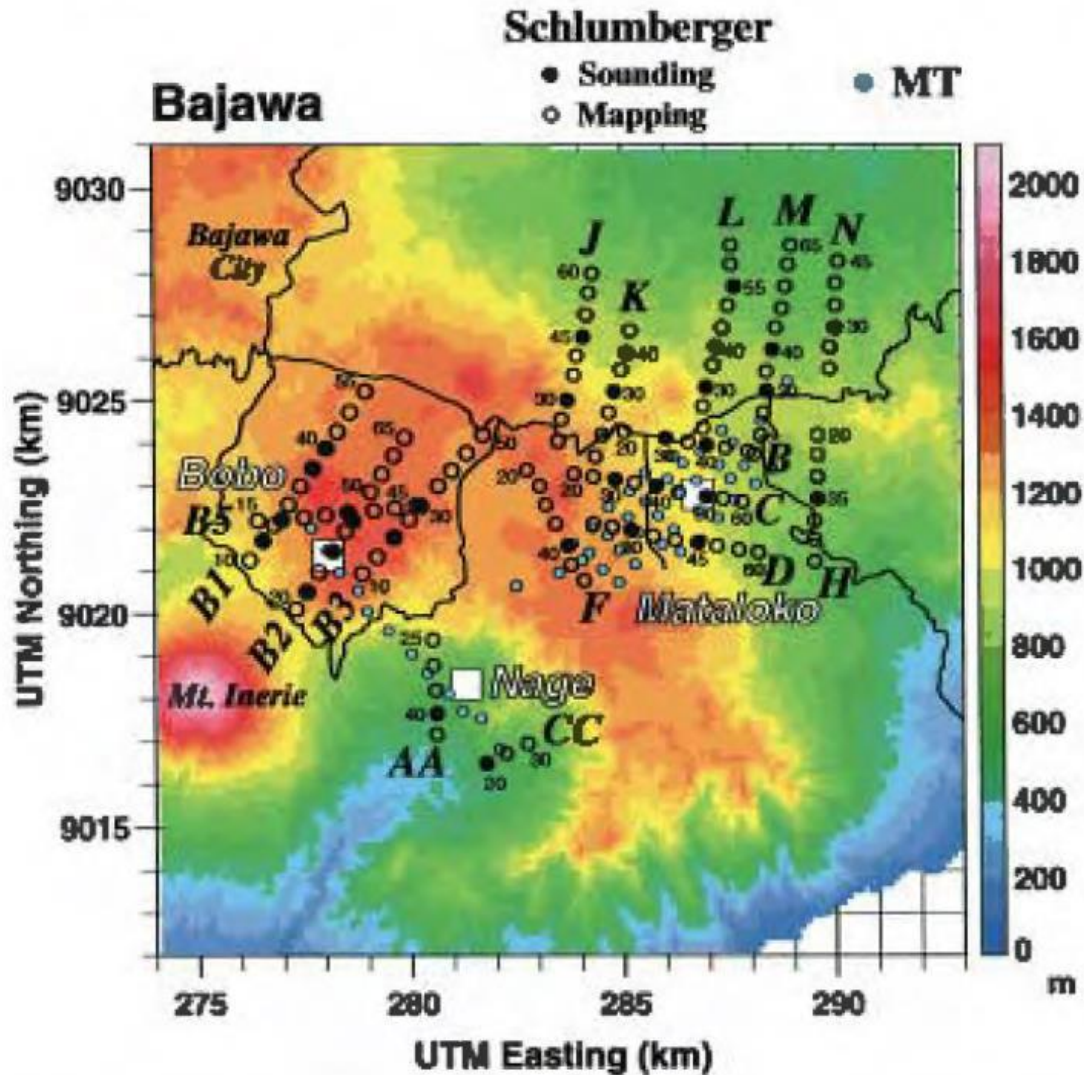
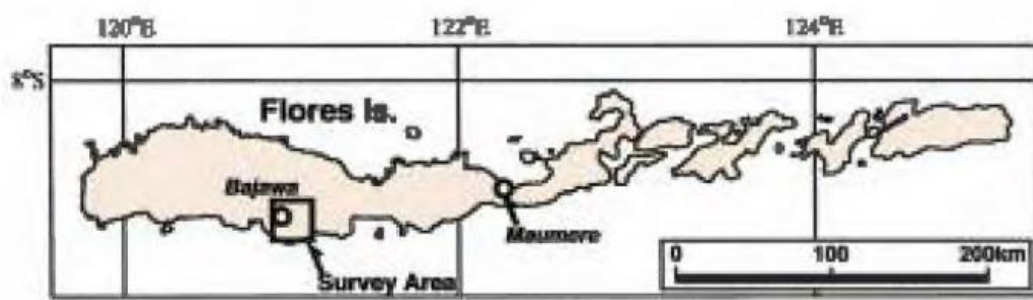


**Factor affect
resistivity**

VERTICAL ELECTRICAL SOUNDING







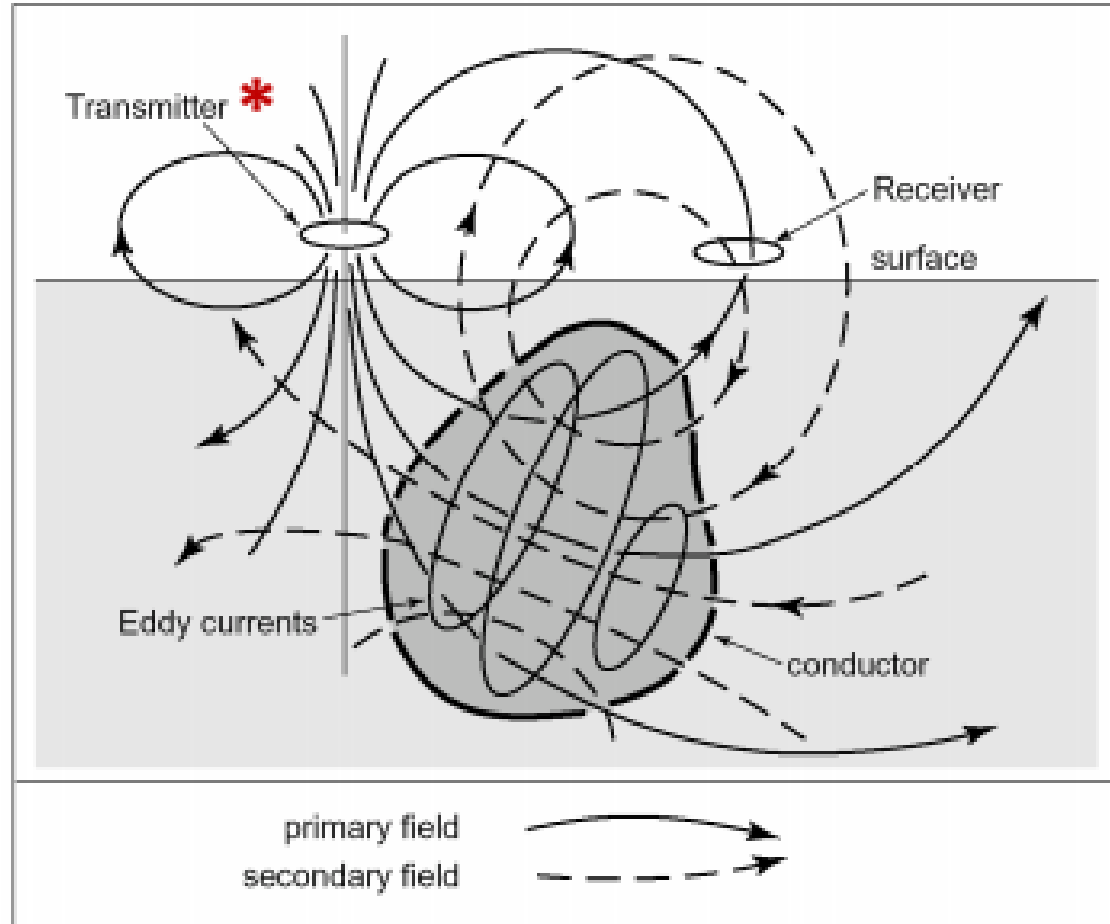
ELECTROMAGNETIC METHOD

Prospeksi geofisika yang memanfaatkan prinsip **induksi EM** untuk mencitrakan struktur **resistivitas listrik** bawah permukaan

Resistivitas (= tahanan jenis): sifat fisis $\rightarrow$ mudah tidaknya batuan mengalirkan arus listrik, ρ (ohm.m)

Konduktivitas: kebalikan resistivitas, $\sigma = 1/\rho$ (S/m)





*no need for transmitter
in MT method

primary time varying
EM fields



Eddy currents in the
conductor (Earth)



secondary (induced)
EM field



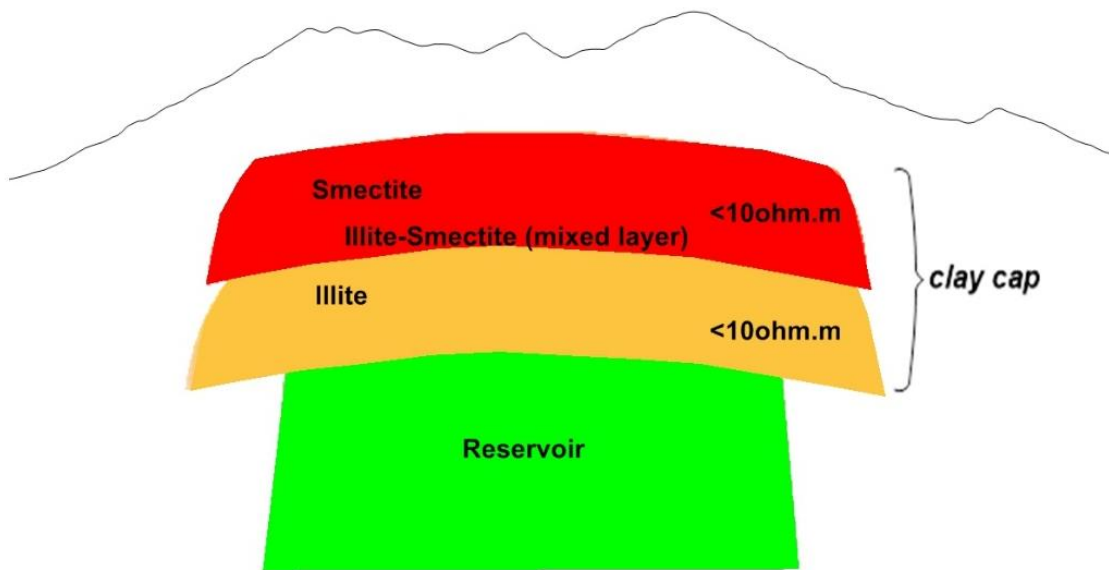
EM fields sensed at
the receiver

Grandis, 2021



- Kebanyakan batuan vulkanik bersifat resistif
- Kehadiran fluida pengisi batuan pada pori/rekahan → menurunkan resistivitas (menaikkan konduktivitas) → konduktor elektrolitik
- Temperatur yang tinggi → menurunkan resistivitas

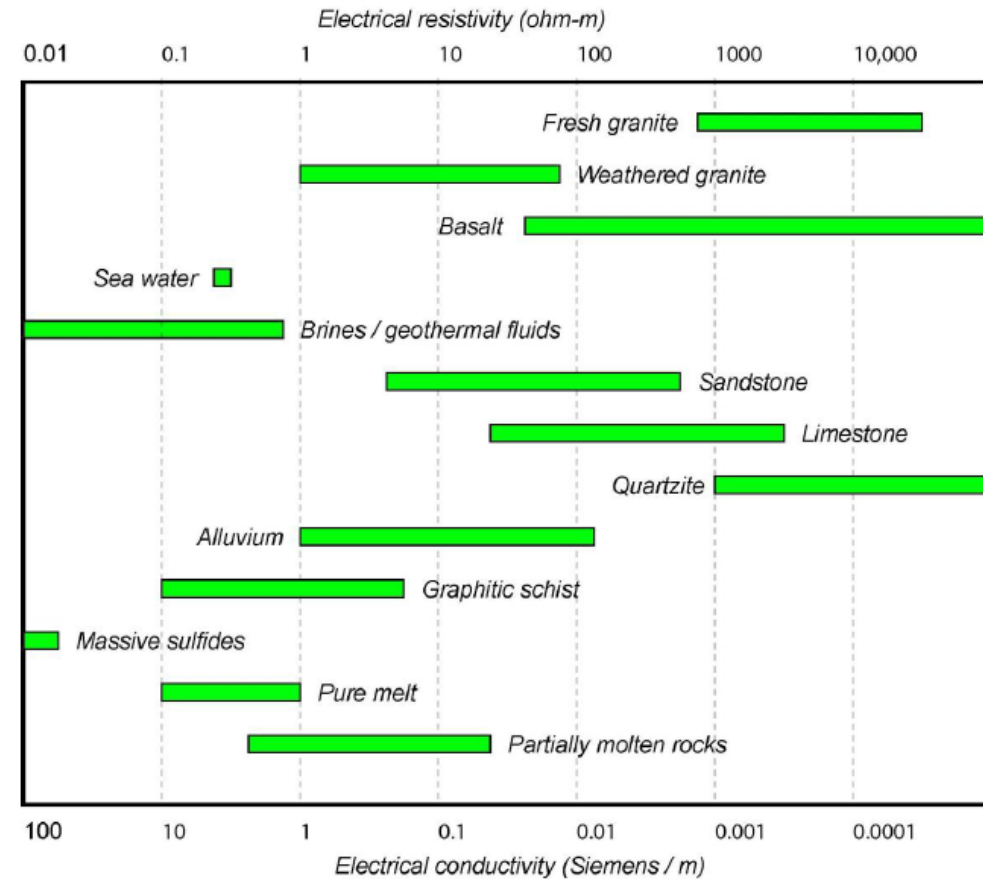
Alterasi hidrotermal → **pembentukan mineral lempung (*illite, smectite, montmorillonite*) konduktif** disertai densifikasi sedimen *host* → peningkatan nilai resistivitas (Pellerin et al., 1996).



ELECTROMAGNETIC METHOD

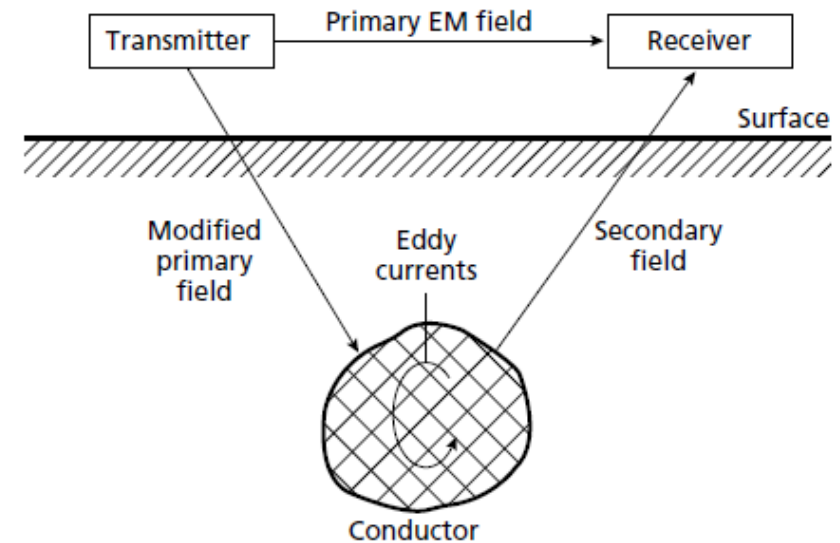
Resistivitas batuan (bulk):

- Resistivitas matriks
- Resistivitas fluida pengisi pori (termasuk salinitas)
- Porositas
- Permeabilitas
- Temperatur
- Kandungan lempung



ELECTROMAGNETIC METHOD

- ❑ Medan EM terdiri dari medan listrik (**E**) dan medan magnet (**B**) yang bervariasi terhadap waktu
- ❑ Penjalaran medan EM pada material bumi akan menginduksi (membangkitkan) sistem arus Eddy → sumber medan EM sekunder → informasi struktur bawah permukaan



ELECTROMAGNETIC METHOD

METODE EM

AKTIF
(Sumber artifisial)

Frequency Domain:

- **Controlled-Source Audio-Magnetotellurik (CSAMT)**
- Very Low Frequency (VLF) → semi pasif

Time Domain:

- **Time Domain EM (TDEM) sering disebut juga Transient (EM)**
- Ground Penetrating Radar → dangkal (belasan m)

PASIF
(Sumber alami)

Frequency Domain:

- **Magnetotellurik (MT)**
- **Audio-Magnetotellurik (AMT)**

Sumber artifisial:

- Dipol listrik → CSAMT, TDEM
- Loop (koil) dialiri listrik → TDEM



ELECTROMAGNETIC METHOD

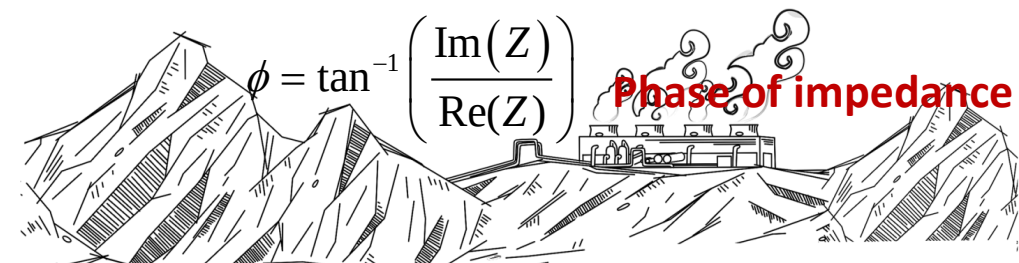
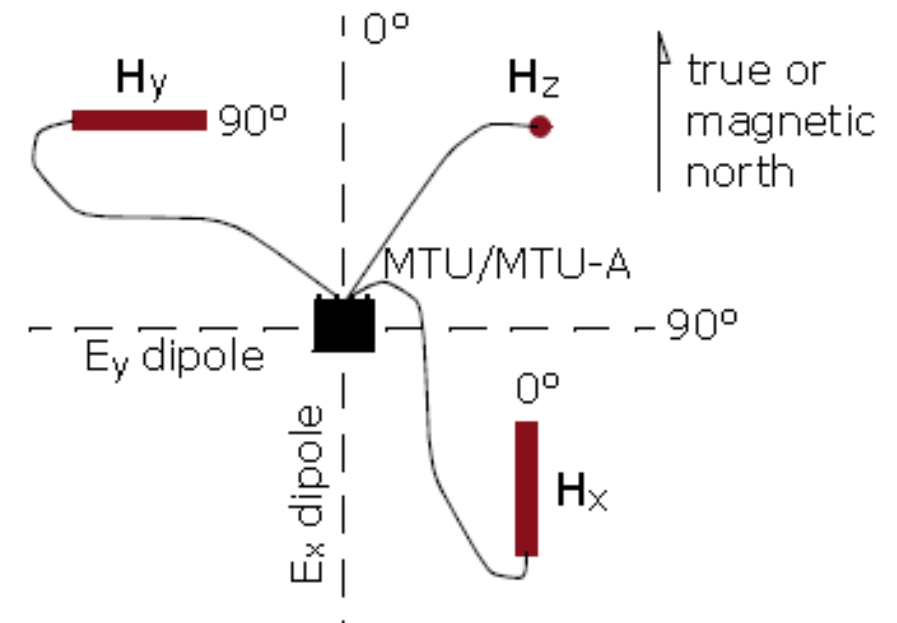
- Inductive passive sounding technique for measuring the natural magnetic and electric fields variation at the surface → conductivity distribution in the subsurface.
- Source electromagnetic (EM) excitation:
 - interaction between solar plasma and Earth's magnetic field (< 1 Hz)
 - activity in the ionosphere (> 1 Hz)

$E = ZH$ Relation between electric and magnetic fields

$$Z = \begin{bmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{bmatrix} \text{ Tensor of impedance}$$

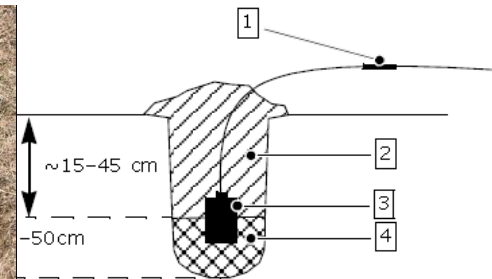
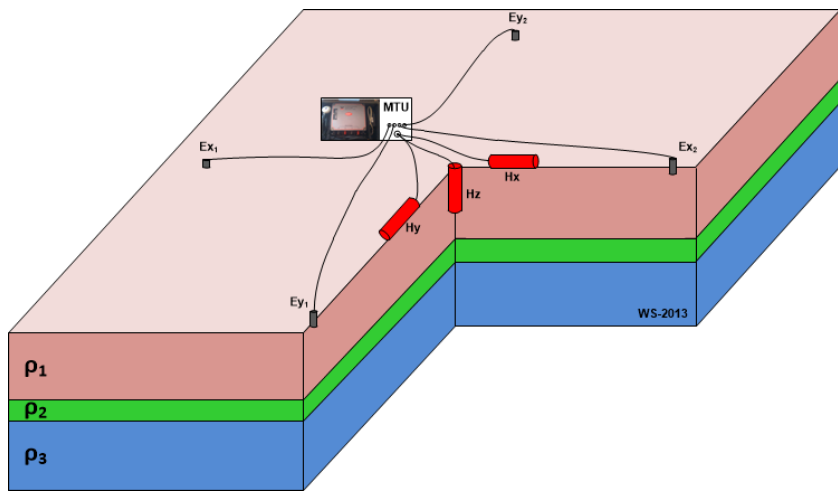
$$\rho_a = \frac{1}{\mu\omega} |Z_{ij}| \text{ Apparent resistivity}$$

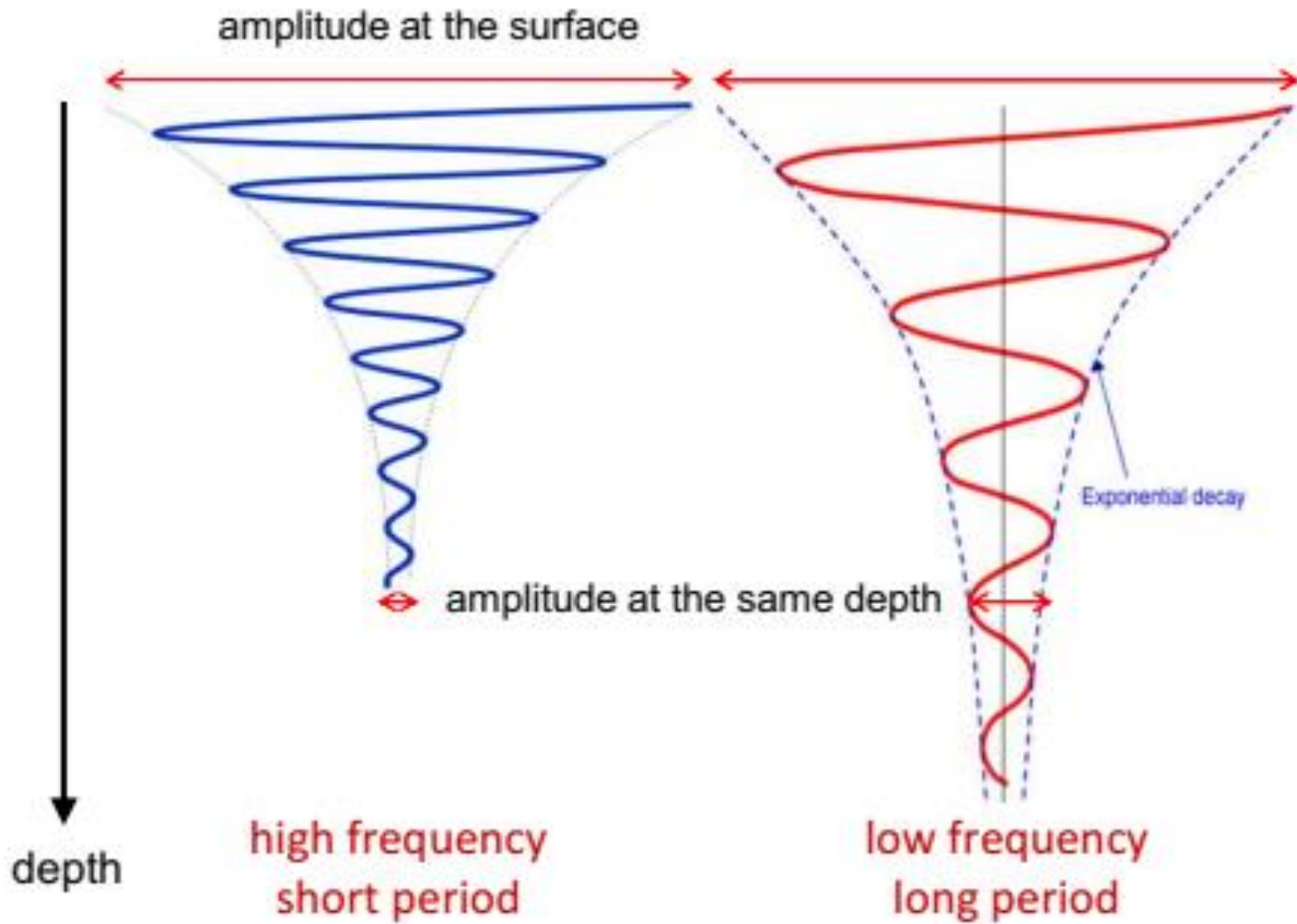
$$\phi = \tan^{-1} \left(\frac{\text{Im}(Z)}{\text{Re}(Z)} \right) \text{ Phase of impedance}$$

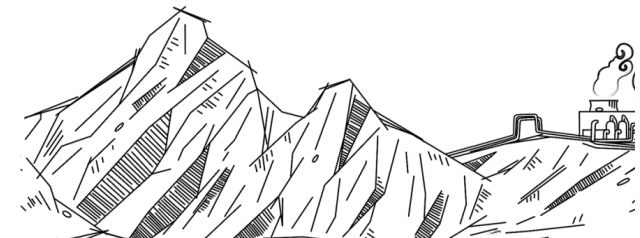
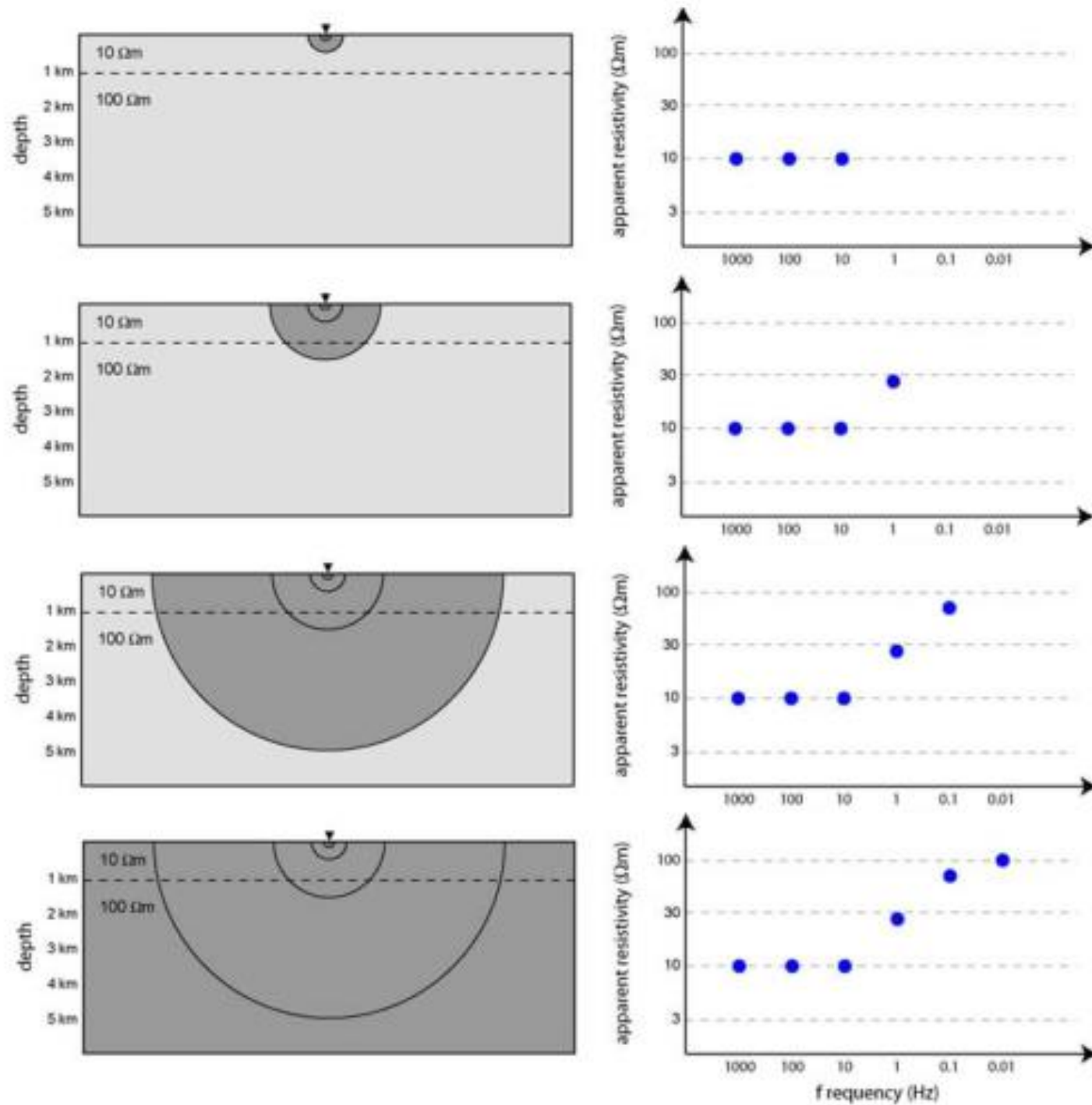


MAGNETOTELLURICS (MT)

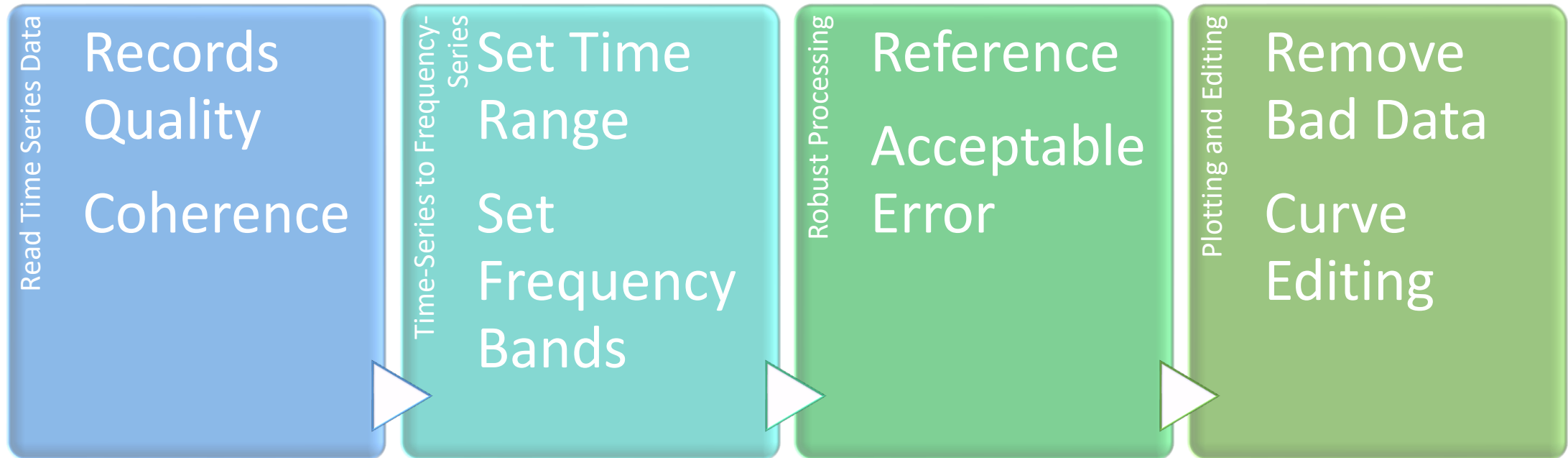
Mengukur **variasi medan listrik** dan **medan magnet** dipermukaan untuk mendapatkan informasi mengenai informasi **distribusi konduktivitas** bawah permukaan bumi.

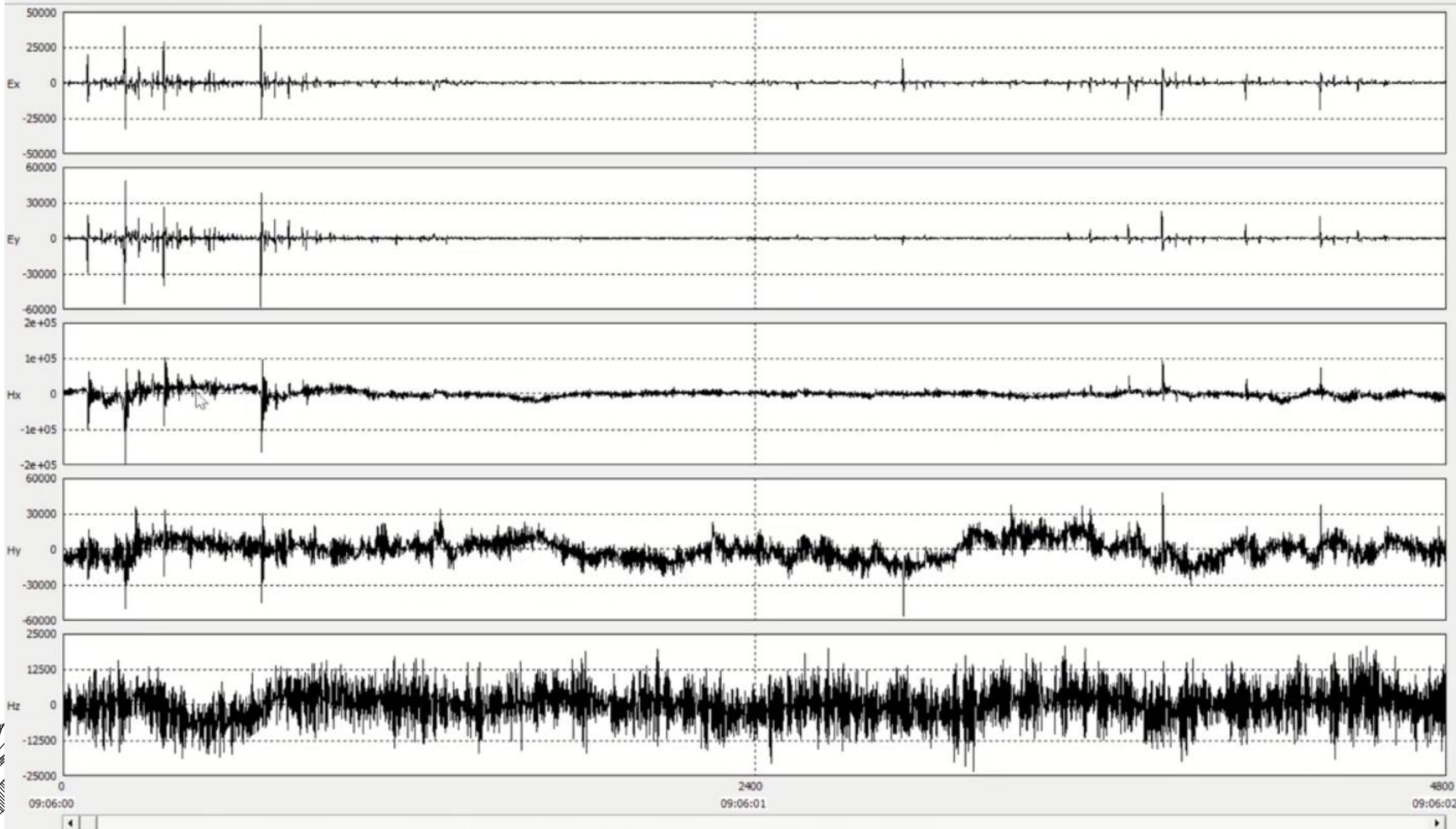


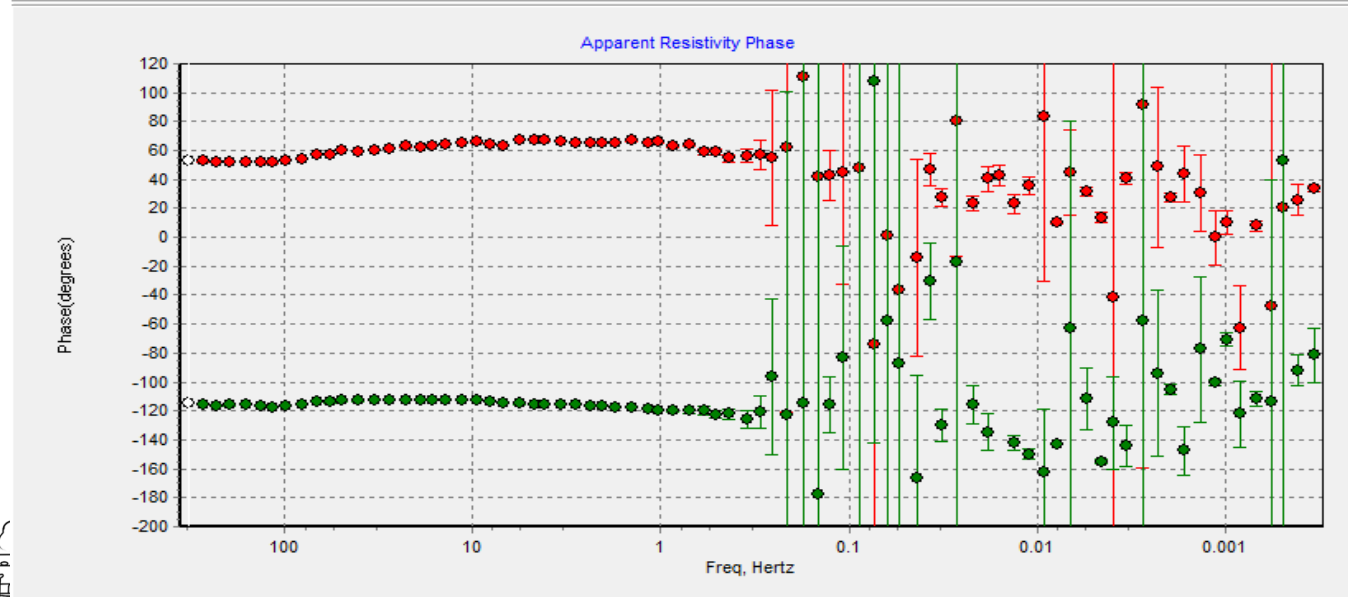
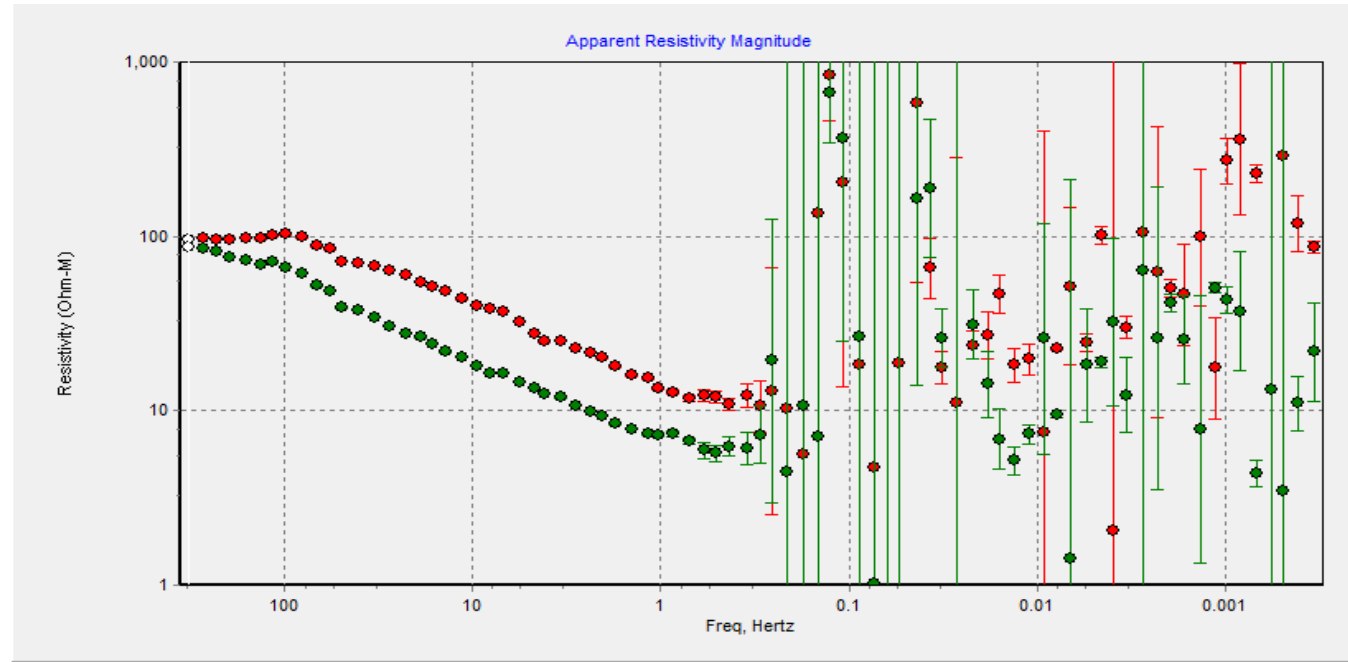


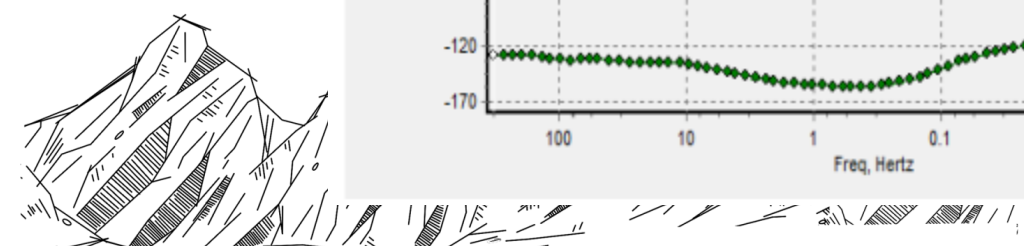
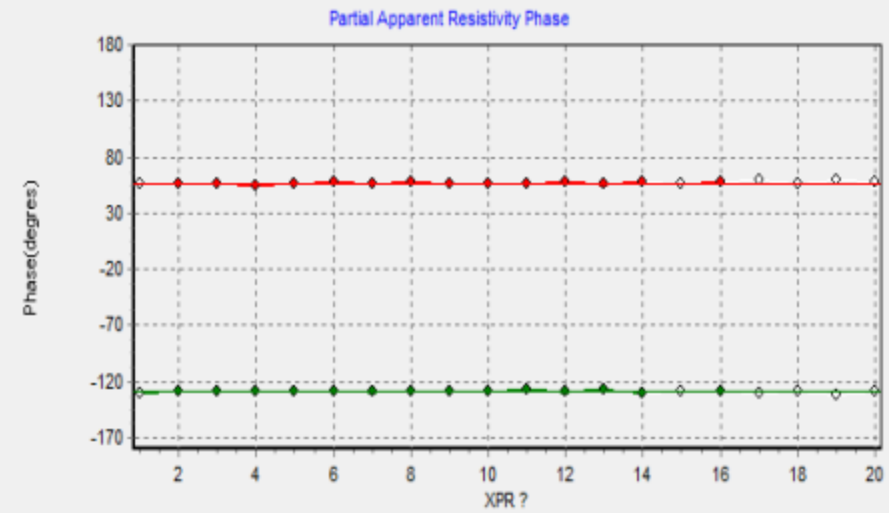
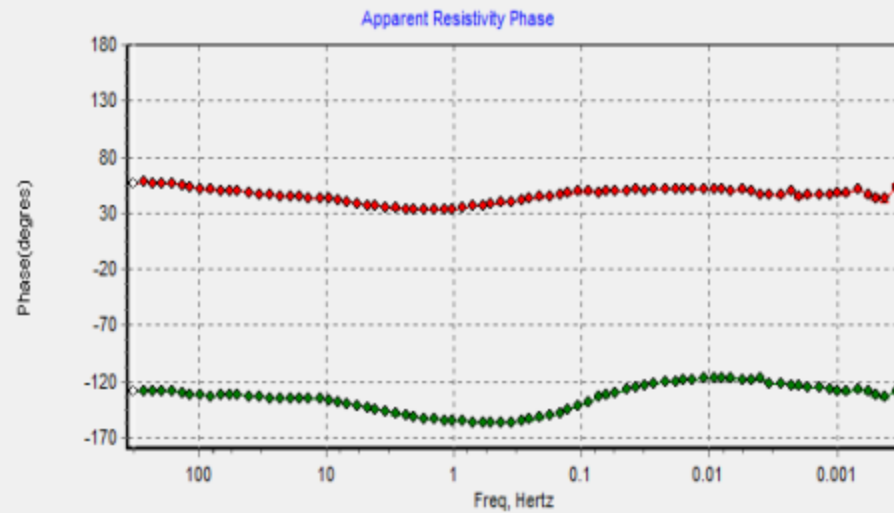
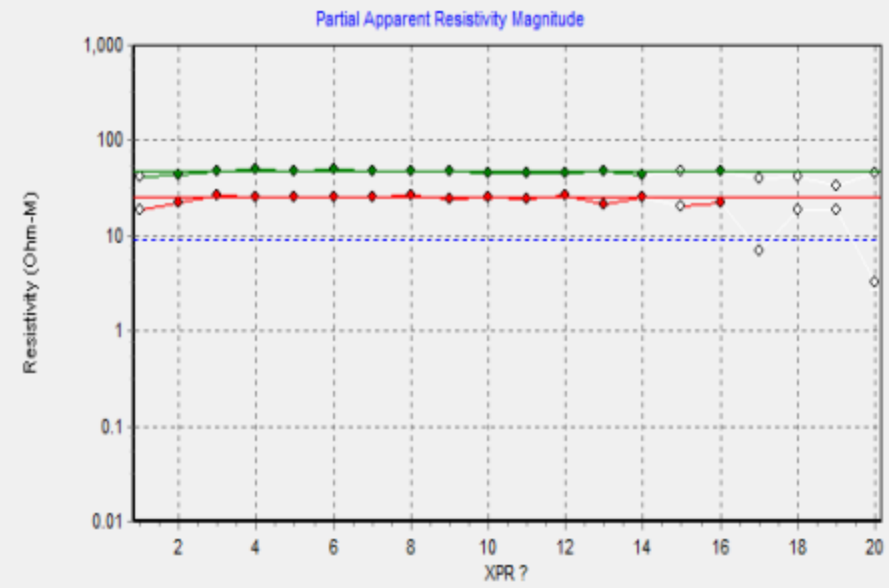
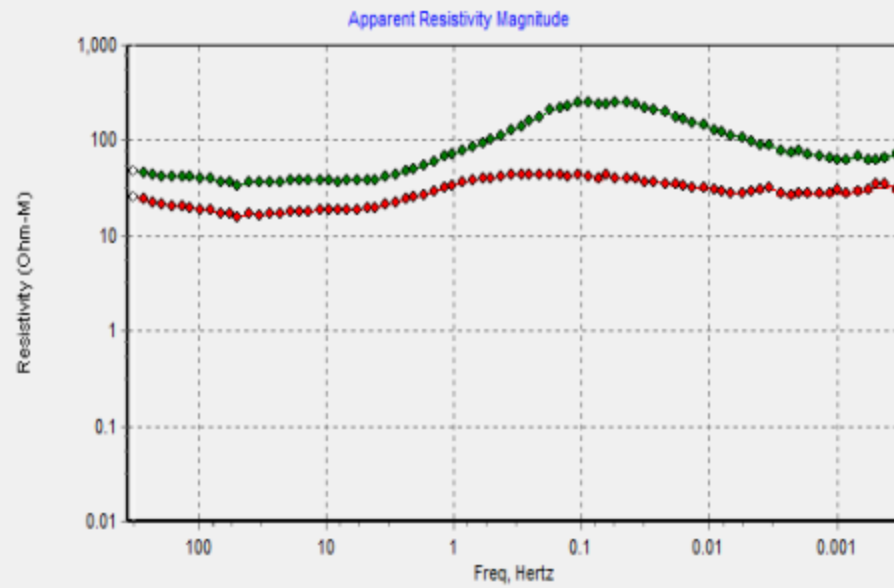


Processing Flow

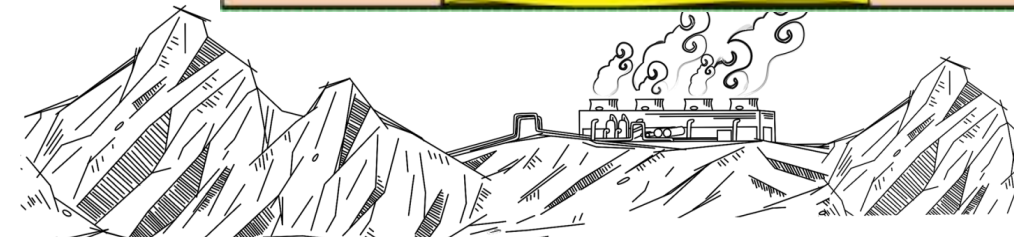
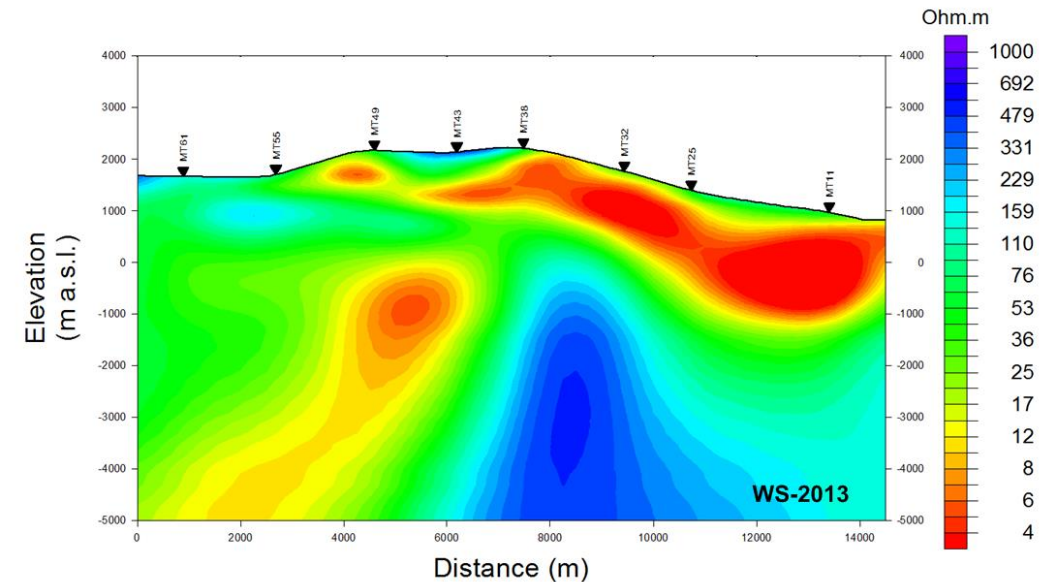
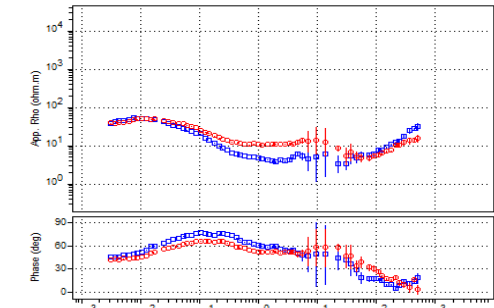
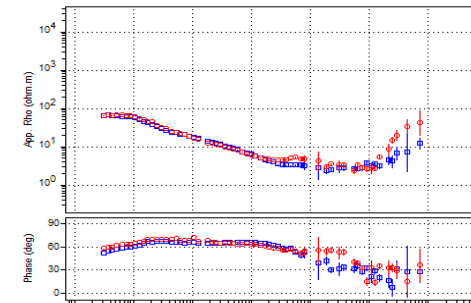
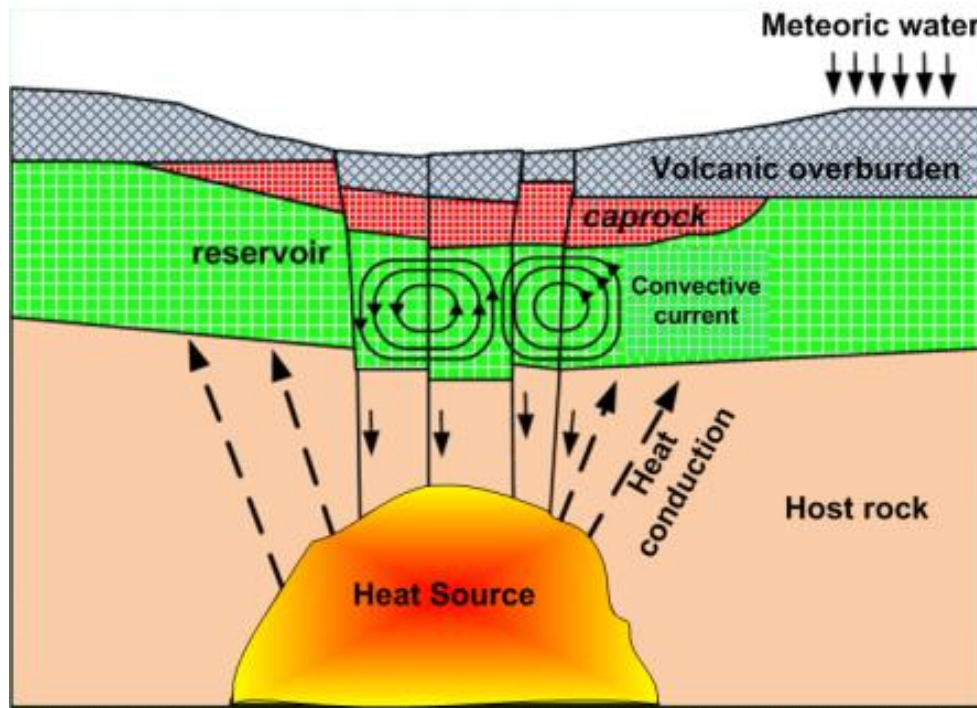




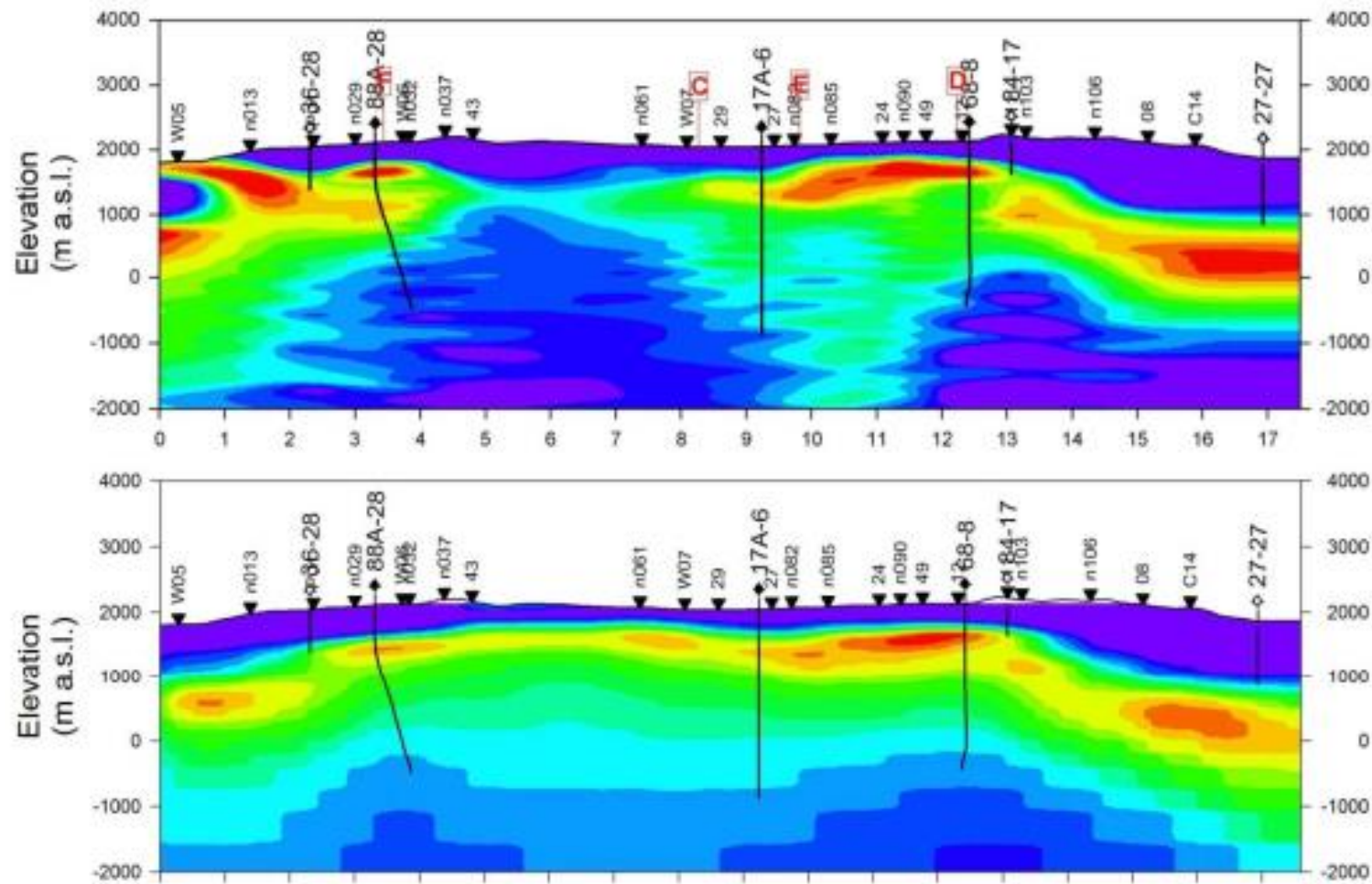




MAGNETOTELLURICS (MT)



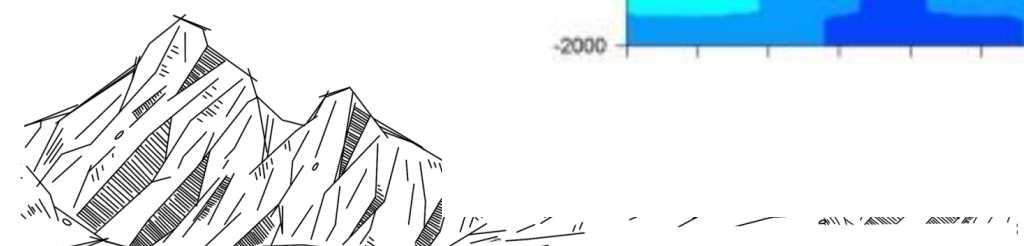
MT MODELLING

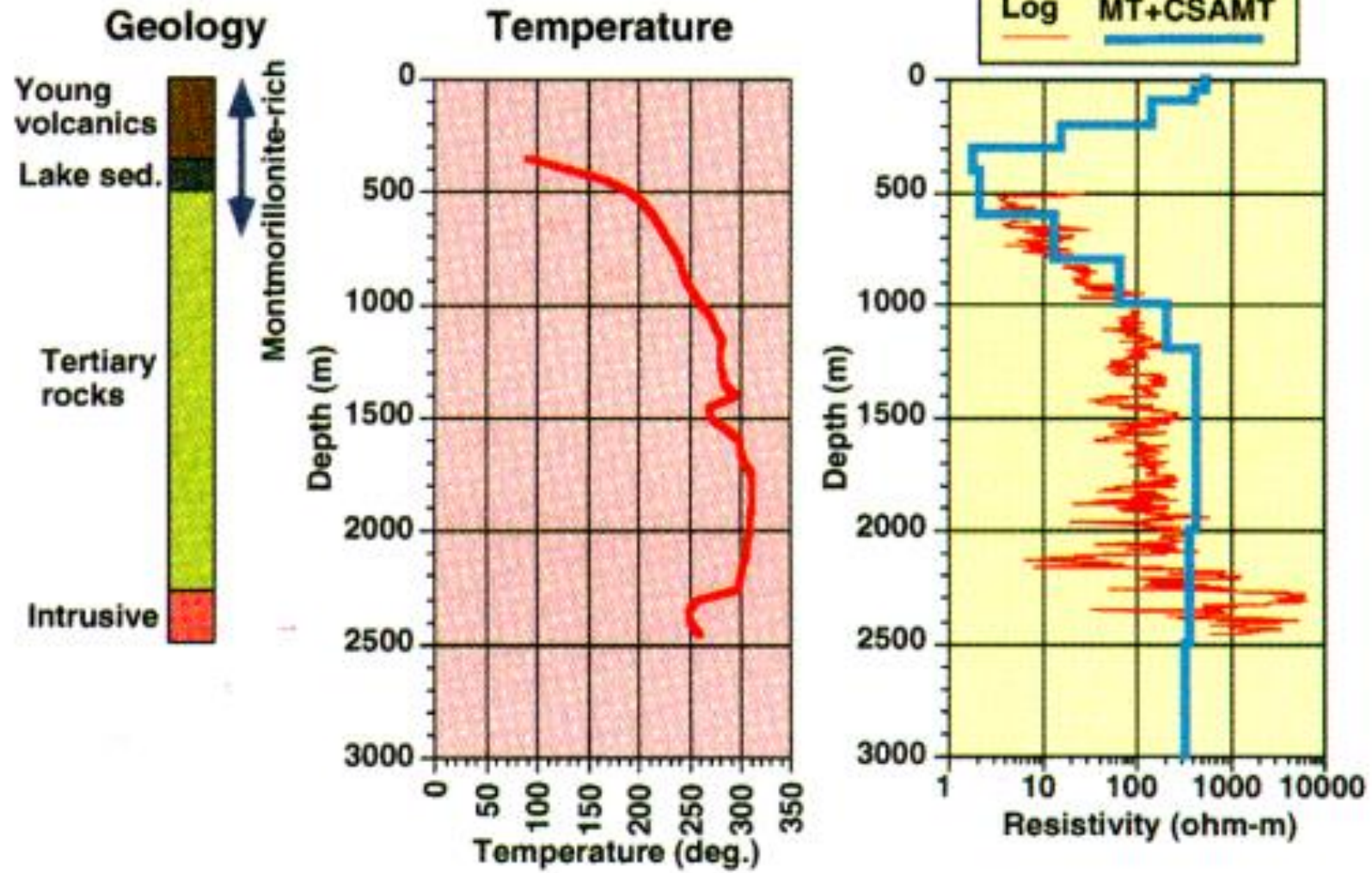


1D
(smooth)

2D

(Cumming & Mackie, 2010)





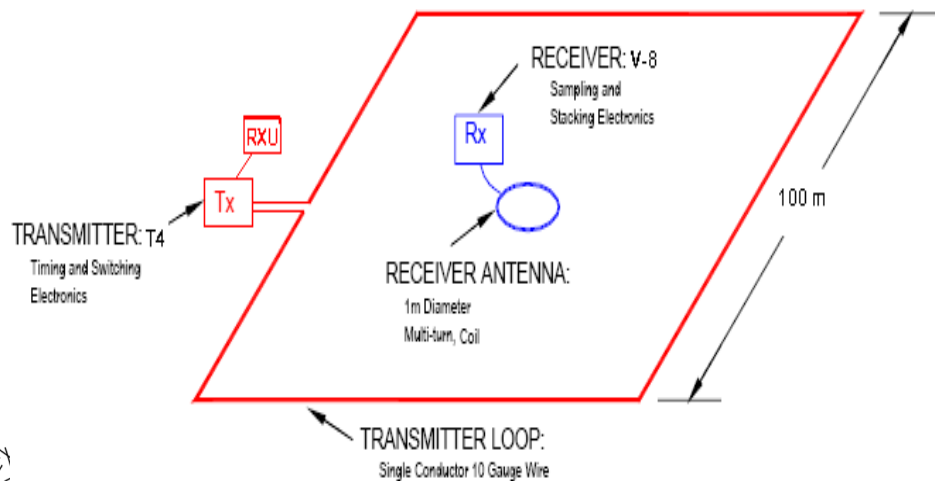
(Uchida et al., 1994)



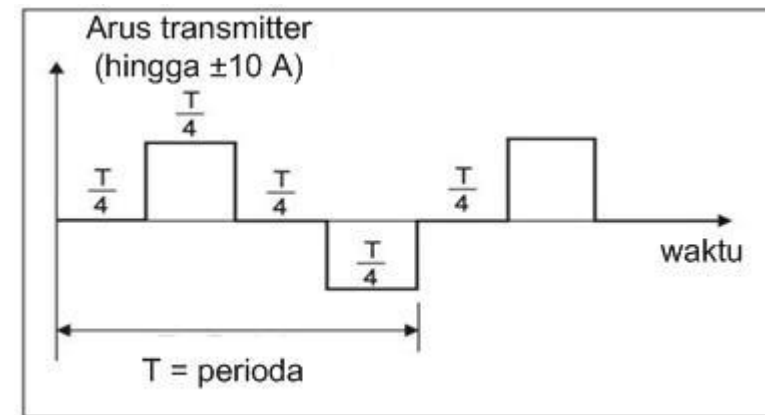
TDEM

Time-domain electromagnetic atau kadang disebut TEM (*transient electromagnetic* → metoda ini menggunakan sumber buatan berupa gelombang elektromagnetik yang dibangkitkan oleh *coil* (kumparan) berdiameter tertentu misalnya 100 m yang dialiri arus yang bervariasi terhadap waktu.

Penggunaan TDEM dengan *loop* arus → aplikasi dangkal (300-500 m) dan juga digunakan untuk proses **koreksi statik data MT**. Keberadaan anomali konduktif dangkal menyebabkan bergesernya kurva resistivitas semu MT (efek statik).



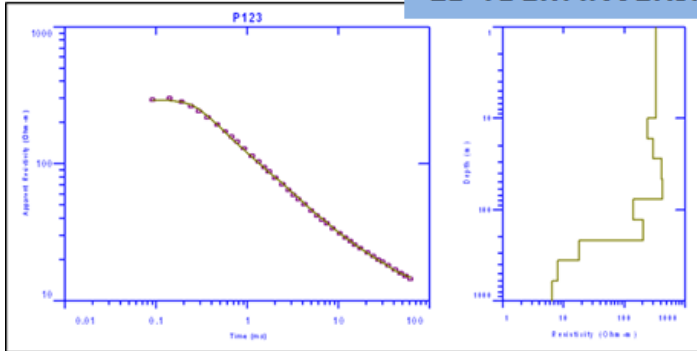
Setup alat TDEM di lapangan



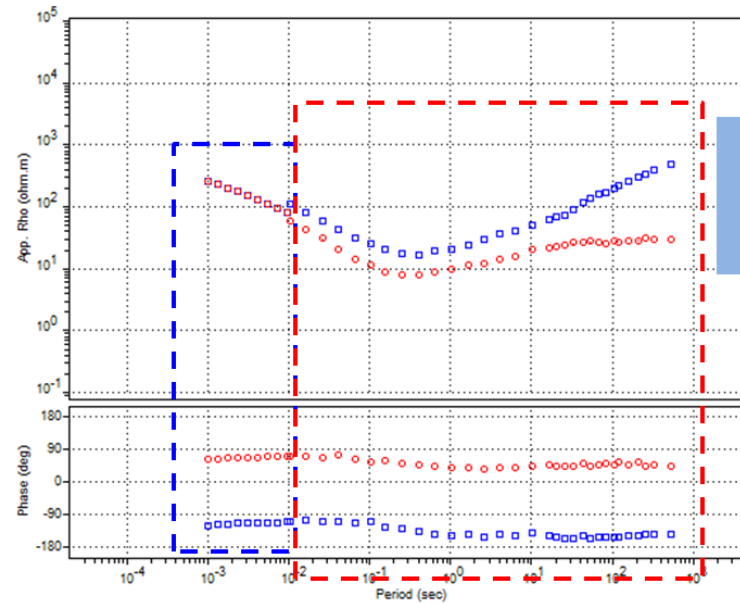
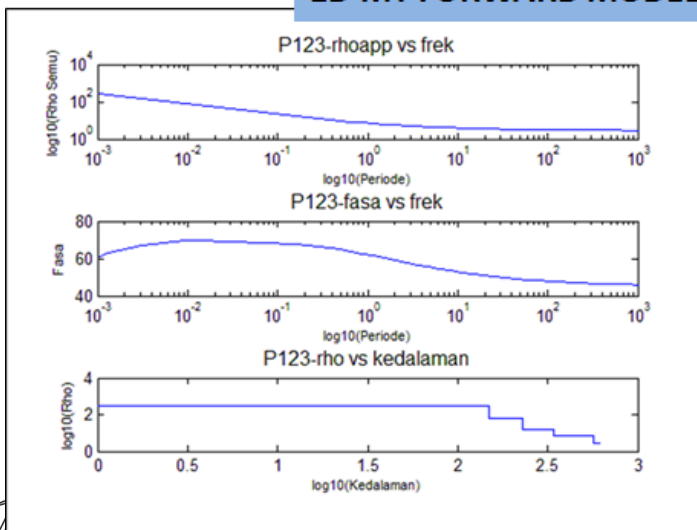
Bentuk muka gelombang pada *transmitter* TDEM

TDEM

1D TDEM INVERSION

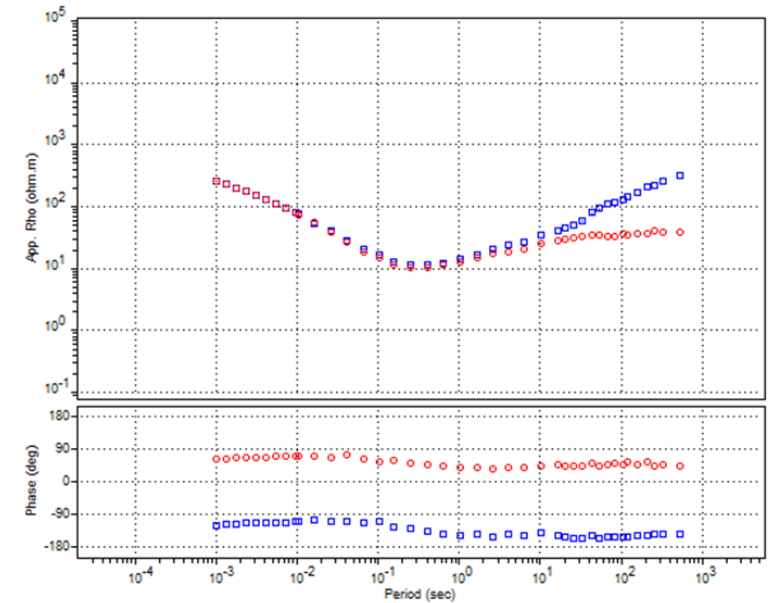


1D MT FORWARD MODELING



TDEM BASED MT APP RESISTIVITY AND PHASE IMPEDANCE

MT APP RESISTIVITY AND PHASE IMPEDANCE



Thank You

